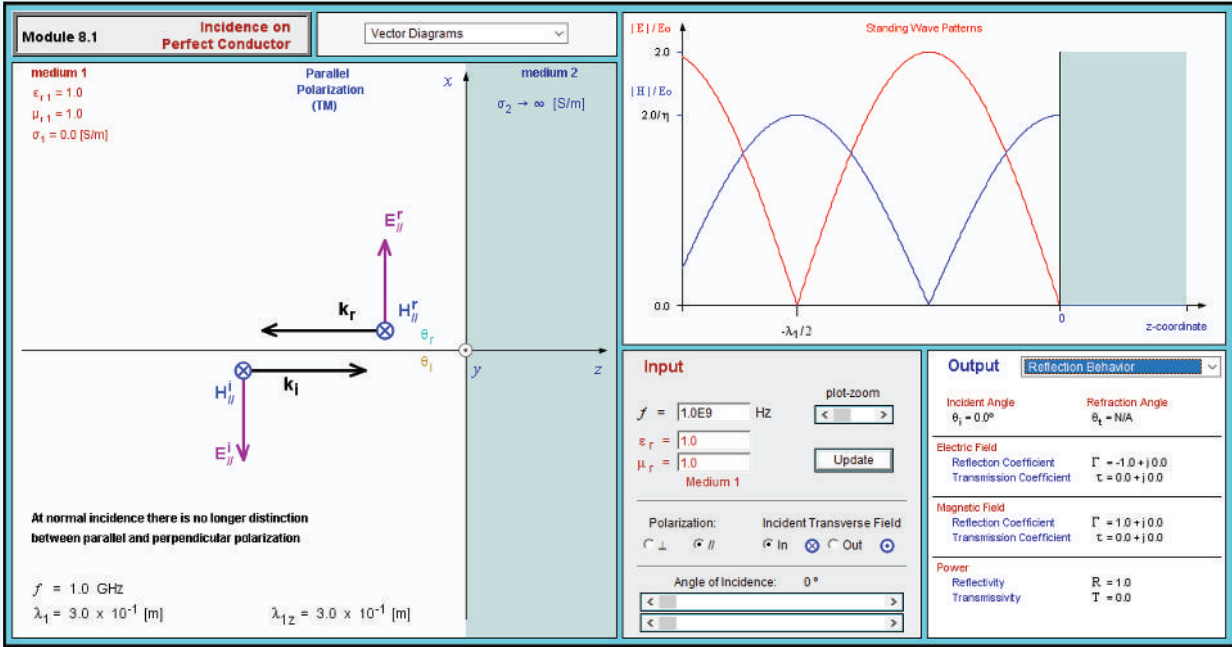


Module 8.1 Incidence on Perfect Conductor Observe the standing-wave pattern created by the combination of a wave incident normally onto the plane surface of a conductor and its reflection.



Concept Question 8-3: Explain on the basis of boundary conditions why it is necessary that $\Gamma = -1$ at the boundary between a dielectric and a perfect conductor.

Exercise 8-1: To eliminate reflections of normally incident plane waves, a dielectric slab of thickness d and relative permittivity ϵ_{r2} is to be inserted between two semi-infinite media with relative permittivities $\epsilon_{r1} = 1$ and $\epsilon_{r3} = 16$. Use the quarter-wave transformer technique to select d and ϵ_{r2} . Assume $f = 3$ GHz.

Answer: $\epsilon_{r2} = 4$ and $d = (1.25 + 2.5n)$ (cm), with $n = 0, 1, 2, \dots$ (See [EM](#).)

Exercise 8-2: Express the normal-incidence reflection coefficient at the boundary between two nonmagnetic, conducting media in terms of their complex permittivities.

Answer: For incidence in medium 1 ($\epsilon_1, \mu_0, \sigma_1$) onto medium 2 ($\epsilon_2, \mu_0, \sigma_2$),

$$\Gamma = \frac{\sqrt{\epsilon_{c1}} - \sqrt{\epsilon_{c2}}}{\sqrt{\epsilon_{c1}} + \sqrt{\epsilon_{c2}}},$$

with $\epsilon_{c1} = (\epsilon_1 - j\sigma_1/\omega)$ and $\epsilon_{c2} = (\epsilon_2 - j\sigma_2/\omega)$. (See [EM](#).)

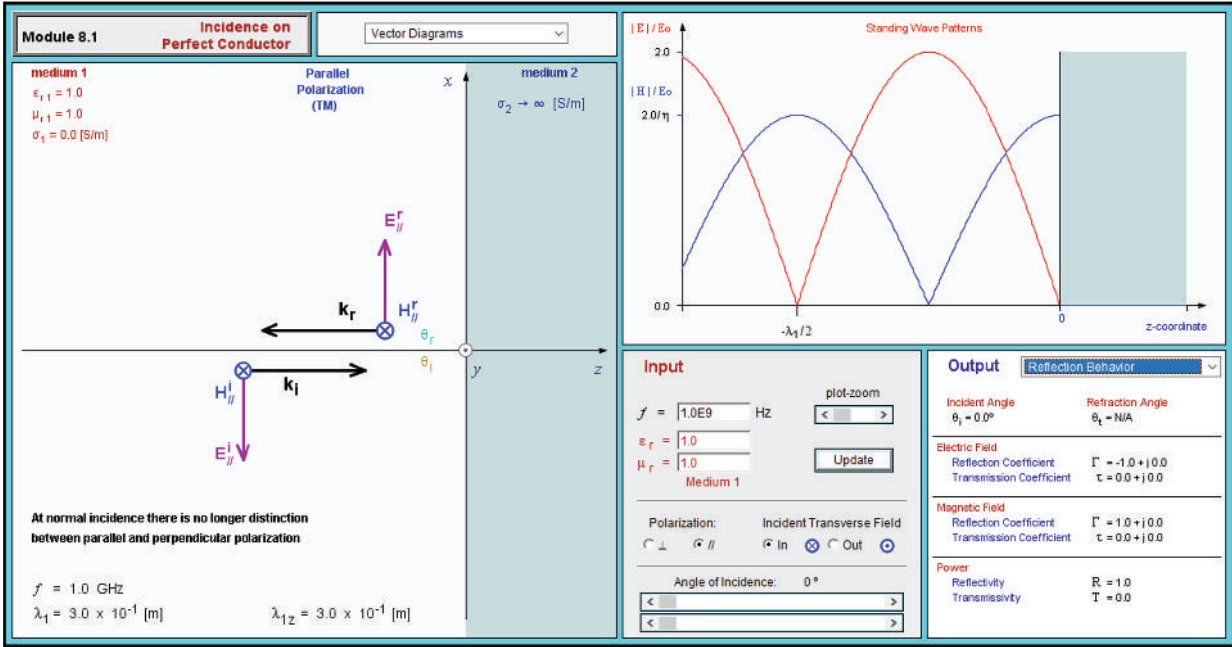
Exercise 8-3: Obtain expressions for the average power densities in media 1 and 2 for the fields described by Eqs. (8.22a) through (8.23b), assuming medium 1 is slightly lossy with η_{c1} approximately real.

Answer: (See [EM](#).)

$$S_{av1} = \hat{\mathbf{z}} \frac{|E_0^i|^2}{2\eta_{c1}} (e^{-2\alpha_1 z} - |\Gamma|^2 e^{2\alpha_1 z}),$$

$$S_{av2} = \hat{\mathbf{z}} |\tau|^2 \frac{|E_0^i|^2}{2} e^{-2\alpha_2 z} \Re \left(\frac{1}{\eta_{c2}^*} \right).$$

模块 8.1 理想导体上的入射 观察由垂直入射到导体平面上的波与其反射相结合而产生的驻波图案。



概念问题 8-3：根据边界条件解释为什么 $\Gamma = -1$ 必须位于电介质和理想导体之间的边界处。

练习 8-1：为了消除正常入射平面波的反射，将厚度为 d 且相对介电常数为 ϵ_{r2} 的电介质板插入两个相对介电常数为 $\epsilon_{r1} = 1$ 和 $\epsilon_{r3} = 16$ 的半无限介质之间。16. 使用四分之一波长变换器技术选择 d 和 ϵ_{r2} 。假设 $f = 3$ GHz。

答案： $\epsilon_{r2} = 4$ 和 $d = (1.25 + 2.5n)$ (cm)，其中 $n = 0, 1, 2, \dots$ 。（参见 [EM](#)。）

练习 8-2：用复介电常数表示两种非磁性导电介质边界的法向入射反射系数。

答案：对于介质 1 ($\epsilon_1, \mu_0, \sigma_1$) 中的入射到介质 2 ($\epsilon_2, \mu_0, \sigma_2$) 中，

$$\Gamma = \frac{\sqrt{\epsilon_{c1}} - \sqrt{\epsilon_{c2}}}{\sqrt{\epsilon_{c1}} + \sqrt{\epsilon_{c2}}},$$

与 $\epsilon_{c1} = (\epsilon_1 - j\sigma_1/\omega)$ 和 $\epsilon_{c2} = (\epsilon_2 - j\sigma_2/\omega)$ 。（参见 [EM](#)。）

练习 8-3：获得方程 1 和 2 所描述的场中的平均功率密度表达式。(8.22a) 到 (8.23b)，假设介质 1 略有损耗，且 η_{c1} 近似为实数。

答案：（参见 [EM](#)。）

$$S_{av1} = \hat{\mathbf{z}} \frac{|E_0^i|^2}{2\eta_{c1}} (e^{-2\alpha_1 z} - |\Gamma|^2 e^{2\alpha_1 z}),$$

$$S_{av2} = \hat{\mathbf{z}} |\tau|^2 \frac{|E_0^i|^2}{2} e^{-2\alpha_2 z} \Re \left(\frac{1}{\eta_{c2}^*} \right).$$

Technology Brief 15: Lasers

Lasers are used in CD and DVD players, bar-code readers, eye surgery, and multitudes of other systems and applications (Fig. TF15-1).

► A laser—acronym for **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation—is a source of **monochromatic** (single wavelength), **coherent** (uniform wavefront), narrow-beam light. ◀

This is in contrast with other sources of light (such as the sun or a light bulb), which usually encompass waves of many different wavelengths with random phase (incoherent). A laser source generating microwaves is called a **maser**. The first maser was built in 1953 by Charles Townes, and the first laser was constructed in 1960 by Theodore Maiman.

Basic Principles

Despite its complex quantum-mechanical structure, an atom can be conveniently modeled as a nucleus (containing protons and neutrons) surrounded by a cloud of electrons. Associated with the atom or molecule of any given material is a specific set of **quantized** (discrete) **energy states** (orbits) that the electrons can occupy. Supply of energy (in the form of heat, exposure to intense light, or other means) by an external source can cause an electron to move from a lower energy state to a higher energy (**excited**) state. Exciting the atoms is called **pumping** because it leads to

increasing the population of electrons in higher states (Fig. TF15-2(a)). **Spontaneous emission** of a photon (light energy) occurs when the electron in the excited state moves to a lower state (Fig. TF15-2(b)), and **stimulated emission** (Fig. TF15-2(c)) happens when an emitted photon “entices” an electron in an excited state of another atom to move to a lower state, thereby emitting a second photon of identical energy, wavelength, and wavefront (phase).

Principle of Operation

► Highly amplified stimulated emission is called **lasing**. ◀

The lasing medium can be solid, liquid, or gas. Laser operation is illustrated in Fig. TF15-3 for a ruby crystal surrounded by a flash tube (similar to a camera flash). A perfectly reflecting mirror is placed on one end of the crystal and a partially reflecting mirror on the other end. Light from the flash tube excites the atoms; some undergo spontaneous emission, generating photons that cause others to undergo stimulated emission; photons moving along the axis of the crystal bounce back and forth between the mirrors, causing additional stimulated emission (i.e., amplification), with only a fraction of the photons exiting through the partially reflecting mirror.

► Because all of the stimulated photons are identical, the light wave generated by the laser is of a single wavelength. ◀

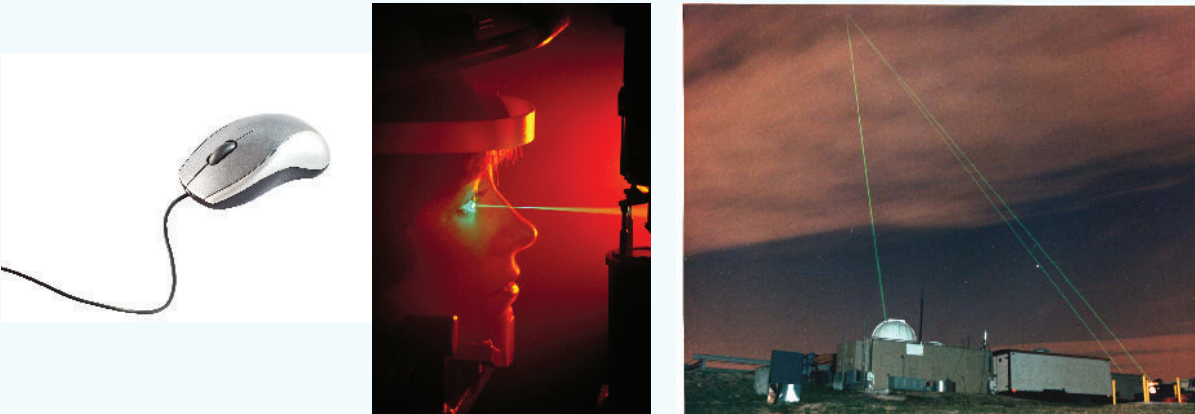


Figure TF15-1 A few examples of laser applications.

技术简介 15：激光器

激光器用于 CD 和 DVD 播放器、条形码阅读器、眼科手术以及众多其他系统和应用 (图 TF15-1)。

► 激光 (受激辐射光放大的缩写) 是单色 (单波长)、相干 (均匀波前) 窄光束光源。 ◀

这与其他光源 (例如太阳或灯泡) 形成对比, 其他光源通常包含具有随机相位 (不相干) 的许多不同波长的波。产生微波的激光源称为微波激光器。第一台微波激光器由 Charles Townes 于 1953 年制造, 第一台激光器由 Theodore Maiman 于 1960 年制造。

基本原则

尽管原子具有复杂的量子力学结构, 但可以方便地将原子建模为被电子云包围的原子核 (包含质子和中子)。与任何给定材料的原子或分子相关的是电子可以占据的一组特定的量子化 (离散) 能态 (轨道)。通过外部源提供能量 (以热量、强光照射或其他方式) 可以使电子从较低能态移动到较高能态 (激发态)。激发原子称为泵浦, 因为它会导致

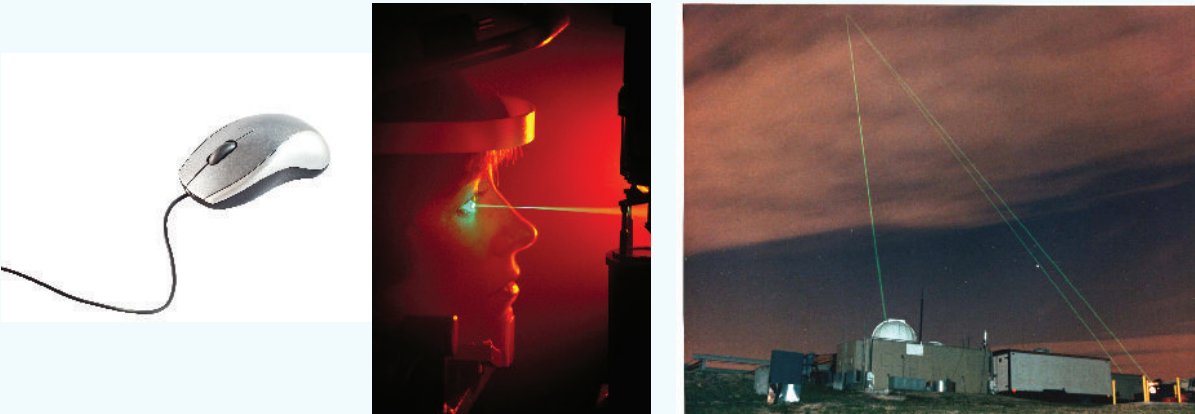
增加更高能态的电子数量 (图 TF15-2(a))。当处于激发态的电子移动到较低的状态时 (图TF15-2 (b)), 会发生光子 (光能) 的自发发射; 当发射的光子 “引诱” 处于激发态的另一个原子的电子移动到较低的状态时, 会发生受激发射 (图TF15-2 (c)), 从而发射具有相同能量、波长和波前 (相位) 的第二个光子。

工作原理

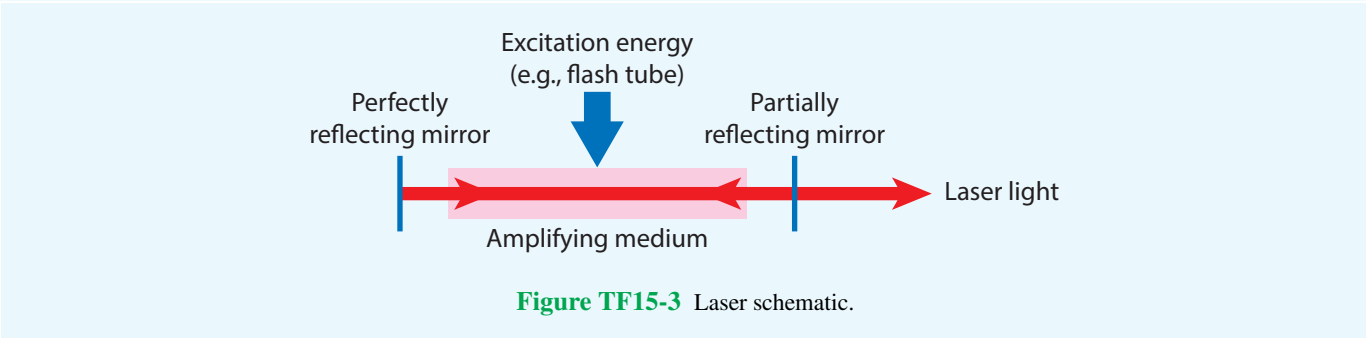
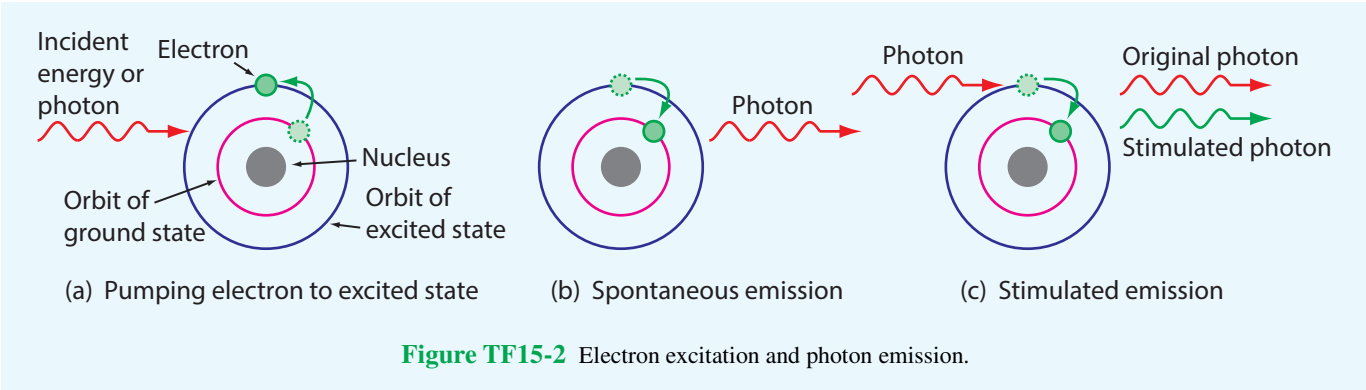
► 高度放大的受激发射称为激光。 ◀

激光介质可以是固体、液体或气体。图 TF15-3 说明了被闪光灯管 (类似于相机闪光灯) 包围的红宝石晶体的激光操作。晶体的一端放置一面全反射镜, 另一端放置一面部分反射镜。闪光灯管发出的光激发原子; 有些发生自发发射, 产生光子, 导致其他发生受激发射; 沿着晶体轴移动的光子在镜子之间来回反弹, 引起额外的受激发射 (即放大), 只有一小部分光子通过部分反射镜射出。

► 由于所有受激光子都是相同的, 因此激光器产生的光波具有单一波长。 ◀



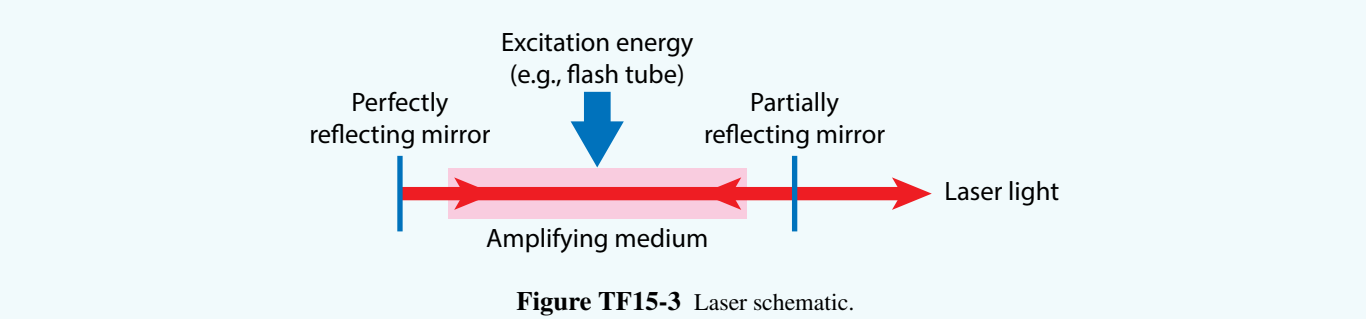
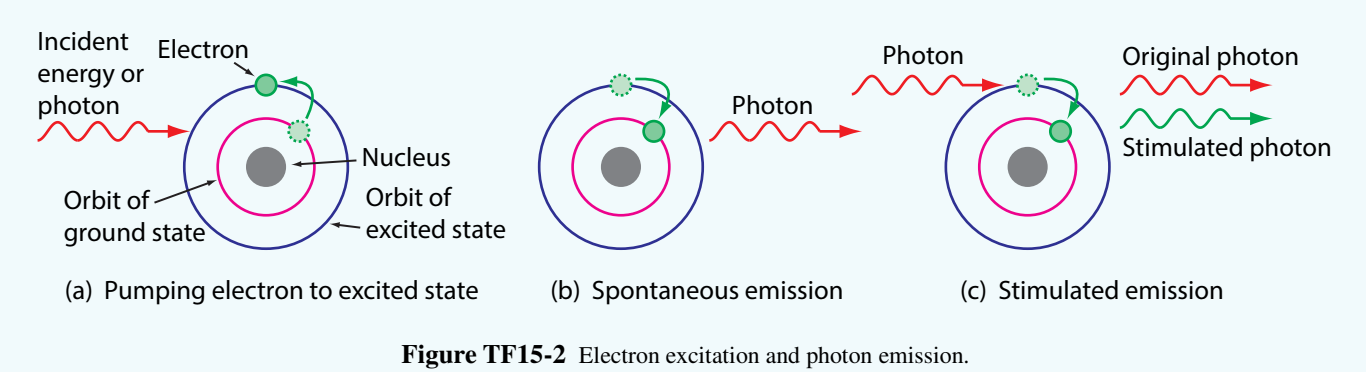
图TF15-1 激光应用的几个示例。



Wavelength (Color) of Emitted Light

The atom of any given material has unique energy states. The difference in energy between the excited high energy state and the stable lower energy state

determines the wavelength of the emitted photons (EM wave). Through proper choice of lasing material, monochromatic waves can be generated with wavelengths in the ultraviolet, visible, infrared or microwave bands.



发射光的波长（颜色）

任何给定材料的原子都具有独特的能态。激发的高能态和稳定的低能态之间的能量差

决定发射光子（电磁波）的波长。通过正确选择激光材料，可以产生波长在紫外线、可见光、红外线或微波波段的单色波。

8-2 Snell's Laws

In the preceding sections, we examined reflection and transmission of plane waves that are normally incident upon a planar interface between two different media. We now consider the oblique-incidence case depicted in **Fig. 8-9**, and for simplicity, we assume all media to be lossless. The $z = 0$ plane forms the boundary between media 1 and 2 with constitutive parameters (ϵ_1, μ_1) and (ϵ_2, μ_2) , respectively. The two lines in **Fig. 8-9** with direction $\hat{\mathbf{k}}_i$ represent rays drawn normal to the wavefront of the incident wave, and those along directions $\hat{\mathbf{k}}_r$ and $\hat{\mathbf{k}}_t$ are similarly associated with the reflected and transmitted waves. The **angles of incidence, reflection, and transmission** (or **refraction**), which are defined with respect to the normal to the boundary (the z axis), are θ_i , θ_r , and θ_t , respectively. These three angles are interrelated by **Snell's laws**, which we derive shortly by considering the propagation of the wavefronts of the three waves. Rays of the incident wave intersect the boundary at O and O' . Here A_iO represents a constant-phase wavefront of the incident wave. Likewise, A_rO' and A_tO' are constant-phase wavefronts of the reflected and transmitted waves, respectively (**Fig. 8-9**). The incident and reflected waves propagate in medium 1 with the same phase velocity $u_{p1} = 1/\sqrt{\mu_1\epsilon_1}$, while the transmitted wave in medium 2 propagates with a velocity $u_{p2} = 1/\sqrt{\mu_2\epsilon_2}$. The time it takes for the incident wave to travel from A_i to O' is the same as the time it takes for the reflected wave to travel from O to A_r , and also the time it takes the transmitted wave to travel from O

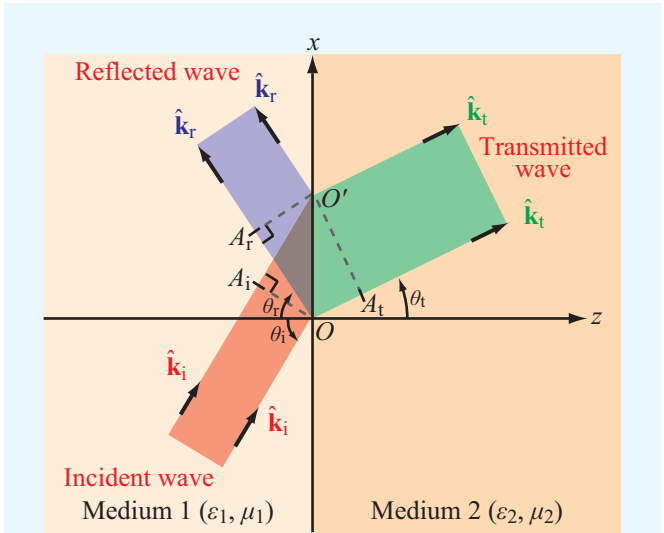


Figure 8-9 Wave reflection and refraction at a planar boundary between different media.

to A_t . Since time equals distance divided by velocity, it follows that

$$\frac{\overline{A_iO'}}{u_{p1}} = \frac{\overline{OA_r}}{u_{p1}} = \frac{\overline{OA_t}}{u_{p2}}. \quad (8.26)$$

From the geometries of the three right triangles in **Fig. 8-9**, we deduce that

$$\overline{A_iO'} = \overline{OO'} \sin \theta_i, \quad (8.27a)$$

$$\overline{OA_r} = \overline{OO'} \sin \theta_r, \quad (8.27b)$$

$$\overline{OA_t} = \overline{OO'} \sin \theta_t. \quad (8.27c)$$

Use of these expressions in Eq. (8.26) leads to

$$\theta_i = \theta_r, \quad \text{(Snell's law of reflection)} \quad (8.28a)$$

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{u_{p2}}{u_{p1}} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}}. \quad (8.28b)$$

(Snell's law of refraction)

► **Snell's law of reflection** states that the angle of reflection equals the angle of incidence, and **Snell's law of refraction** provides a relation between $\sin \theta_t$ and $\sin \theta_i$ in terms of the ratio of the phase velocities. ◀

The **index of refraction** of a medium, n , is defined as the ratio of the phase velocity in free space (i.e., the speed of light c) to the phase velocity in the medium. Thus,

$$n = \frac{c}{u_p} = \sqrt{\frac{\mu \epsilon}{\mu_0 \epsilon_0}} = \sqrt{\mu_r \epsilon_r}. \quad (8.29)$$

(index of refraction)

In view of Eq. (8.29), Eq. (8.28b) may be rewritten as

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{n_1}{n_2} = \sqrt{\frac{\mu_{r1} \epsilon_{r1}}{\mu_{r2} \epsilon_{r2}}}. \quad (8.30)$$

For nonmagnetic materials, $\mu_{r1} = \mu_{r2} = 1$, in which case

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{n_1}{n_2} = \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \frac{\eta_2}{\eta_1} \quad (\text{for } \mu_1 = \mu_2). \quad (8.31)$$

Usually, materials with higher densities have higher permittivities. Air, with $\mu_r = \epsilon_r = 1$, has an index of refraction $n_0 = 1$. Since for nonmagnetic materials $n = \sqrt{\epsilon_r}$, *a material is often*

8-2 斯涅尔定律

在前面的部分中，我们研究了垂直入射到两种不同介质之间的平面界面上的平面波的反射和传输。我们现在考虑图8-9中描述的斜入射情况，为了简单起见，我们假设所有媒体都是无损的。 $z = 0$ 平面分别形成具有本构参数 (ϵ_1, μ_1) 和 (ϵ_2, μ_2) 的介质1和2之间的边界。图 8-9 中方向为 $\hat{\mathbf{k}}_i$ 的两条线表示垂直于入射波波前绘制的射线，而沿方向 $\hat{\mathbf{k}}_r$ 和 $\hat{\mathbf{k}}_t$ 的射线同样与反射波和透射波相关。相对于边界法线 (z 轴) 定义的入射角、反射角和透射角 (或折射角) 分别为 θ_i 、 θ_r 和 θ_t 。这三个角度通过斯涅尔定律相互关联，我们通过考虑三个波的波前传播很快就得出了该定律。入射波的光线在 O 和 O' 处与边界相交。这里 A_iO 表示入射波的恒定相位波前。同样， A_rO' 和 A_tO' 分别是反射波和透射波的恒定相位波前 (图8-9)。入射波和反射波在介质 1 中以相同的相速度 $u_{p1} = 1/\sqrt{\mu_1\epsilon_1}$ 传播，而透射波在介质 2 中以速度 $u_{p2} = 1/\sqrt{\mu_2\epsilon_2}$ 传播。入射波从 A_i 传播到 O' 所需的时间与反射波从 O 传播到 A_r 所需的时间相同，透射波从 O 传播的时间也相同

到 A_t 。由于时间等于距离除以速度，因此得出

$$\frac{A_iO'}{u_{p1}} = \frac{OA_r}{u_{p1}} = \frac{OA_t}{u_{p2}}. \quad (8.26)$$

根据图8-9中三个直角三角形的几何形状，我们推导出

$$A_iO' = OO' \sin \theta_i, \quad (8.27a)$$

$$OA_r = OO' \sin \theta_r, \quad (8.27b)$$

$$OA_t = OO' \sin \theta_t. \quad (8.27c)$$

在等式中使用这些表达式。 (8.26) 导致

$$\theta_i = \theta_r, \quad \text{(Snell's law of reflection)} \quad (8.28a)$$

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{u_{p2}}{u_{p1}} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}}. \quad (8.28b)$$

(Snell's law of refraction)

► 斯涅尔反射定律规定反射角等于入射角，斯涅尔折射定律提供了 $\sin \theta_t$ 和 $\sin \theta_i$ 之间以相速度之比表示的关系。 ◀

介质的折射率 n 定义为自由空间中的相速度 (即光速 c) 与介质中的相速度之比。因此，

$$n = \frac{c}{u_p} = \sqrt{\frac{\mu \epsilon}{\mu_0 \epsilon_0}} = \sqrt{\mu_r \epsilon_r}. \quad (8.29)$$

(index of refraction)

鉴于方程。 (8.29)，等式。 (8.28b) 可以重写为

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{n_1}{n_2} = \sqrt{\frac{\mu_{r1} \epsilon_{r1}}{\mu_{r2} \epsilon_{r2}}}. \quad (8.30)$$

对于非磁性材料， $\mu_{r1} = \mu_{r2} = 1$ ，在这种情况下

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{n_1}{n_2} = \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \frac{\eta_2}{\eta_1} \quad (\text{for } \mu_1 = \mu_2). \quad (8.31)$$

通常，密度较高的材料具有较高的介电常数。 $\mu_r = \epsilon_r = 1$ 的空气中的折射率 $n_0 = 1$ 。由于对于非磁性材料 $n = \sqrt{\epsilon_r}$ ，材料通常是

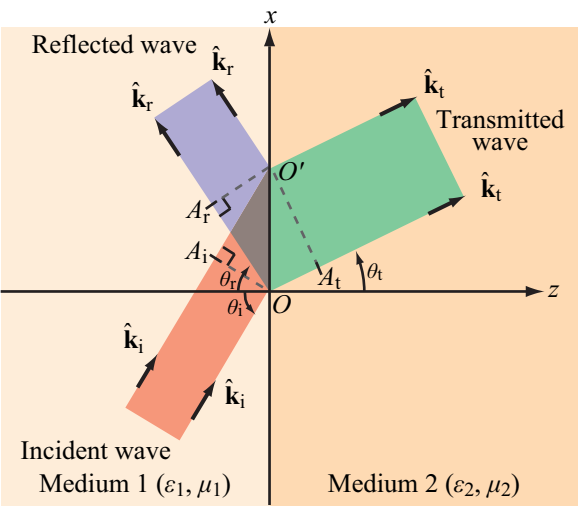


图 8-9 不同介质之间平面边界处的波反射和折射。

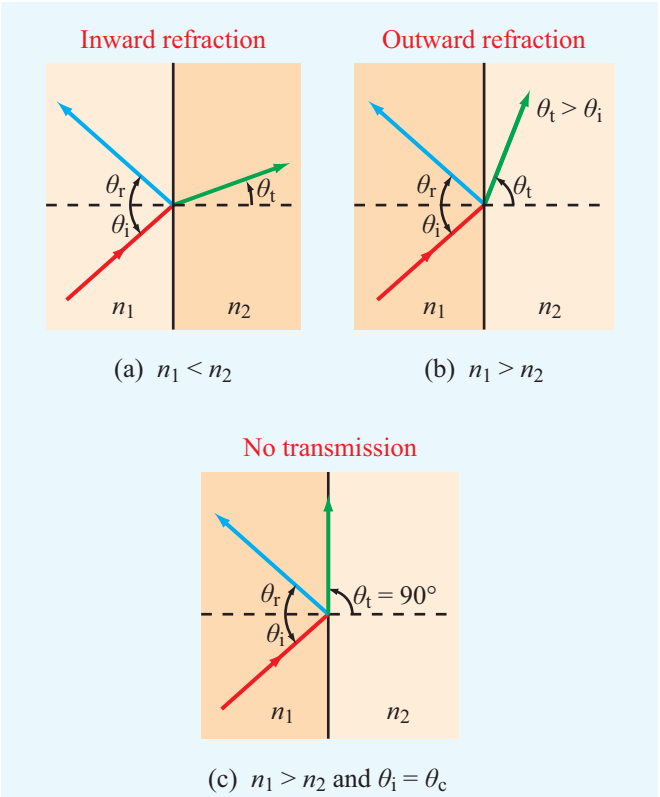


Figure 8-10 Snell's laws state that $\theta_r = \theta_i$ and $\sin \theta_t = (n_1/n_2) \sin \theta_i$. Refraction is (a) inward if $n_1 < n_2$ and (b) outward if $n_1 > n_2$; (c) the refraction angle is 90° if $n_1 > n_2$ and θ_i is equal to or greater than the critical angle $\theta_c = \sin^{-1}(n_2/n_1)$.

referred to as *more dense than another material if it has a greater index of refraction*.

At normal incidence ($\theta_i = 0$), Eq. (8.31) gives $\theta_t = 0$, as expected. At oblique incidence $\theta_i < \theta_c$ when $n_2 > n_1$ and $\theta_t > \theta_i$ when $n_2 < n_1$.

▶ If a wave is incident on a more dense medium (**Fig. 8-10(a)**), the transmitted wave refracts inwardly (toward the z axis), and the opposite is true if a wave is incident on a less dense medium (**Fig. 8-10(b)**). ◀

A case of particular interest is when $\theta_i = \pi/2$, as shown in **Fig. 8-10(c)**; in this case, the refracted wave flows along the surface and no energy is transmitted into medium 2. The value of the angle of incidence θ_i corresponding to $\theta_t = \pi/2$ is called

the **critical angle** θ_c and is obtained from Eq. (8.30) as

$$\sin \theta_c = \frac{n_2}{n_1} \sin \theta_i \Big|_{\theta_i=\pi/2} = \frac{n_2}{n_1} \quad (8.32a)$$

$$= \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}} \quad (\text{for } \mu_1 = \mu_2). \quad (8.32b)$$

(critical angle)

If θ_i exceeds θ_c , the incident wave is totally reflected, and the refracted wave becomes a nonuniform **surface wave** that travels along the boundary between the two media. This wave behavior is called **total internal reflection**.

Example 8-4: Light Beam Passing through a Slab

A dielectric slab with index of refraction n_2 is surrounded by a medium with index of refraction n_1 , as shown in **Fig. 8-11**. If $\theta_i < \theta_c$, show that the emerging beam is parallel to the incident beam.

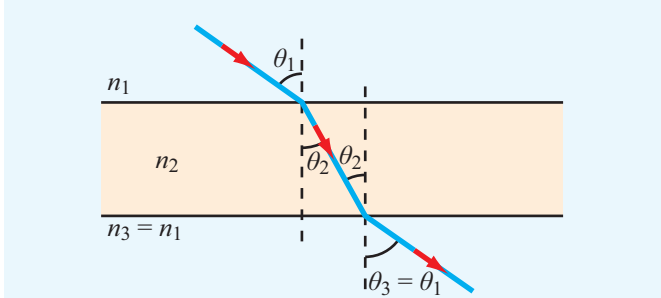


Figure 8-11 The exit angle θ_3 is equal to the incidence angle θ_1 if the dielectric slab has parallel boundaries and is surrounded by media with the same index of refraction on both sides (Example 8-4).

Solution: At the slab's upper surface, Snell's law gives

$$\sin \theta_2 = \frac{n_1}{n_2} \sin \theta_1. \quad (8.33)$$

Similarly, at the slab's lower surface,

$$\sin \theta_3 = \frac{n_2}{n_3} \sin \theta_2 = \frac{n_2}{n_1} \sin \theta_2. \quad (8.34)$$

Substituting Eq. (8.33) into Eq. (8.34) gives

$$\sin \theta_3 = \left(\frac{n_2}{n_1} \right) \left(\frac{n_1}{n_2} \right) \sin \theta_1 = \sin \theta_1.$$

临界角 θ_c 由方程式获得(8.30) 作为

$$\sin \theta_c = \frac{n_2}{n_1} \sin \theta_i \Big|_{\theta_i=\pi/2} = \frac{n_2}{n_1} \quad (8.32a)$$

$$= \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}} \quad (\text{for } \mu_1 = \mu_2). \quad (8.32b)$$

(critical angle)

如果 θ_i 超过 θ_c ，则入射波被全反射，折射波成为沿着两种介质之间的边界传播的不均匀表面波。这种波行为称为全内反射。

例 8-4：光束穿过平板

折射率为 n_2 的电介质板被折射率为 n_1 的介质包围，如图8-11所示。如果 $\theta_i < \theta_c$ ，则表明出射光束与入射光束平行。

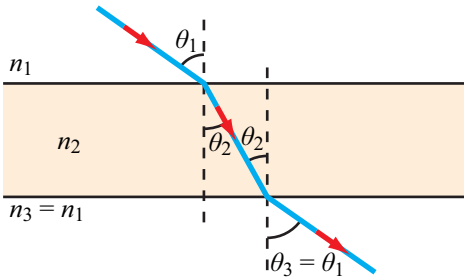


图 8-11 如果电介质板具有平行边界并且两侧都被折射率相同的介质包围，则出射角 θ_3 等于入射角 θ_1 (示例 8-4) 。

解：在板的上表面，斯涅尔定律给出

$$\sin \theta_2 = \frac{n_1}{n_2} \sin \theta_1. \quad (8.33)$$

同样，在板的下表面，

$$\sin \theta_3 = \frac{n_2}{n_3} \sin \theta_2 = \frac{n_2}{n_1} \sin \theta_2. \quad (8.34)$$

代入方程式 (8.33) 代入等式。 (8.34) 给出

$$\sin \theta_3 = \left(\frac{n_2}{n_1} \right) \left(\frac{n_1}{n_2} \right) \sin \theta_1 = \sin \theta_1.$$

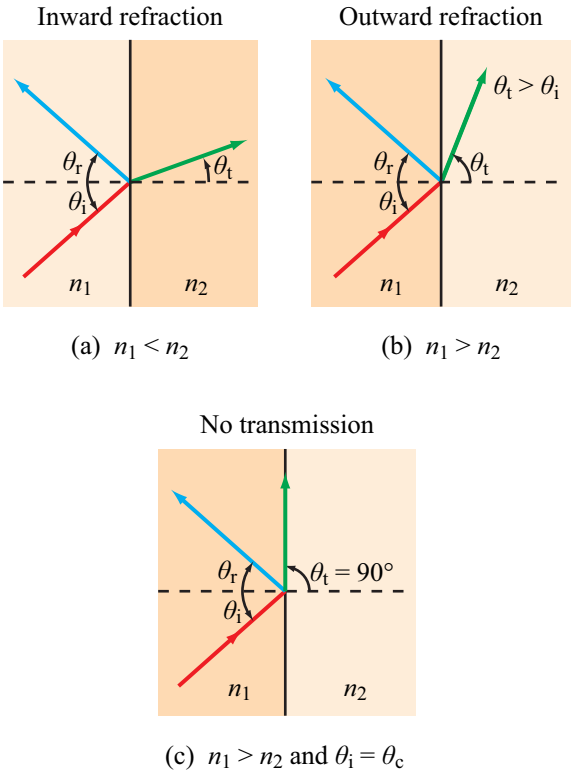


图 8-10 斯涅尔定律指出 $\theta_r = \theta_i$ 和 $\sin \theta_t = (n_1/n_2) \sin \theta_i$ 。如果 $n_1 < n_2$ ，则折射为 (a) 向内；如果 $n_1 > n_2$ ，则折射为 (b) 向外； (c)如果 $n_1 > n_2$ 和 θ_i 等于或大于临界角 $\theta_c = \sin^{-1}(n_2/n_1)$ ，则折射角为 90° 。

如果它具有更大的折射率，则被称为比另一种材料更致密。

在法线入射 ($\theta_i = 0$) 下，方程正如预期的那样， (8.31) 给出 $\theta_t = 0$ 。当 $n_2 > n_1$ 时倾斜入射 $\theta_t < \theta_i$ ，当 $n_2 < n_1$ 时倾斜入射 $\theta_t > \theta_i$ 。

▶ 如果波入射到密度较大的介质上 (图8-10 (a))，则透射波向内折射 (朝向 z 轴)，如果波入射到密度较小的介质上则相反 (图8-10 (b))。 ◀

特别令人感兴趣的情况是当 $\theta_i = \pi/2$ 时，如图8-10(c)所示；在这种情况下，折射波沿表面流动，没有能量传输到介质 2 中。对应于 $\theta_i = \pi/2$ 的入射角 θ_i 的值称为

Hence, $\theta_3 = \theta_1$. The slab displaces the beam's position, but the beam's direction remains unchanged.

Exercise 8-4: In the visible part of the electromagnetic spectrum, the index of refraction of water is 1.33. What is the critical angle for light waves generated by an upward-looking underwater light source?

Answer: $\theta_c = 48.8^\circ$. (See [EM](#).)

Exercise 8-5: If the light source of Exercise 8-4 is situated at a depth of 1 m below the water surface and if its beam is isotropic (radiates in all directions), how large a circle would it illuminate when observed from above?

Answer: Circle's diameter = 2.28 m. (See [EM](#).)

8-3 Fiber Optics

By successive total internal reflections, as illustrated in [Fig. 8-12\(a\)](#), light can be guided through thin dielectric rods made of glass or transparent plastic known as **optical fibers**. Because the light is confined to traveling within the rod, the only loss in power is due to reflections at the sending and receiving ends of the fiber and absorption by the fiber material (because it is not a perfect dielectric). Optical fibers are useful for the transmission of wide-band signals as well as many imaging applications.

An optical fiber usually consists of a cylindrical **fiber core** with an index of refraction n_f , surrounded by another cylinder of lower index of refraction, n_c , called the **cladding** ([Fig. 8-12\(b\)](#)). The cladding layer serves to optically isolate the fiber when a large number of fibers are packed in close proximity, thereby avoiding leakage of light from one fiber into another. To ensure total internal reflection, the incident angle θ_3 in the fiber core must be equal to or greater than the

critical angle θ_c for a wave in the fiber medium (with n_f) incident upon the cladding medium (with n_c). From Eq. (8.32a), we have

$$\sin \theta_c = \frac{n_c}{n_f}.$$
 (8.35)

To meet the total reflection requirement $\theta_3 \geq \theta_c$, it is necessary that $\sin \theta_3 \geq n_c/n_f$. The angle θ_2 is the complement of angle θ_3 ; hence, $\cos \theta_2 = \sin \theta_3$. The necessary condition therefore may be written as

$$\cos \theta_2 \geq \frac{n_c}{n_f}.$$
 (8.36)

Moreover, θ_2 is related to the incidence angle on the face of the fiber, θ_1 , by Snell's law:

$$\sin \theta_2 = \frac{n_0}{n_f} \sin \theta_1,$$
 (8.37)

where n_0 is the index of refraction of the medium surrounding the fiber ($n_0 = 1$ for air and $n_0 = 1.33$ if the fiber is in water) or

$$\cos \theta_2 = \left[1 - \left(\frac{n_0}{n_f} \right)^2 \sin^2 \theta_1 \right]^{1/2}.$$
 (8.38)

Using Eq. (8.38) on the left-hand side of Eq. (8.36) and then solving for $\sin \theta_1$ gives

$$\sin \theta_1 \leq \frac{1}{n_0} (n_f^2 - n_c^2)^{1/2}.$$
 (8.39)

8-3.1 Maximum Acceptance Cone

The **acceptance angle** θ_a is defined as the maximum value of θ_1 for which the condition of total internal reflection remains satisfied:

$$\sin \theta_a = \frac{1}{n_0} (n_f^2 - n_c^2)^{1/2}.$$
 (8.40)

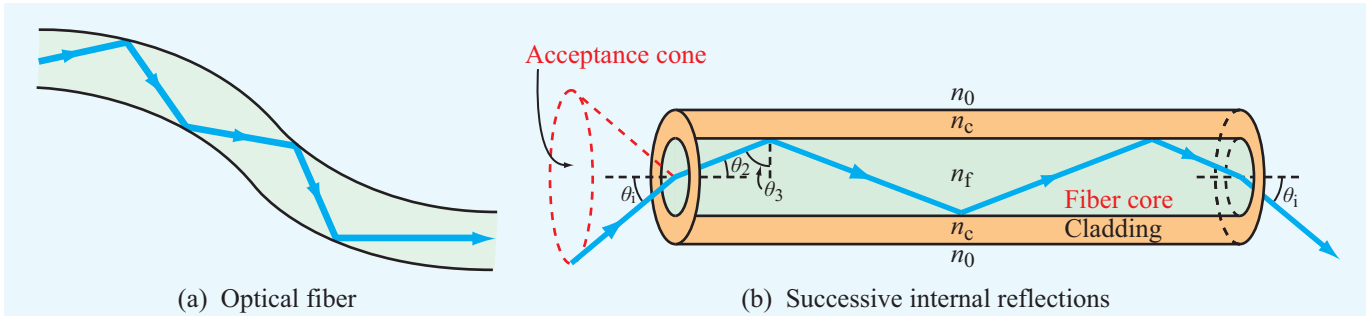


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

因此, $\theta_3 = \theta_1$. 板移动了光束的位置, 但光束的方向保持不变。

练习8-4: 在电磁波谱的可见光部分, 水的折射率为1.33。向上看的水下光源产生的光波的临界角是多少?

答案: $\theta_c = 48.8^\circ$ 。(参见^{EM}。)

练习8-5: 如果练习8-4的光源位于水面以下1m的深度, 并且其光束是各向同性的(向各个方向辐射), 那么从上方观察时, 它会照亮多大的圆?

答案: 圆的直径= 2.28 m。(参见^{EM}。)

8-3 光纤

通过连续的全内反射, 如图 8-12(a) 所示, 光可以被引导通过由玻璃或透明塑料(称为光纤)制成的细电棒。由于光被限制在棒内传播, 唯一的功率损失是由于光纤发送端和接收端的反射以及光纤材料的吸收(因为它不是理想的电介质)。光纤可用于传输宽带信号以及许多成像应用。

光纤通常由折射率为 n_f 的圆柱形纤芯组成, 周围环绕着另一个折射率较低的圆柱体 n_c , 称为包层(图 8-12(b))。当大量光纤紧密排列时, 包层用于光学隔离光纤, 从而避免光从一根光纤泄漏到另一根光纤中。为了确保全内反射, 纤芯中的入射角 θ_3 必须等于或大于

光纤介质中的波 (n_f) 入射到包层介质 (n_c) 上的临界角 θ_c 。从方程。(8.32a), 我们有

$$\sin \theta_c = \frac{n_c}{n_f}.$$
 (8.35)

为了满足总反射要求 $\theta_3 \geq \theta_c$, 必须有 $\sin \theta_3 \geq n_c/n_f$ 。角 θ_2 是角 θ_3 的余角; 因此, $\cos \theta_2 = \sin \theta_3$ 。因此必要条件可以写为

$$\cos \theta_2 \geq \frac{n_c}{n_f}.$$
 (8.36)

此外, 根据斯涅耳定律, θ_2 与光纤表面的入射角 θ_1 相关:

$$\sin \theta_2 = \frac{n_0}{n_f} \sin \theta_1,$$
 (8.37)

其中 n_0 是光纤周围介质的折射率(对于空气, $n_0 = 1$, 如果光纤在水中, 则 $n_0 = 1.33$) 或

$$\cos \theta_2 = \left[1 - \left(\frac{n_0}{n_f} \right)^2 \sin^2 \theta_1 \right]^{1/2}.$$
 (8.38)

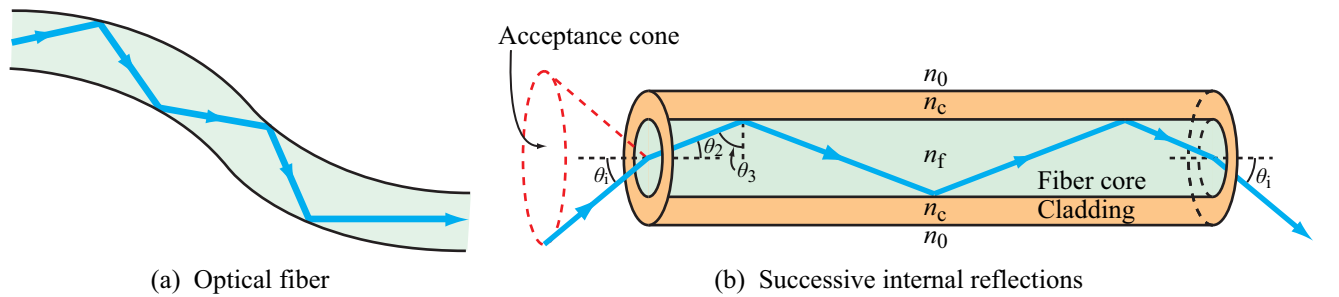
使用方程式 (8.38) 在等式的左侧。(8.36) 然后求解 $\sin \theta_1$ 给出

$$\sin \theta_1 \leq \frac{1}{n_0} (n_f^2 - n_c^2)^{1/2}.$$
 (8.39)

8-3.1 最大接收锥角

接受角 θ_a 定义为满足全内反射条件的 θ_1 的最大值:

$$\sin \theta_a = \frac{1}{n_0} (n_f^2 - n_c^2)^{1/2}.$$
 (8.40)



菲图8-12 只要反射角超过全内反射的临界角, 波就可以沿着光纤传导。

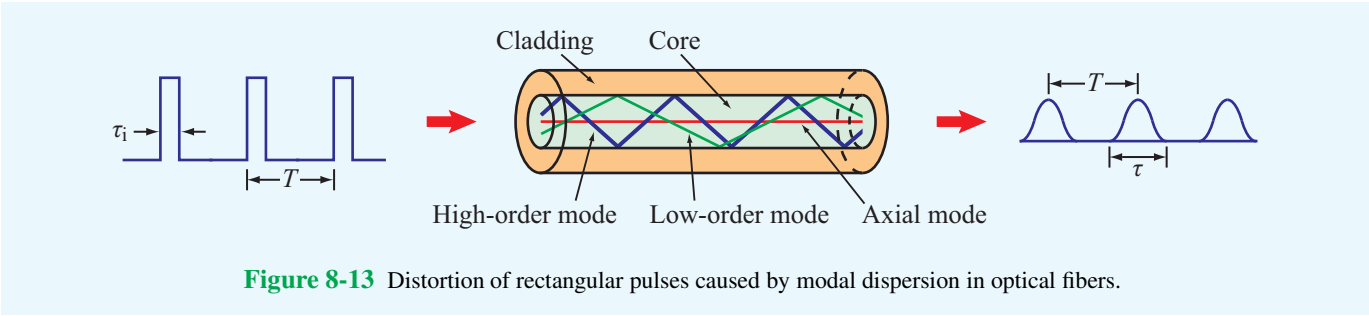


Figure 8-13 Distortion of rectangular pulses caused by modal dispersion in optical fibers.

The angle θ_a is equal to half the angle of the acceptance cone of the fiber. Any ray of light incident upon the face of the core fiber at an incidence angle within the acceptance cone can propagate down the core. This means that there can be a large number of ray paths, called **modes**, by which light energy can travel in the core. Rays characterized by large angles θ_i travel longer paths than rays that propagate along the axis of the fiber, as illustrated by the three modes shown in **Fig. 8-13**. Consequently, different modes have different transit times between the two ends of the fiber. This property of optical fibers is called **modal dispersion** and has the undesirable effect of changing the shape of pulses used for the transmission of digital data. When a rectangular pulse of light incident upon the face of the fiber gets broken up into many modes and the different modes do not arrive at the other end of the fiber at the same time, the pulse gets distorted—both in shape and length. In the example shown in **Fig. 8-13**, the narrow rectangular pulses at the input side of the optical fiber are of width τ_i separated by a time duration T . After propagating through the fiber core, modal dispersion causes the pulses to look more like spread-out sine waves with spread-out temporal width τ . If the output pulses spread out so much that $\tau > T$, the output signals will smear out, making it impossible to decipher the transmitted message from the output signal. Hence, to ensure that the transmitted pulses remain distinguishable at the output side of the fiber, it is necessary that τ be shorter than T . As a safety margin, it is common practice to design the transmission system such that $T \geq 2\tau$.

The spread-out width τ is equal to the time delay Δt between the arrival of the slowest ray and the arrival of the fastest ray. The slowest ray is the one traveling the longest distance and corresponds to the ray incident upon the input face of the fiber at the acceptance angle θ_a . From the geometry in **Fig. 8-12(b)** and Eq. (8.36), this ray corresponds to $\cos \theta_2 = n_c/n_f$. For an optical fiber of length l , the length of the path traveled by such a ray is

$$l_{\max} = \frac{l}{\cos \theta_2} = l \frac{n_f}{n_c}, \quad (8.41)$$

and its travel time in the fiber at velocity $u_p = c/n_f$ is

$$t_{\max} = \frac{l_{\max}}{u_p} = \frac{ln_f^2}{cn_c}. \quad (8.42)$$

The minimum time of travel is realized by the axial ray and is given by

$$t_{\min} = \frac{l}{u_p} = \frac{l}{c} n_f. \quad (8.43)$$

The total time delay is therefore

$$\tau = \Delta t = t_{\max} - t_{\min} = \frac{ln_f}{c} \left(\frac{n_f - 1}{n_c} \right) \quad (\text{s}). \quad (8.44)$$

As we stated before, to retrieve the desired information from the transmitted signals, it is advisable that T , which is the interpulse period of the input train of pulses, be no shorter than 2τ . This in turn means that the **maximum data rate** (in bits per second), or equivalently the number of pulses per second, that can be transmitted through the fiber is limited to

$$f_p = \frac{1}{T} = \frac{1}{2\tau} = \frac{cn_c}{2ln_f(n_f - n_c)} \quad (\text{bits/s}). \quad (8.45)$$

Example 8-5: Transmission Data Rate on Optical Fibers

A 1-km long optical fiber (in air) is made of a fiber core with an index of refraction of 1.52 and a cladding with an index of refraction of 1.49. Determine

- (a) the acceptance angle θ_a and
- (b) the maximum usable data rate of signals that can be transmitted through the fiber.

Solution: (a) From Eq. (8.40),

$$\sin \theta_a = \frac{1}{n_0} (n_f^2 - n_c^2)^{1/2} = [(1.52)^2 - (1.49)^2]^{1/2} = 0.3,$$

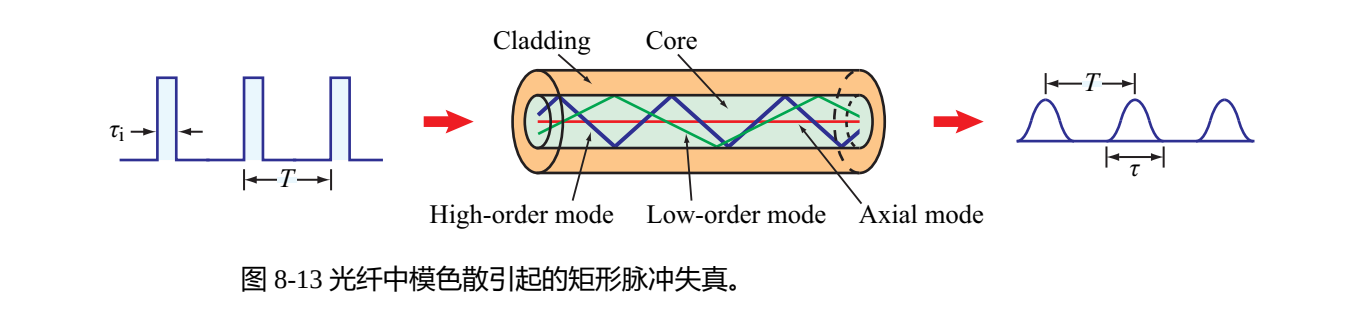


图 8-13 光纤中模色散引起的矩形脉冲失真。

角度 θ_a 等于光纤接收锥角的一半。任何以接收锥内的入射角入射到纤芯光纤表面的光线都可以沿着纤芯传播。这意味着可以有大量的光线路径（称为模式），光能可以通过这些路径在核心中传播。以大角度 θ_i 为特征的光线比沿光纤轴传播的光线传播的路径更长，如图 8-13 中的三种模式所示。因此，不同的模式在光纤两端之间具有不同的传输时间。光纤的这种特性称为模式色散，会产生改变用于传输数字数据的脉冲形状的不良影响。当入射到光纤表面的矩形光脉冲被分解成多种模式并且不同模式不同时到达光纤的另一端时，脉冲在形状和长度上都会发生扭曲。在图 8-13 所示的示例中，光纤输入侧的窄矩形脉冲的宽度为 τ_i ，间隔时间为 T 。通过光纤纤芯传播后，模态色散使脉冲看起来更像具有扩展时间宽度 τ 的扩展正弦波。如果输出脉冲扩散得如此之大，以至于 $\tau > T$ ，输出信号将变得模糊，从而无法从输出信号中破译传输的消息。因此，为了确保传输的脉冲在光纤的输出侧保持可区分， τ 必须短于 T 。作为安全裕度，通常的做法是将传输系统设计为 $T \geq 2\tau$ 。

其在光纤中以速度 $u_p = c/n_f$ 的传播时间为

$$t_{\max} = \frac{l_{\max}}{u_p} = \frac{ln_f^2}{cn_c}. \quad (8.42)$$

最小传播时间由轴向射线实现，由下式给出

$$t_{\min} = \frac{l}{u_p} = \frac{l}{c} n_f. \quad (8.43)$$

因此总延迟时间为

$$\tau = \Delta t = t_{\max} - t_{\min} = \frac{ln_f}{c} \left(\frac{n_f - 1}{n_c} \right) \quad (\text{s}). \quad (8.44)$$

正如我们之前所说，为了从传输的信号中检索所需的信息，建议输入脉冲串的脉冲间周期 T 不小于 2τ 。这又意味着可以通过光纤传输的最大数据速率（以每秒位数为单位）或等效的每秒脉冲数限制为

$$f_p = \frac{1}{T} = \frac{1}{2\tau} = \frac{cn_c}{2ln_f(n_f - n_c)} \quad (\text{位/秒}). \quad (8.45)$$

例 8-5：光纤上的传输数据速率

1 公里长的光纤（在空气中）由折射率为 1.52 的纤芯和折射率为 1.49 的包层组成。决定

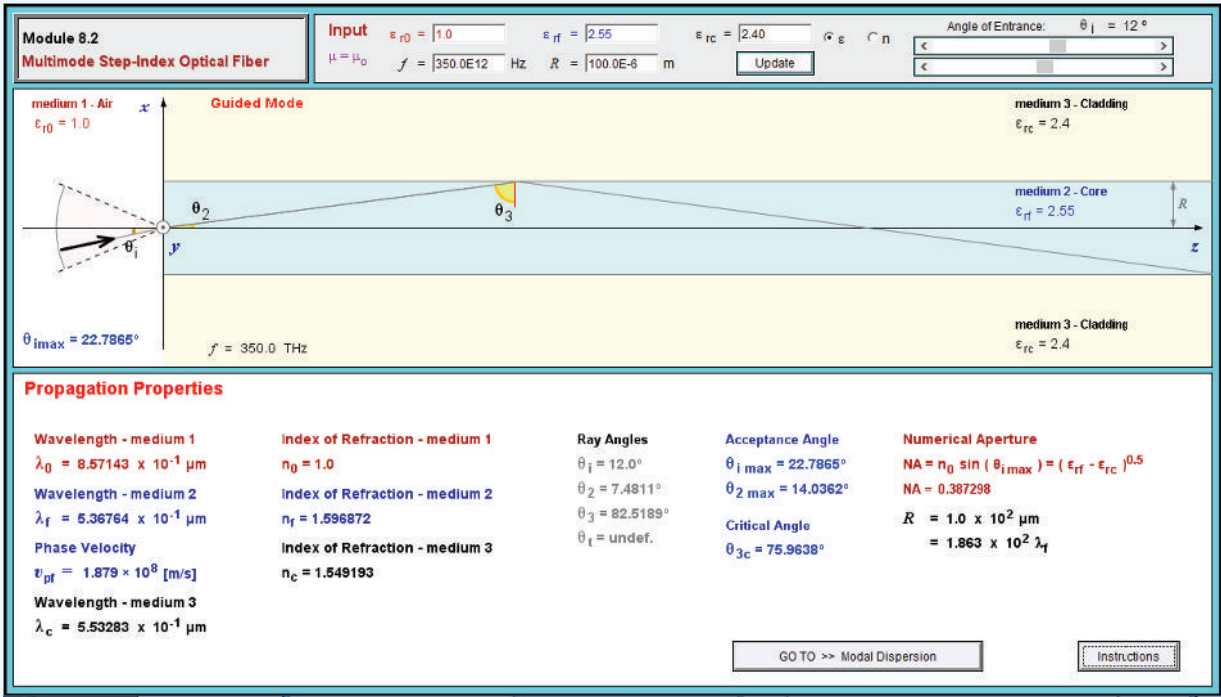
- (a) 接收角 θ_a 和 (b) 可通过光纤传输的信号的最大可用数据速率。

解： (a) 根据方程。(8.40),

$$\sin \theta_a = \frac{1}{n_0} (n_f^2 - n_c^2)^{1/2} = [(1.52)^2 - (1.49)^2]^{1/2} = 0.3,$$

$$l_{\max} = \frac{l}{\cos \theta_2} = l \frac{n_f}{n_c}, \quad (8.41)$$

Module 8.2 Multimode Step-Index Optical Fiber Choose the indices on the fiber core and cladding and then observe the zigzag pattern of the wave propagation inside the fiber.



which corresponds to $\theta_a = 17.5^\circ$.

(b) From Eq. (8.45),

$$f_p = \frac{cn_c}{2ln_f(n_f - n_c)} = \frac{3 \times 10^8 \times 1.49}{2 \times 10^3 \times 1.52(1.52 - 1.49)} = 4.9 \text{ (Mb/s)}.$$

8-3.2 Confined Acceptance Cone

The expression for the data rate given by Eq. (8.45) pertains to the case where the light coupled to the fiber is confined to the acceptance cone defined by the acceptance angle θ_a given by Eq. (8.40). If the entrance cone is larger than the acceptance cone, then some of the light will leak out of the fiber through the cladding, which not only reduces the power carried by the light but may also cause interference with signals carried by neighboring fibers.

On the other hand, if the entrance cone can be made to be smaller than the acceptance cone, the data rate can be safely increased. Let us denote $\theta_1(\text{max})$ as the maximum

angle of the entrance cone. Per Eq. (8.38), $\theta_2(\text{max})$ is related to $\theta_1(\text{max})$ by

$$\cos \theta_2(\text{max}) = \left[1 - \left(\frac{n_0}{n_f} \right)^2 \sin^2 \theta_1(\text{max}) \right]^{1/2}.$$

Repetition of the steps between Eqs. (8.41) and (8.45) leads to

$$f_p = \frac{c}{2ln_f} \left[\frac{\cos \theta_2(\text{max})}{1 - \cos \theta_2(\text{max})} \right] \quad (\theta_1(\text{max}) < \theta_a).$$

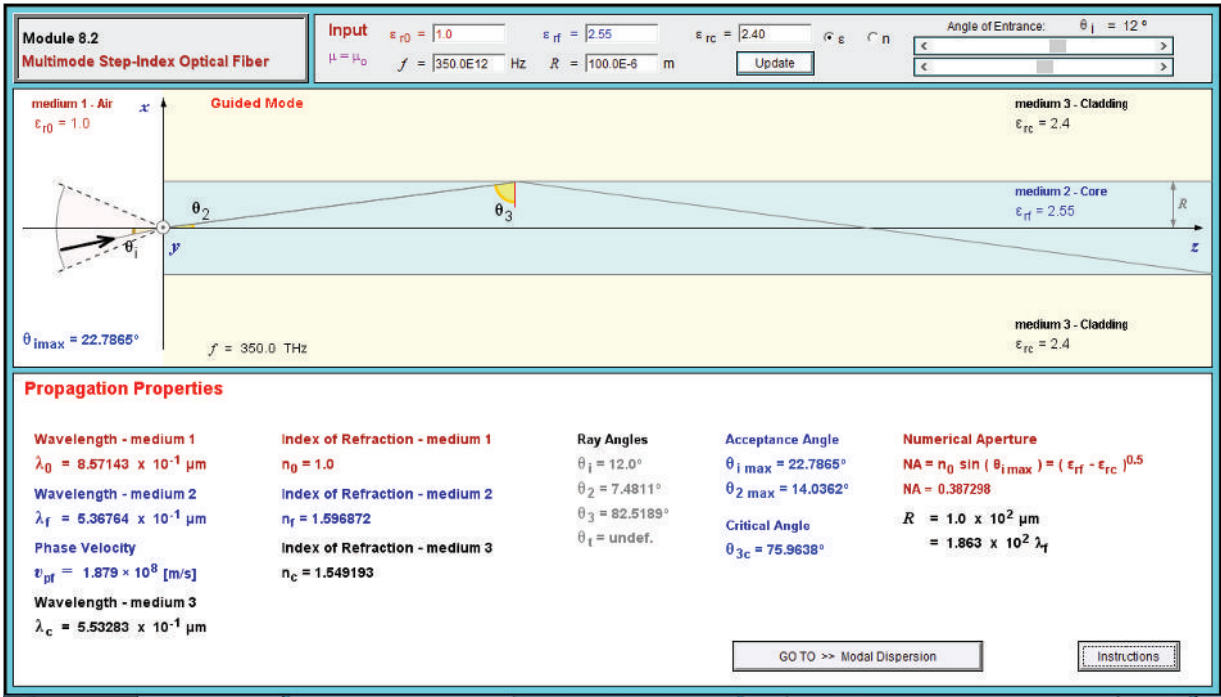
Exercise 8-6: If the index of refraction of the cladding material in Example 8-5 is increased to 1.50, what would be the new maximum usable data rate?

Answer: 7.4 (Mb/s). (See [EM](#).)

8-4 Wave Reflection and Transmission at Oblique Incidence

In this section, we develop a rigorous theory of reflection and refraction of plane waves obliquely incident upon planar

模块 8.2 多模阶跃折射率光纤 选择纤芯和包层的折射率，然后观察光纤内波传播的锯齿形图案。



对应于 $\theta_a = 17.5^\circ$ 。

(b) 从方程式(8.45),

$$f_p = \frac{cn_c}{2ln_f(n_f - n_c)} = \frac{3 \times 10^8 \times 1.49}{2 \times 10^3 \times 1.52(1.52 - 1.49)} = 4.9 \text{ (Mb/s)}.$$

8-3.2 受限接受锥

数据速率的表达式由方程式给出。（8.45）适用于耦合到光纤的光被限制在由方程（8.45）给出的接收角 θ_a 定义的接收锥内的情况。（8.40）。如果入口锥体大于接收锥体，则一些光将通过包层泄漏出光纤，这不仅会降低光携带的功率，而且还可能对相邻光纤携带的信号造成干扰。

另一方面，如果可以使入口锥体小于接收锥体，则可以安全地提高数据速率。让我们将 $\theta_1(\text{max})$ 表示为最大值

入口锥体的角度。根据方程式(8.38)， $\theta_2(\text{max})$ 与 $\theta_1(\text{max})$ 的关系为

$$\cos \theta_2(\text{max}) = \left[1 - \left(\frac{n_0}{n_f} \right)^2 \sin^2 \theta_1(\text{max}) \right]^{1/2}.$$

重复方程式之间的步骤。（8.41）和（8.45）领先到

$$f_p = \frac{c}{2ln_f} \left[\frac{\cos \theta_2(\text{max})}{1 - \cos \theta_2(\text{max})} \right] \quad (\theta_1(\text{max}) < \theta_a).$$

练习8-6：如果示例8-5中的包层材料的折射率增加到1.50，新的最大可用数据速率是多少？

答案：7.4 (Mb/s)。（参见^{EM}。）

8-4 斜入射波的反射和透射

在本节中，我们发展了一个关于倾斜入射到平面上的平面波的反射和折射的严格理论。

boundaries between different media. Our treatment parallels that in Section 8-1 for the normal-incidence case and goes beyond that in Section 8-2 on Snell's laws, which yielded information on only the angles of reflection and refraction.

For normal incidence, the reflection and transmission coefficients Γ and τ at a boundary between two media are independent of the polarization of the incident wave, because both the electric and magnetic fields of a normally incident plane wave are tangential to the boundary regardless of the wave polarization. This is not the case for obliquely incident waves traveling at an angle $\theta_i \neq 0$ with respect to the normal to the interface.

► The **plane of incidence** is defined as the plane containing the normal to the boundary and the direction of propagation of the incident wave. ◀

A wave of arbitrary polarization may be described as the superposition of two orthogonally polarized waves: one with its electric field parallel to the plane of incidence (**parallel polarization**) and the other with its electric field perpendicular to the plane of incidence (**perpendicular polarization**). These two polarization configurations are shown in Fig. 8-14, where the plane of incidence is coincident with the x - z plane. Polarization with $\mathbf{E}$ perpendicular to the plane of incidence is also called **transverse electric (TE)** polarization because $\mathbf{E}$ is perpendicular to the plane of incidence and with $\mathbf{E}$ parallel to the plane of incidence is called **transverse magnetic (TM)** polarization because in that case it is the magnetic field that is perpendicular to the plane of incidence.

For the general case of a wave with an arbitrary polarization, it is common practice to decompose the incident wave ($\mathbf{E}^i, \mathbf{H}^i$) into a perpendicularly polarized component ($\mathbf{E}_\perp^i, \mathbf{H}_\perp^i$) and a parallel polarized component ($\mathbf{E}_\parallel^i, \mathbf{H}_\parallel^i$). Then, after determining the reflected waves ($\mathbf{E}_\perp^r, \mathbf{H}_\perp^r$) and ($\mathbf{E}_\parallel^r, \mathbf{H}_\parallel^r$) due to the two incident components, the reflected waves are added together to give the total reflected wave ($\mathbf{E}^r, \mathbf{H}^r$) corresponding to the original incident wave. A similar process can be used to determine the total transmitted wave ($\mathbf{E}^t, \mathbf{H}^t$).

8-4.1 Perpendicular Polarization

Figure 8-15 shows a perpendicularly polarized incident plane wave propagating along the x_1 direction in dielectric medium 1. The electric field phasor $\tilde{\mathbf{E}}_\perp^i$ points along the y direction, and the associated magnetic field phasor $\tilde{\mathbf{H}}_\perp^i$ is along the y_1 axis. The directions of $\tilde{\mathbf{E}}_\perp^i$ and $\tilde{\mathbf{H}}_\perp^i$ are such that $\tilde{\mathbf{E}}_\perp^i \times \tilde{\mathbf{H}}_\perp^i$ points along the propagation direction $\hat{\mathbf{x}}_1$. The electric and magnetic

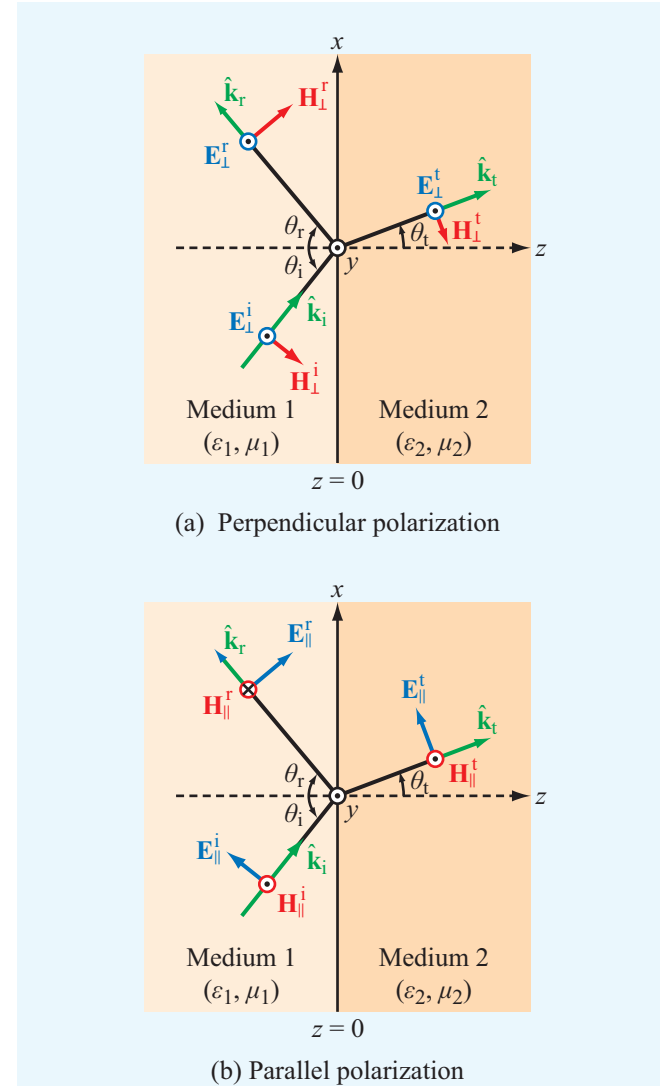


Figure 8-14 The plane of incidence is the plane containing the direction of wave travel, $\hat{\mathbf{k}}_i$, and the surface normal to the boundary. In the present case, the plane of incidence containing $\hat{\mathbf{k}}_i$ and $\hat{\mathbf{z}}$ coincides with the plane of the paper. A wave is (a) perpendicularly polarized when its electric field vector is perpendicular to the plane of incidence and (b) parallel polarized when its electric field vector lies in the plane of incidence.

fields of such a plane wave are given by

$$\tilde{\mathbf{E}}_\perp^i = \hat{\mathbf{y}} E_{\perp 0}^i e^{-jk_1 x_1}, \quad (8.46a)$$

$$\tilde{\mathbf{H}}_\perp^i = \hat{\mathbf{y}}_1 \frac{E_{\perp 0}^i}{\eta_1} e^{-jk_1 x_1}, \quad (8.46b)$$

不同媒体之间的界限。我们的处理方法与第 8-1 节中针对正入射情况的处理方法相似，并且超出了第 8-2 节中关于斯涅耳定律的处理方法，后者仅提供了有关反射和折射角度的信息。

对于法向入射，两种介质之间边界处的反射和透射系数 Γ 和 τ 与入射波的偏振无关，因为无论波偏振如何，法向入射平面波的电场和磁场都与边界相切。对于相对于界面法线以角度 $\theta_i \neq 0$ 传播的倾斜入射波，情况并非如此。

► 入射平面定义为包含边界法线和入射波传播方向的平面。◀

任意偏振波可以描述为两个正交偏振波的叠加：一个的电场平行于入射平面（平行偏振），另一个的电场垂直于入射平面（垂直偏振）。这两种偏振配置如图 8-14 所示，其中入射平面与 x - z 平面重合。E 垂直于入射平面的偏振也称为横向电 (TE) 偏振，因为 E 垂直于入射平面，而 E 平行于入射平面的偏振称为横向磁 (TM) 偏振，因为在这种情况下，磁场垂直于入射平面。

对于具有任意偏振的波的一般情况，通常的做法是将入射波 ($\mathbf{E}^i, \mathbf{H}^i$) 分解为垂直偏振分量 ($\mathbf{E}_\perp^i, \mathbf{H}_\perp^i$) 和平行偏振分量 ($\mathbf{E}_\parallel^i, \mathbf{H}_\parallel^i$)。然后，在确定由两个入射分量引起的反射波 ($\mathbf{E}_\perp^r, \mathbf{H}_\perp^r$) 和 ($\mathbf{E}_\parallel^r, \mathbf{H}_\parallel^r$) 后，将反射波相加，得到与原始入射波相对应的总反射波 ($\mathbf{E}^r, \mathbf{H}^r$)。可以使用类似的过程来确定总透射波 ($\mathbf{E}^t, \mathbf{H}^t$)。

8-4.1 垂直偏振

图 8-15 显示了在电介质 1 中沿 x_1 方向传播的垂直偏振入射平面波。电场相量 $\tilde{\mathbf{E}}_\perp^i$ 沿 y 方向指向，相关的磁场相量 $\tilde{\mathbf{H}}_\perp^i$ 沿 y_1 轴指向。 $\tilde{\mathbf{E}}_\perp^i$ 和 $\tilde{\mathbf{H}}_\perp^i$ 的方向使得 $\tilde{\mathbf{E}}_\perp^i \times \tilde{\mathbf{H}}_\perp^i$ 沿着传播方向 $\hat{\mathbf{x}}_1$ 指向。电和磁

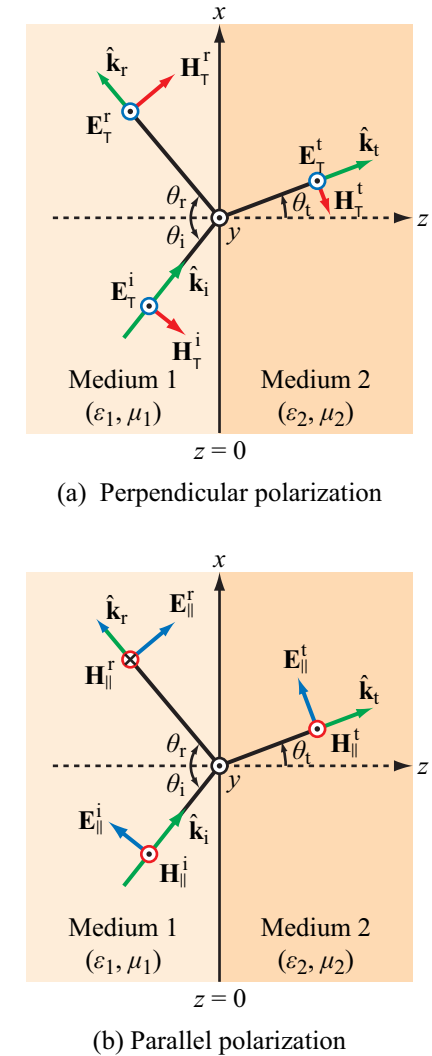


图 8-14 入射平面是包含波传播方向 $\hat{\mathbf{k}}_i$ 和垂直于边界的表面的平面。在本例中，包含 $\hat{\mathbf{k}}_i$ 和 $\hat{\mathbf{z}}$ 的入射平面与纸平面重合。当波的电场矢量垂直于入射平面时，波是 (a) 垂直偏振的；(b) 当其电场矢量位于入射平面时，波是平行偏振的。

这种平面波的场由下式给出

$$\tilde{\mathbf{E}}_\perp^i = \hat{\mathbf{y}} E_{\perp 0}^i e^{-jk_1 x_1}, \quad (8.46a)$$

$$\tilde{\mathbf{H}}_\perp^i = \hat{\mathbf{y}}_1 \frac{E_{\perp 0}^i}{\eta_1} e^{-jk_1 x_1}, \quad (8.46b)$$

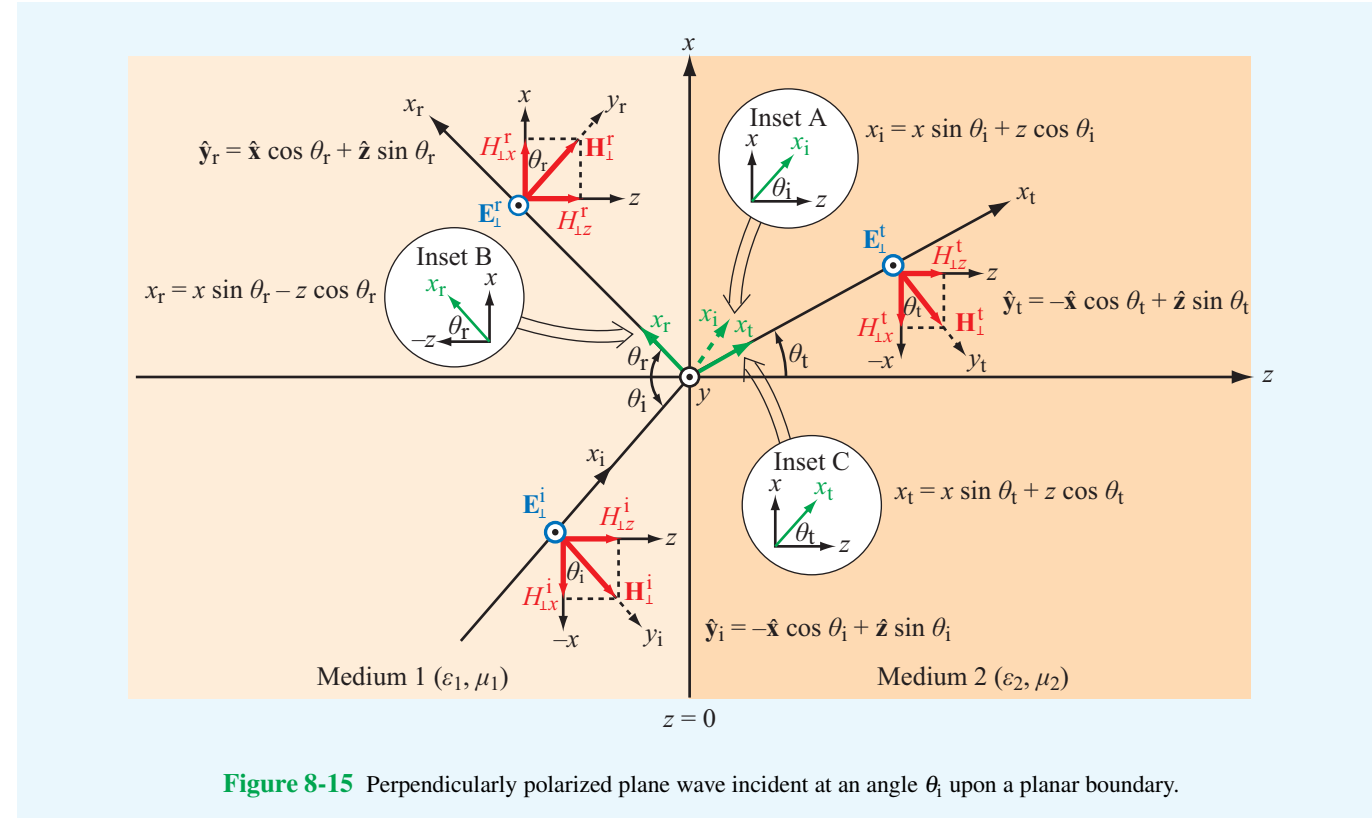


Figure 8-15 Perpendicularly polarized plane wave incident at an angle θ_i upon a planar boundary.

where $E_{\perp 0}^i$ is the amplitude of the electric field phasor at $x_i = 0$ and $k_1 = \omega\sqrt{\mu_1\epsilon_1}$ and $\eta_1 = \sqrt{\mu_1/\epsilon_1}$ are the wavenumber and intrinsic impedance of medium 1. From Fig. 8-15, the distance x_i and the unit vector $\hat{\mathbf{y}}_i$ may be expressed in terms of the (x, y, z) global coordinate system as

$$x_i = x \sin \theta_i + z \cos \theta_i, \quad (8.47a)$$

$$\hat{\mathbf{y}}_i = -\hat{\mathbf{x}} \cos \theta_i + \hat{\mathbf{z}} \sin \theta_i. \quad (8.47b)$$

Substituting Eqs. (8.47a) and (8.47b) into Eqs. (8.46a) and (8.46b) gives

$$\tilde{\mathbf{E}}_{\perp}^i = \hat{\mathbf{y}} E_{\perp 0}^i e^{-jk_1(x \sin \theta_i + z \cos \theta_i)}, \quad (8.48a)$$

$$\tilde{\mathbf{H}}_{\perp}^i = (-\hat{\mathbf{x}} \cos \theta_i + \hat{\mathbf{z}} \sin \theta_i) \frac{E_{\perp 0}^i}{\eta_1} e^{-jk_1(x \sin \theta_i + z \cos \theta_i)}. \quad (8.48b)$$

With the aid of the directional relationships given in Fig. 8-15 for the reflected and transmitted waves, these fields are given by the following:

Reflected Wave

$$\tilde{\mathbf{E}}_{\perp}^r = \hat{\mathbf{y}} E_{\perp 0}^r e^{-jk_1 x_r} = \hat{\mathbf{y}} E_{\perp 0}^r e^{-jk_1(x \sin \theta_r - z \cos \theta_r)}, \quad (8.49a)$$

$$\begin{aligned} \tilde{\mathbf{H}}_{\perp}^r &= \hat{\mathbf{y}}_r \frac{E_{\perp 0}^r}{\eta_1} e^{-jk_1 x_r} \\ &= (\hat{\mathbf{x}} \cos \theta_r + \hat{\mathbf{z}} \sin \theta_r) \frac{E_{\perp 0}^r}{\eta_1} e^{-jk_1(x \sin \theta_r - z \cos \theta_r)}, \end{aligned} \quad (8.49b)$$

Transmitted Wave

$$\tilde{\mathbf{E}}_{\perp}^t = \hat{\mathbf{y}} E_{\perp 0}^t e^{-jk_2 x_t} = \hat{\mathbf{y}} E_{\perp 0}^t e^{-jk_2(x \sin \theta_t + z \cos \theta_t)}, \quad (8.49c)$$

$$\begin{aligned} \tilde{\mathbf{H}}_{\perp}^t &= \hat{\mathbf{y}}_t \frac{E_{\perp 0}^t}{\eta_2} e^{-jk_2 x_t} \\ &= (-\hat{\mathbf{x}} \cos \theta_t + \hat{\mathbf{z}} \sin \theta_t) \frac{E_{\perp 0}^t}{\eta_2} e^{-jk_2(x \sin \theta_t + z \cos \theta_t)}, \end{aligned} \quad (8.49d)$$

where θ_r and θ_t are the reflection and transmission angles shown in Fig. 8-15 and k_2 and η_2 are the wavenumber and intrinsic impedance of medium 2. Our goal is to describe the reflected and transmitted fields in terms of the parameters that characterize the incident wave, namely the incidence angle θ_i and the amplitude $E_{\perp 0}^i$. The four expressions given by Eqs. (8.49a) through (8.49d) contain four unknowns: $E_{\perp 0}^r$, $E_{\perp 0}^t$, θ_r , and θ_t . Even though angles θ_r and θ_t are related to θ_i by Snell's laws (Eqs. (8.28a) and (8.28b)), here we choose to

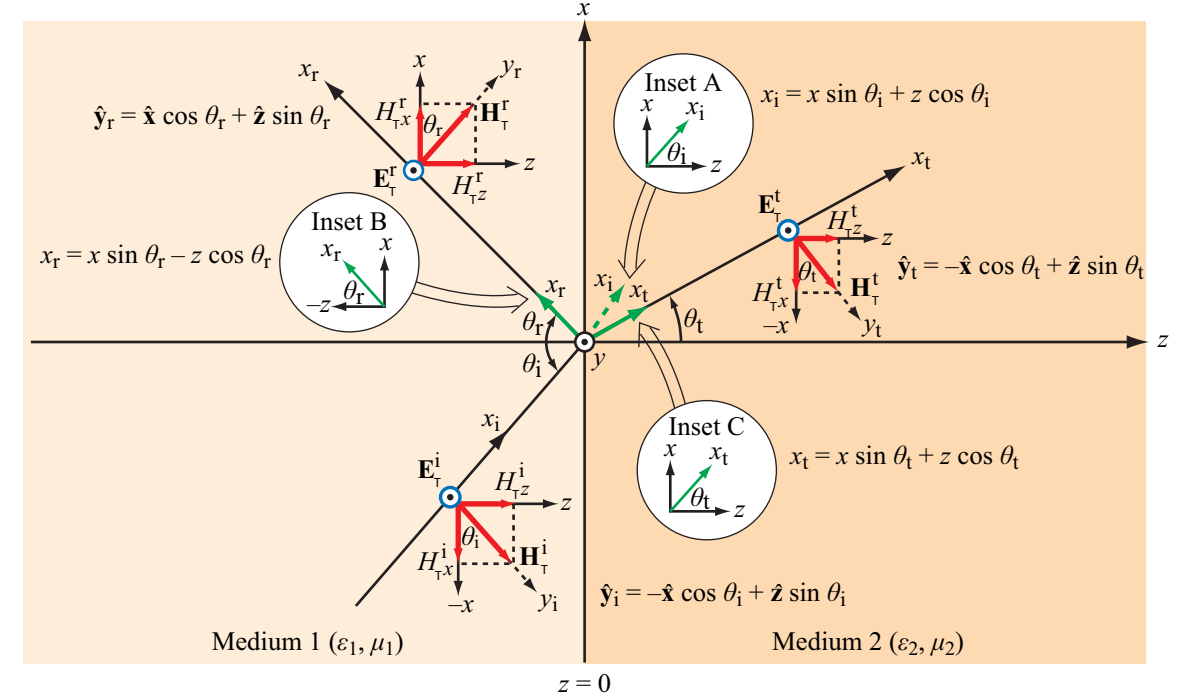


图 8-15 垂直偏振平面波以角度 θ_i 入射到平面边界上。

其中 $E_{\perp 0}^i$ 是 $x_i = 0$ 处电场相量的幅度, $k_1 = \omega\sqrt{\mu_1\epsilon_1}$ 和 $\eta_1 = \sqrt{\mu_1/\epsilon_1}$ 是介质1的波数和本征阻抗。从图8-15中, 距离 x_i 和单位向量 $\hat{\mathbf{y}}_i$ 可以用 (x, y, z) 全局坐标系表示为

$$x_i = x \sin \theta_i + z \cos \theta_i, \quad (8.47a)$$

$$\hat{\mathbf{y}}_i = -\hat{\mathbf{x}} \cos \theta_i + \hat{\mathbf{z}} \sin \theta_i. \quad (8.47b)$$

代入等式。 (8.47a) 和 (8.47b) 代入等式。 (8.46a) 和 (8.46b) 给出

事件波

$$\tilde{\mathbf{E}}_{\perp}^i = \hat{\mathbf{y}} E_{\perp 0}^i e^{-jk_1(x \sin \theta_i + z \cos \theta_i)}, \quad (8.48a)$$

$$\tilde{\mathbf{H}}_{\perp}^i = (-\hat{\mathbf{x}} \cos \theta_i + \hat{\mathbf{z}} \sin \theta_i) \frac{E_{\perp 0}^i}{\eta_1} e^{-jk_1(x \sin \theta_i + z \cos \theta_i)}. \quad (8.48b)$$

借助图 8-15 中给出的反射波和透射波的方向关系, 这些场由下式给出:

反射波

$$\tilde{\mathbf{E}}_{\perp}^r = \hat{\mathbf{y}} E_{\perp 0}^r e^{-jk_1 x_r} = \hat{\mathbf{y}} E_{\perp 0}^r e^{-jk_1(x \sin \theta_r - z \cos \theta_r)}, \quad (8.49a)$$

$$\begin{aligned} \tilde{\mathbf{H}}_{\perp}^r &= \hat{\mathbf{y}}_r \frac{E_{\perp 0}^r}{\eta_1} e^{-jk_1 x_r} \\ &= (\hat{\mathbf{x}} \cos \theta_r + \hat{\mathbf{z}} \sin \theta_r) \frac{E_{\perp 0}^r}{\eta_1} e^{-jk_1(x \sin \theta_r - z \cos \theta_r)}, \end{aligned} \quad (8.49b)$$

传输波

$$\tilde{\mathbf{E}}_{\perp}^t = \hat{\mathbf{y}} E_{\perp 0}^t e^{-jk_2 x_t} = \hat{\mathbf{y}} E_{\perp 0}^t e^{-jk_2(x \sin \theta_t + z \cos \theta_t)}, \quad (8.49c)$$

$$\begin{aligned} \tilde{\mathbf{H}}_{\perp}^t &= \hat{\mathbf{y}}_t \frac{E_{\perp 0}^t}{\eta_2} e^{-jk_2 x_t} \\ &= (-\hat{\mathbf{x}} \cos \theta_t + \hat{\mathbf{z}} \sin \theta_t) \frac{E_{\perp 0}^t}{\eta_2} e^{-jk_2(x \sin \theta_t + z \cos \theta_t)}, \end{aligned} \quad (8.49d)$$

其中 θ_r 和 θ_t 是图8-15中所示的反射和透射角, k_2 和 η_2 是介质2的波数和固有阻抗。我们的目标是用表征入射波的参数 (即入射角 θ_i 和振幅 $E_{\perp 0}^i$) 来描述反射和透射场。等式给出的四个表达式。 (8.49a) 到 (8.49d) 包含四个未知数: $E_{\perp 0}^r$, $E_{\perp 0}^t$, θ_r 和 θ_t 。尽管根据斯涅耳定律, 角度 θ_r 和 θ_t 与 θ_i 相关 (方程 (8.28a) 和 (8.28b)), 但这里我们选择

treat them as unknown for the time being because we intend to show that Snell’s laws also can be derived by applying field boundary conditions at $z = 0$. The total electric field in medium 1 is the sum of the incident and reflected electric fields: $\tilde{\mathbf{E}}_{\perp}^1 = \tilde{\mathbf{E}}_{\perp}^i + \tilde{\mathbf{E}}_{\perp}^r$; a similar statement holds true for the total magnetic field in medium 1: $\tilde{\mathbf{H}}_{\perp}^1 = \tilde{\mathbf{H}}_{\perp}^i + \tilde{\mathbf{H}}_{\perp}^r$. Boundary conditions state that the tangential components of $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ each must be continuous across the boundary between the two media. Field components tangential to the boundary extend along $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$. Since the electric fields in media 1 and 2 have $\hat{\mathbf{y}}$ components only, the boundary condition for $\tilde{\mathbf{E}}$ is

$$(\tilde{E}_{\perp y}^i + \tilde{E}_{\perp y}^r)\Big|_{z=0} = \tilde{E}_{\perp y}^t\Big|_{z=0}.$$
(8.50)

Upon using Eqs. (8.48a), (8.49a), and (8.49c) in Eq. (8.50) and then setting $z = 0$, we have

$$E_{\perp 0}^i e^{-jk_1 x \sin \theta_i} + E_{\perp 0}^r e^{-jk_1 x \sin \theta_i} = E_{\perp 0}^t e^{-jk_2 x \sin \theta_t}.$$
(8.51)

Since the magnetic fields in media 1 and 2 have no $\hat{\mathbf{y}}$ components, the boundary condition for $\tilde{\mathbf{H}}$ is

$$(\tilde{H}_{\perp x}^i + \tilde{H}_{\perp x}^r)\Big|_{z=0} = \tilde{H}_{\perp x}^t\Big|_{z=0},$$
(8.52)

or

$$\begin{aligned} -\frac{E_{\perp 0}^i}{\eta_1} \cos \theta_i e^{-jk_1 x \sin \theta_i} + \frac{E_{\perp 0}^r}{\eta_1} \cos \theta_r e^{-jk_1 x \sin \theta_r} \\ = -\frac{E_{\perp 0}^t}{\eta_2} \cos \theta_t e^{-jk_2 x \sin \theta_t}. \end{aligned}$$
(8.53)

To satisfy Eqs. (8.51) and (8.53) for all possible values of x (i.e., all along the boundary), it follows that the arguments of all three exponentials must be equal. That is,

$$k_1 \sin \theta_i = k_1 \sin \theta_r = k_2 \sin \theta_t,$$
(8.54)

which is known as the *phase-matching condition*. The first equality in Eq. (8.54) leads to

$\theta_r = \theta_i,$

(Snell’s law of reflection)

(8.55)

while the second equality leads to

$$\frac{\sin \theta_i}{\sin \theta_t} = \frac{k_1}{k_2} = \frac{\omega \sqrt{\mu_1 \epsilon_1}}{\omega \sqrt{\mu_2 \epsilon_2}} = \frac{n_1}{n_2}.$$

(Snell’s law of refraction)

(8.56)

The results expressed by Eqs. (8.55) and (8.56) are identical with those derived previously in Section 8-2 through consideration of the ray path traversed by the incident, reflected, and transmitted wavefronts.

In view of Eq. (8.54), the boundary conditions given by Eqs. (8.51) and (8.53) reduce to

$$E_{\perp 0}^i + E_{\perp 0}^r = E_{\perp 0}^t,$$
(8.57a)

$$\frac{\cos \theta_i}{\eta_1} (-E_{\perp 0}^i + E_{\perp 0}^r) = -\frac{\cos \theta_t}{\eta_2} E_{\perp 0}^t.$$
(8.57b)

These two equations can be solved simultaneously to yield the following expressions for the reflection and transmission coefficients in the perpendicular polarization case:

$$\Gamma_{\perp} = \frac{E_{\perp 0}^r}{E_{\perp 0}^i} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t},$$
(8.58a)

$$\tau_{\perp} = \frac{E_{\perp 0}^t}{E_{\perp 0}^i} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}.$$
(8.58b)

These two coefficients are known formally as the *Fresnel reflection and transmission coefficients for perpendicular polarization* and are related by

$$\tau_{\perp} = 1 + \Gamma_{\perp}.$$
(8.59)

If medium 2 is a perfect conductor ($\eta_2 = 0$), Eqs. (8.58a and b) reduce to $\Gamma_{\perp} = -1$ and $\tau_{\perp} = 0$, respectively, which means that the incident wave is totally reflected by the conducting medium.

For nonmagnetic dielectrics with $\mu_1 = \mu_2 = \mu_0$ and with the help of Eq. (8.56), the expression for $\Gamma_{\perp}$ can be written as

$$\Gamma_{\perp} = \frac{\cos \theta_i - \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}} \quad (\text{for } \mu_1 = \mu_2).$$
(8.60)

Since $(\epsilon_2/\epsilon_1) = (n_2/n_1)^2$, this expression also can be written in terms of the indices of refraction n_1 and n_2 .

暂时将它们视为未知，因为我们打算证明斯涅耳定律也可以通过在 $z = 0$ 处应用场边界条件来导出。介质 1 中的总电场是入射电场和反射电场的总和： $\tilde{\mathbf{E}}_{\perp}^1 = \tilde{\mathbf{E}}_{\perp}^i + \tilde{\mathbf{E}}_{\perp}^r$ ；类似的说法也适用于介质 1 中的总磁场： $\tilde{\mathbf{H}}_{\perp}^1 = \tilde{\mathbf{H}}_{\perp}^i + \tilde{\mathbf{H}}_{\perp}^r$ 。边界条件规定 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 的切向分量必须在两种介质之间的边界上连续。与边界相切的场分量沿 $\hat{\mathbf{x}}$ 和 $\hat{\mathbf{y}}$ 延伸。由于介质 1 和 2 中的电场仅具有 $\hat{\mathbf{y}}$ 分量，因此 $\tilde{\mathbf{E}}$ 的边界条件为

$$(\tilde{E}_{\perp y}^i + \tilde{E}_{\perp y}^r)\Big|_{z=0} = \tilde{E}_{\perp y}^t\Big|_{z=0}.$$
(8.50)

使用方程式后。式 (8.48a)、(8.49a) 和 (8.49c) (8.50) 然后设置 $z = 0$ ，我们有

$$E_{\perp 0}^i e^{-jk_1 x \sin \theta_i} + E_{\perp 0}^r e^{-jk_1 x \sin \theta_r} = E_{\perp 0}^t e^{-jk_2 x \sin \theta_t}.$$
(8.51)

由于介质 1 和 2 中的磁场没有 $\hat{\mathbf{y}}$ 分量，因此 $\tilde{\mathbf{H}}$ 的边界条件为

$$(\tilde{H}_{\perp x}^i + \tilde{H}_{\perp x}^r)\Big|_{z=0} = \tilde{H}_{\perp x}^t\Big|_{z=0},$$
(8.52)

或者

$$\begin{aligned} -\frac{E_{\perp 0}^i}{\eta_1} \cos \theta_i e^{-jk_1 x \sin \theta_i} + \frac{E_{\perp 0}^r}{\eta_1} \cos \theta_r e^{-jk_1 x \sin \theta_r} \\ = -\frac{E_{\perp 0}^t}{\eta_2} \cos \theta_t e^{-jk_2 x \sin \theta_t}. \end{aligned}$$
(8.53)

满足方程。 (8.51) 和 (8.53) 对于 x 的所有可能值（即，沿着边界），所有三个指数的参数必须相等。那是，

$$k_1 \sin \theta_i = k_1 \sin \theta_r = k_2 \sin \theta_t,$$
(8.54)

这称为相位匹配条件。方程中的第一个等式。 (8.54) 导致

$\theta_r = \theta_i,$

(Snell’s law of reflection)

(8.55)

而第二个等式导致

$$\frac{\sin \theta_i}{\sin \theta_t} = \frac{k_1}{k_2} = \frac{\omega \sqrt{\mu_1 \epsilon_1}}{\omega \sqrt{\mu_2 \epsilon_2}} = \frac{n_1}{n_2}.$$

(Snell’s law of refraction)

(8.56)

结果由方程式表示。（8.55）和（8.56）与前面在第 8-2 节中通过考虑入射、反射和透射波前穿过的光线路径而推导出来的相同。

鉴于方程。（8.54），由方程给出的边界条件。（8.51）和 (8.53) 减少为

$$E_{\perp 0}^i + E_{\perp 0}^r = E_{\perp 0}^t,$$
(8.57a)

$$\cos \theta_i (-E_{\perp 0}^i + E_{\perp 0}^r) = -\cos \theta_t E_{\perp 0}^t.$$
(8.57b)

可以同时求解这两个方程，得出垂直偏振情况下反射和透射系数的以下表达式：

$$\Gamma_{\perp} = \frac{E_{\perp 0}^r}{E_{\perp 0}^i} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t},$$
(8.58a)

$$\tau_{\perp} = \frac{E_{\perp 0}^t}{E_{\perp 0}^i} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}.$$
(8.58b)

这两个系数的正式名称为垂直偏振的非涅尔反射和透射系数，其关系式为

$$\tau_{\perp} = 1 + \Gamma_{\perp}.$$
(8.59)

如果介质 2 是理想导体 ($\eta_2 = 0$)，则等式： (8.58a和b) 分别减少到 $\Gamma_{\perp} = -1$ 和 $\tau_{\perp} =0$ ，这意味着入射波被导电介质完全反射。

对于具有 $\mu_1 = \mu_2 = \mu_0$ 的非磁性电介质，并借助方程式。（8.56）， $\Gamma_{\perp}$ 的表达式可以写为

$$\Gamma_{\perp} = \frac{\cos \theta_i - \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}} \quad (\text{for } \mu_1 = \mu_2).$$
(8.60)

由于 $(\epsilon_2/\epsilon_1) = (n_2/n_1)^2$ ，该表达式也可以用折射率 n_1 和 n_2 来编写。

Example 8-6: Wave Incident Obliquely on a Soil Surface

Using the coordinate system of **Fig. 8-15**, a plane wave radiated by a distant antenna is incident in air upon a plane soil surface located at $z = 0$. The electric field of the incident wave is given by

$$\mathbf{E}^i = \hat{\mathbf{y}}100\cos(\omega t - \pi x - 1.73\pi z) \quad (\text{V/m}), \quad (8.61)$$

and the soil medium may be assumed to be a lossless dielectric with a relative permittivity of 4.

- (a) Determine k_1 , k_2 , and the incidence angle θ_i .
 (b) Obtain expressions for the total electric fields in air and in the soil.
 (c) Determine the average power density carried by the wave traveling in soil.

Solution: (a) We begin by converting Eq. (8.61) into phasor form, akin to the expression given by Eq. (8.46a):

$$\tilde{\mathbf{E}}^i = \hat{\mathbf{y}}100e^{-j\pi x - j1.73\pi z} = \hat{\mathbf{y}}100e^{-jk_1 x_1} \quad (\text{V/m}), \quad (8.62)$$

where x_1 is the axis along which the wave is traveling and

$$k_1 x_1 = \pi x + 1.73\pi z. \quad (8.63)$$

Using Eq. (8.47a), we have

$$k_1 x_1 = k_1 x \sin \theta_i + k_1 z \cos \theta_i. \quad (8.64)$$

Hence,

$$\begin{aligned} k_1 \sin \theta_i &= \pi, \\ k_1 \cos \theta_i &= 1.73\pi, \end{aligned}$$

which together give

$$\begin{aligned} k_1 &= \sqrt{\pi^2 + (1.73\pi)^2} = 2\pi \quad (\text{rad/m}), \\ \theta_i &= \tan^{-1}\left(\frac{\pi}{1.73\pi}\right) = 30^\circ. \end{aligned}$$

The wavelength in medium 1 (air) is

$$\lambda_1 = \frac{2\pi}{k_1} = 1 \text{ m},$$

and the wavelength in medium 2 (soil) is

$$\lambda_2 = \frac{\lambda_1}{\sqrt{\epsilon_{r2}}} = \frac{1}{\sqrt{4}} = 0.5 \text{ m}.$$

The corresponding wavenumber in medium 2 is

$$k_2 = \frac{2\pi}{\lambda_2} = 4\pi \quad (\text{rad/m}).$$

Since $\tilde{\mathbf{E}}^i$ is along $\hat{\mathbf{y}}$, it is perpendicularly polarized ($\hat{\mathbf{y}}$ is perpendicular to the plane of incidence containing the surface normal $\hat{\mathbf{z}}$ and the propagation direction $\hat{\mathbf{x}}_1$).

(b) Given that $\theta_i = 30^\circ$, the transmission angle θ_t is obtained with the help of Eq. (8.56):

$$\sin \theta_t = \frac{k_1}{k_2} \sin \theta_i = \frac{2\pi}{4\pi} \sin 30^\circ = 0.25$$

or

$$\theta_t = 14.5^\circ.$$

With $\epsilon_1 = \epsilon_0$ and $\epsilon_2 = \epsilon_{r2}\epsilon_0 = 4\epsilon_0$, the reflection and transmission coefficients for perpendicular polarization are determined with the help of Eqs. (8.59) and (8.60),

$$\begin{aligned} \Gamma_\perp &= \frac{\cos \theta_i - \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}} = -0.38, \\ \tau_\perp &= 1 + \Gamma_\perp = 0.62. \end{aligned}$$

Using Eqs. (8.48a) and (8.49a) with $E_{\perp 0}^i = 100 \text{ V/m}$ and $\theta_i = \theta_r$, the total electric field in medium 1 is

$$\begin{aligned} \tilde{\mathbf{E}}_\perp^1 &= \tilde{\mathbf{E}}_\perp^i + \tilde{\mathbf{E}}_\perp^r \\ &= \hat{\mathbf{y}}E_{\perp 0}^i e^{-jk_1(x \sin \theta_i + z \cos \theta_i)} + \hat{\mathbf{y}}\Gamma E_{\perp 0}^i e^{-jk_1(x \sin \theta_i - z \cos \theta_i)} \\ &= \hat{\mathbf{y}}100e^{-j(\pi x + 1.73\pi z)} - \hat{\mathbf{y}}38e^{-j(\pi x - 1.73\pi z)}, \end{aligned}$$

and the corresponding instantaneous electric field in medium 1 is

$$\begin{aligned} \mathbf{E}_\perp^1(x, z, t) &= \Re \left[\tilde{\mathbf{E}}_\perp^1 e^{j\omega t} \right] \\ &= \hat{\mathbf{y}}[100\cos(\omega t - \pi x - 1.73\pi z) \\ &\quad - 38\cos(\omega t - \pi x + 1.73\pi z)] \quad (\text{V/m}). \end{aligned}$$

In medium 2, using Eq. (8.49c) with $E_{\perp 0}^t = \tau_\perp E_{\perp 0}^i$ gives

$$\tilde{\mathbf{E}}_\perp^t = \hat{\mathbf{y}}\tau E_{\perp 0}^i e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} = \hat{\mathbf{y}}62e^{-j(\pi x + 3.87\pi z)}$$

and, correspondingly,

$$\begin{aligned} \mathbf{E}_\perp^t(x, z, t) &= \Re \left[\tilde{\mathbf{E}}_\perp^t e^{j\omega t} \right] \\ &= \hat{\mathbf{y}}62\cos(\omega t - \pi x - 3.87\pi z) \quad (\text{V/m}). \end{aligned}$$

(c) In medium 2, $\eta_2 = \eta_0/\sqrt{\epsilon_{r2}} \approx 120\pi/\sqrt{4} = 60\pi \text{ } (\Omega)$, and the average power density carried by the wave is

$$S_{\text{av}}^t = \frac{|E_{\perp 0}^t|^2}{2\eta_2} = \frac{(62)^2}{2 \times 60\pi} = 10.2 \quad (\text{W/m}^2).$$

例8-6：波浪倾斜入射到土壤表面

使用图 8-15 的坐标系，由远处天线辐射的平面波在空气中入射到位于 $z = 0$ 处的平面土壤表面上。入射波的电场由下式给出

$$\mathbf{E}^i = \hat{\mathbf{y}}100\cos(\omega t - \pi x - 1.73\pi z) \quad (\text{V/m}), \quad (8.61)$$

土壤介质可假设为相对介电常数为 4 的无损电介质。

- (a) 确定 k_1 、 k_2 和入射角 θ_i 。(b) 获得空气和土壤中总电场的表达式。(c) 确定在土壤中传播的波所携带的平均功率密度。

解决方案：(a) 我们首先将方程转换为：(8.61) 转化为相量形式，类似于方程给出的表达式。(8.46a)：

$$\tilde{\mathbf{E}}^i = \hat{\mathbf{y}}100e^{-j\pi x - j1.73\pi z} = \hat{\mathbf{y}}100e^{-jk_1 x_1} \quad (\text{V/m}), \quad (8.62)$$

其中 x_1 是波传播的轴，

$$k_1 x_1 = \pi x + 1.73\pi z. \quad (8.63)$$

使用方程式 (8.47a)，我们有

$$k_1 x_1 = k_1 x \sin \theta_i + k_1 z \cos \theta_i. \quad (8.64)$$

因此，

$$\begin{aligned} k_1 \sin \theta_i &= \pi, \\ k_1 \cos \theta_i &= 1.73\pi, \end{aligned}$$

一起给予

$$\begin{aligned} k_1 &= \sqrt{\pi^2 + (1.73\pi)^2} = 2\pi \quad (\text{rad/m}), \\ \theta_i &= \tan^{-1}\left(\frac{\pi}{1.73\pi}\right) = 30^\circ. \end{aligned}$$

介质 1 (空气) 中的波长为

$$\lambda_1 = \frac{2\pi}{k_1} = 1 \text{ m},$$

介质 2 (土壤) 中的波长为

$$\lambda_2 = \frac{\lambda_1}{\sqrt{\epsilon_{r2}}} = \frac{1}{\sqrt{4}} = 0.5 \text{ m}.$$

介质2中对应的波数为

$$k_2 = \frac{2\pi}{\lambda_2} = 4\pi \quad (\text{rad/m}).$$

由于 $\tilde{\mathbf{E}}^i$ 沿 $\hat{\mathbf{y}}$ ，因此它是垂直偏振的 ($\hat{\mathbf{y}}$ 垂直于包含表面法线 $\hat{\mathbf{z}}$ 和传播方向 $\hat{\mathbf{x}}_1$ 的入射平面)。

(b) 给定 $\theta_i = 30^\circ$ ，则借助方程(1)获得传输角 θ_t 。(8.56)：

$$\sin \theta_t = \frac{k_1}{k_2} \sin \theta_i = \frac{2\pi}{4\pi} \sin 30^\circ = 0.25$$

or

$$\theta_t = 14.5^\circ.$$

对于 $\epsilon_1 = \epsilon_0$ 和 $\epsilon_2 = \epsilon_{r2}\epsilon_0 = 4\epsilon_0$ ，垂直偏振的反射和透射系数可借助方程 (1) 确定。(8.59) 和 (8.60)，

$$\begin{aligned} \Gamma_\perp &= \frac{\cos \theta_i - \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}} = -0.38, \\ \tau_\perp &= 1 + \Gamma_\perp = 0.62. \end{aligned}$$

使用方程式。(8.48a) 和 (8.49a) 当 $E_{\perp 0}^i = 100 \text{ V/m}$ 和 $\theta_i = \theta_r$ 时，介质 1 中的总电场为

$$\begin{aligned} \tilde{\mathbf{E}}_\perp^1 &= \tilde{\mathbf{E}}_\perp^i + \tilde{\mathbf{E}}_\perp^r \\ &= \hat{\mathbf{y}}E_{\perp 0}^i e^{-jk_1(x \sin \theta_i + z \cos \theta_i)} + \hat{\mathbf{y}}\Gamma E_{\perp 0}^i e^{-jk_1(x \sin \theta_i - z \cos \theta_i)} \\ &= \hat{\mathbf{y}}100e^{-j(\pi x + 1.73\pi z)} - \hat{\mathbf{y}}38e^{-j(\pi x - 1.73\pi z)}, \end{aligned}$$

介质1中相应的瞬时电场为

$$\begin{aligned} \mathbf{E}_\perp^1(x, z, t) &= \Re \left[\tilde{\mathbf{E}}_\perp^1 e^{j\omega t} \right] \\ &= \hat{\mathbf{y}}[100\cos(\omega t - \pi x - 1.73\pi z) \\ &\quad - 38\cos(\omega t - \pi x + 1.73\pi z)] \quad (\text{V/m}). \end{aligned}$$

在介质 2 中，使用等式：(8.49c) 与 $E_{\perp 0}^t = \tau_\perp E_{\perp 0}^i$ 给出

$$\tilde{\mathbf{E}}_\perp^t = \hat{\mathbf{y}}\tau E_{\perp 0}^i e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} = \hat{\mathbf{y}}62e^{-j(\pi x + 3.87\pi z)}$$

并且，相应地，

$$\begin{aligned} \mathbf{E}_\perp^t(x, z, t) &= \Re \left[\tilde{\mathbf{E}}_\perp^t e^{j\omega t} \right] \\ &= \hat{\mathbf{y}}62\cos(\omega t - \pi x - 3.87\pi z) \quad (\text{V/m}). \end{aligned}$$

(c) 在介质 2 中， $\eta_2 = \eta_0/\sqrt{\epsilon_{r2}} \approx 120\pi/\sqrt{4} = 60\pi \text{ } (\Omega)$ ，波携带的平均功率密度为

$$S_{\text{av}}^t = \frac{|E_{\perp 0}^t|^2}{2\eta_2} = \frac{(62)^2}{2 \times 60\pi} = 10.2 \quad (\text{W/m}^2).$$

Technology Brief 16: Bar-Code Readers

A **bar code** consists of a sequence of parallel bars of certain widths, usually printed in black against a white background, configured to represent a particular **binary code** of information about a product and its manufacturer. **Laser scanners** can read the code and transfer the information to a computer, a cash register, or a display screen. For both stationary scanners built into checkout counters at grocery stores and handheld units that can be pointed at the bar-coded object like a gun, the basic operation of a bar-code reader is the same.

Basic Operation

The scanner uses a laser beam of light pointed at a multifaceted **rotating mirror**, spinning at a high speed on the order of 6,000 revolutions per minute (**Fig. TF16-1**). The rotating mirror creates a **fan beam** to illuminate the bar code on the object. Moreover, by exposing the laser light to its many facets, it deflects the beam into many different directions, allowing the object to be scanned over a wide range of positions and orientations. The goal is to have one of those directions be such that the beam reflected

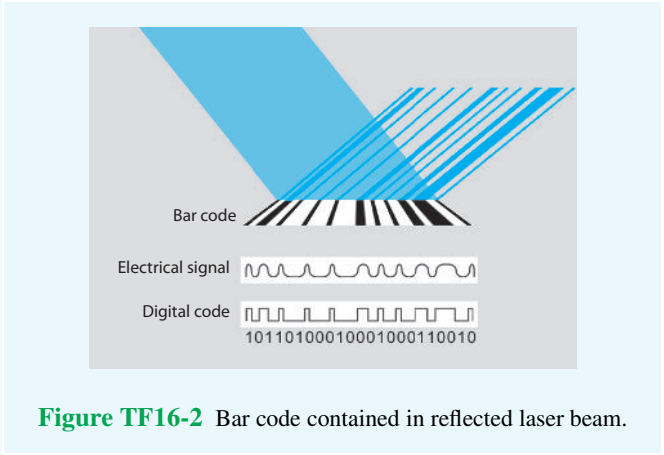


Figure TF16-2 Bar code contained in reflected laser beam.

by the bar code ends up traveling in the direction of, and captured by, the light detector (**sensor**), which then reads the coded sequence (white bars reflect laser light and black ones do not) and converts it into a binary sequence of ones and zeros (**Fig. TF16-2**). To eliminate interference by ambient light, a **glass filter** is used as shown in **Fig. TF16-1** to block out all light except for a narrow wavelength band centered at the wavelength of the laser light.

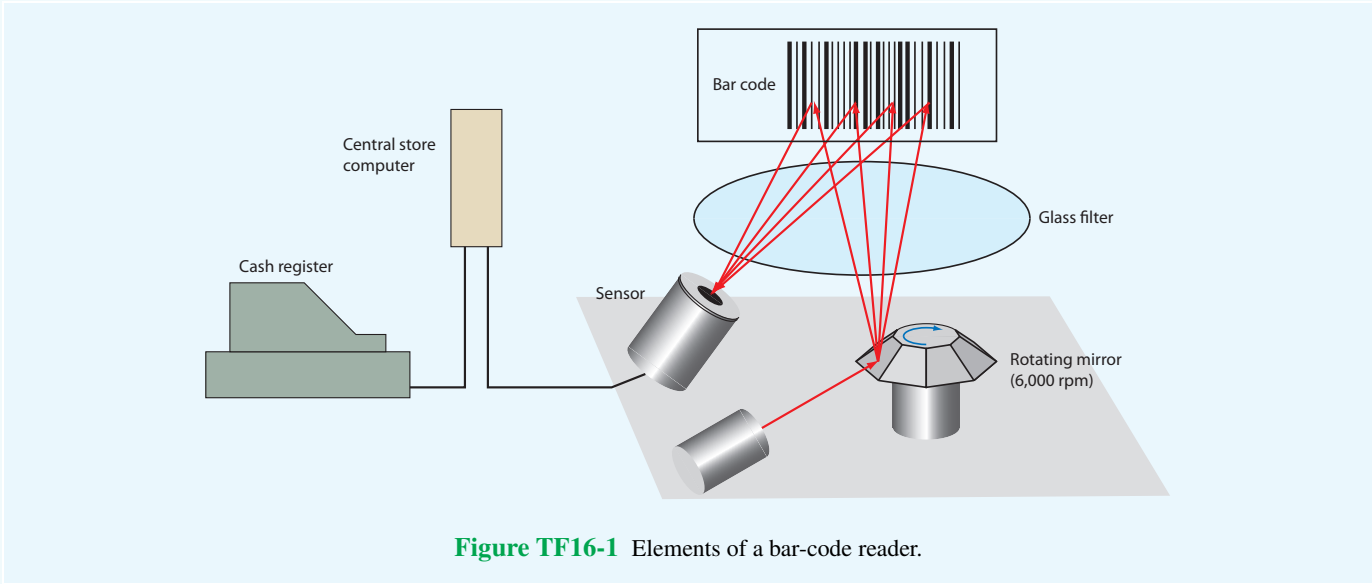


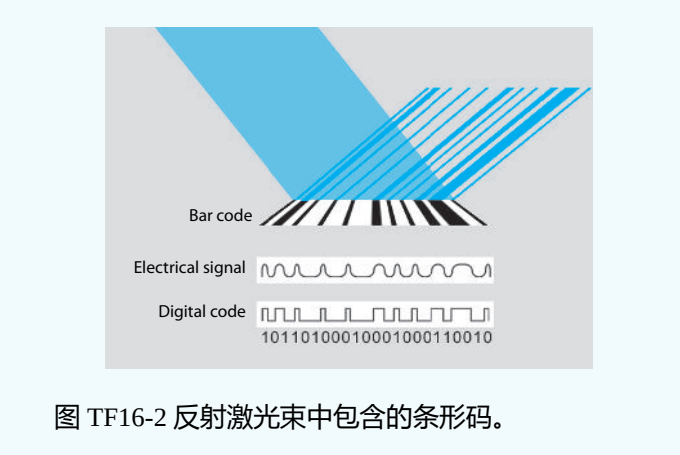
Figure TF16-1 Elements of a bar-code reader.

技术简介 16：条码 读取器

条形码由一系列具有一定宽度的平行条组成，通常以白色背景上的黑色印刷，配置为表示有关产品及其制造商的信息的特定二进制代码。激光扫描仪可以读取代码并将信息传输到计算机、收银机或显示屏。对于内置于杂货店收银台的固定扫描仪和可以像枪一样指向条形码物体的手持设备，条形码阅读器的基本操作是相同的。

基本操作

扫描仪使用激光束指向多面旋转镜，该镜以每分钟 6,000 转的速度高速旋转（图 TF16-1）。旋转镜产生扇形光束来照亮物体上的条形码。此外，通过将激光照射到其多个面，它可以将光束偏转到许多不同的方向，从而可以在广泛的位置和方向上扫描物体。目标是使这些方向之一使光束反射



条形码最终沿光检测器（传感器）的方向行进并被光检测器（传感器）捕获，然后读取编码序列（白条反射激光，黑条不反射激光）并将其转换为 1 和 0 的二进制序列（图 TF16-2）。为了消除环境光的干扰，如图TF16-1所示，使用玻璃滤光片来阻挡除激光波长为中心的窄波长带之外的所有光。

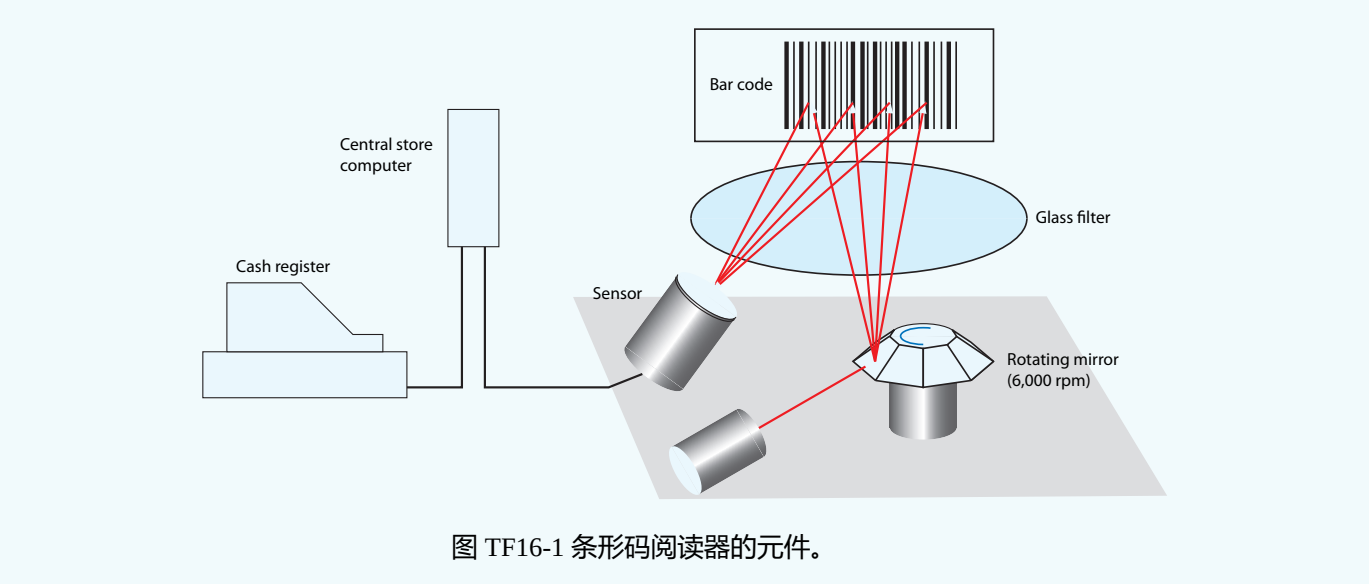


图 TF16-1 条形码阅读器的元件。

8-4.2 Parallel Polarization

If we interchange the roles played by $\mathbf{E}$ and $\mathbf{H}$ in the perpendicular polarization scenario covered in the preceding subsection, while keeping in mind the requirement that $\mathbf{E} \times \mathbf{H}$ must point in the direction of propagation for each of the incident, reflected, and transmitted waves, we end up with the parallel polarization scenario shown in **Fig. 8-16**. Now the electric fields lie in the plane of incidence, while the associated magnetic fields are perpendicular to the plane of incidence. With reference to the directions indicated in **Fig. 8-16**, the fields of the incident, reflected, and transmitted waves are given by

Incident Wave

$$\begin{aligned} \tilde{\mathbf{E}}_{\parallel}^i &= \hat{\mathbf{y}}_i E_{\parallel 0}^i e^{-jk_1 x_i} \\ &= (\hat{\mathbf{x}} \cos \theta_i - \hat{\mathbf{z}} \sin \theta_i) E_{\parallel 0}^i e^{-jk_1 (x \sin \theta_i + z \cos \theta_i)}, \end{aligned} \quad (8.65a)$$

$$\tilde{\mathbf{H}}_{\parallel}^i = \hat{\mathbf{y}} \frac{E_{\parallel 0}^i}{\eta_1} e^{-jk_1 x_i} = \hat{\mathbf{y}} \frac{E_{\parallel 0}^i}{\eta_1} e^{-jk_1 (x \sin \theta_i + z \cos \theta_i)}, \quad (8.65b)$$

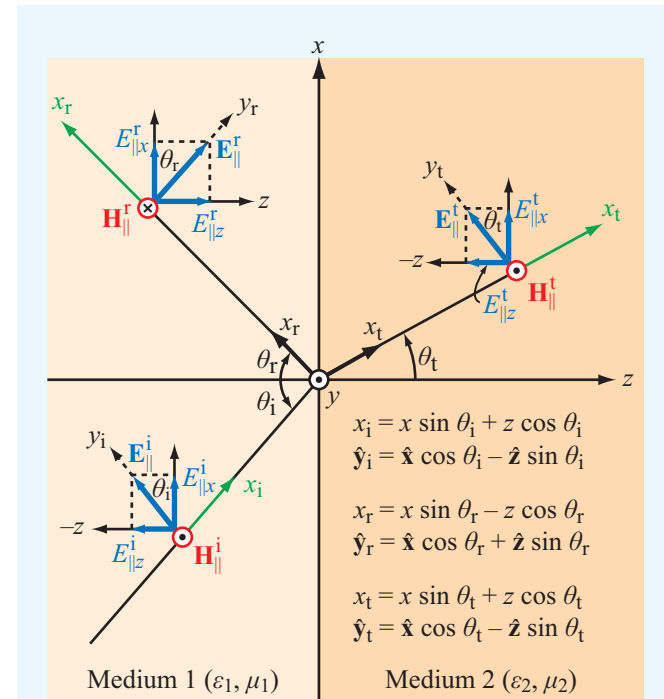


Figure 8-16 Parallel-polarized plane wave incident at an angle θ_i upon a planar boundary.

Reflected Wave

$$\begin{aligned} \tilde{\mathbf{E}}_{\parallel}^r &= \hat{\mathbf{y}}_r E_{\parallel 0}^r e^{-jk_1 x_r} \\ &= (\hat{\mathbf{x}} \cos \theta_r + \hat{\mathbf{z}} \sin \theta_r) E_{\parallel 0}^r e^{-jk_1 (x \sin \theta_r - z \cos \theta_r)}, \end{aligned} \quad (8.65c)$$

$$\tilde{\mathbf{H}}_{\parallel}^r = -\hat{\mathbf{y}} \frac{E_{\parallel 0}^r}{\eta_1} e^{-jk_1 x_r} = -\hat{\mathbf{y}} \frac{E_{\parallel 0}^r}{\eta_1} e^{-jk_1 (x \sin \theta_r - z \cos \theta_r)}, \quad (8.65d)$$

Transmitted Wave

$$\begin{aligned} \tilde{\mathbf{E}}_{\parallel}^t &= \hat{\mathbf{y}}_t E_{\parallel 0}^t e^{-jk_2 x_t} \\ &= (\hat{\mathbf{x}} \cos \theta_t - \hat{\mathbf{z}} \sin \theta_t) E_{\parallel 0}^t e^{-jk_2 (x \sin \theta_t + z \cos \theta_t)}, \end{aligned} \quad (8.65e)$$

$$\tilde{\mathbf{H}}_{\parallel}^t = \hat{\mathbf{y}} \frac{E_{\parallel 0}^t}{\eta_2} e^{-jk_2 x_t} = \hat{\mathbf{y}} \frac{E_{\parallel 0}^t}{\eta_2} e^{-jk_2 (x \sin \theta_t + z \cos \theta_t)}. \quad (8.65f)$$

By matching the tangential components of $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ in both media at $z = 0$, we again obtain the relations defining Snell's laws, as well as the following expressions for the **Fresnel reflection and transmission coefficients for parallel polarization**:

$$\Gamma_{\parallel} = \frac{E_{\parallel 0}^r}{E_{\parallel 0}^i} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}, \quad (8.66a)$$

$$\tau_{\parallel} = \frac{E_{\parallel 0}^t}{E_{\parallel 0}^i} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}. \quad (8.66b)$$

The preceding expressions can be shown to yield the relation

$$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}. \quad (8.67)$$

We noted earlier in connection with the perpendicular-polarization case that, when the second medium is a perfect conductor with $\eta_2 = 0$, the incident wave gets totally reflected at the boundary. The same is true for the parallel polarization case; setting $\eta_2 = 0$ in Eqs. (8.66a and b) gives $\Gamma_{\parallel} = -1$ and $\tau_{\parallel} = 0$.

For nonmagnetic materials, Eq. (8.66a) becomes

$$\Gamma_{\parallel} = \frac{-(\epsilon_2/\epsilon_1) \cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{(\epsilon_2/\epsilon_1) \cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}} \quad (8.68)$$

(for $\mu_1 = \mu_2$).

8-4.2 平行偏振

如果我们交换 $\mathbf{E}$ 和 $\mathbf{H}$ 在上一小节介绍的垂直偏振场景中扮演的角色，同时牢记 $\mathbf{E} \times \mathbf{H}$ 必须指向每个入射波、反射波和透射波的传播方向的要求，我们最终会得到图 8-16 所示的平行偏振场景。现在电场位于入射平面内，而相关的磁场则垂直于入射平面。参考图 8-16 所示的方向，入射波、反射波和透射波的场由下式给出

$$\begin{aligned} \tilde{\mathbf{E}}_{\parallel}^i &= \hat{\mathbf{y}}_i E_{\parallel 0}^i e^{-jk_1 x_i} \\ &= (\hat{\mathbf{x}} \cos \theta_i - \hat{\mathbf{z}} \sin \theta_i) E_{\parallel 0}^i e^{-jk_1 (x \sin \theta_i + z \cos \theta_i)}, \end{aligned} \quad (8.65a)$$

$$\tilde{\mathbf{H}}_{\parallel}^i = \hat{\mathbf{y}} \frac{E_{\parallel 0}^i}{\eta_1} e^{-jk_1 x_i} = \hat{\mathbf{y}} \frac{E_{\parallel 0}^i}{\eta_1} e^{-jk_1 (x \sin \theta_i + z \cos \theta_i)}, \quad (8.65b)$$

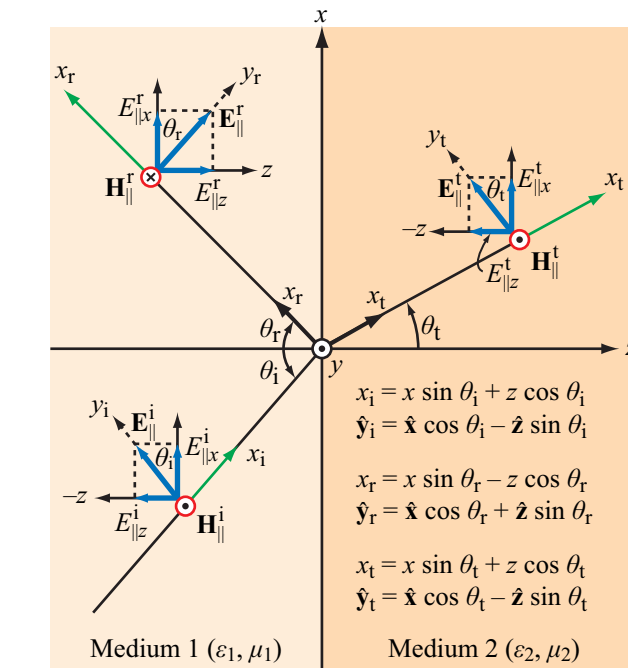


图 8-16 平行偏振平面波以角度 θ_i 入射到平面边界上。

反射波

$$\begin{aligned} \tilde{\mathbf{E}}_{\parallel}^r &= \hat{\mathbf{y}}_r E_{\parallel 0}^r e^{-jk_1 x_r} \\ &= (\hat{\mathbf{x}} \cos \theta_r + \hat{\mathbf{z}} \sin \theta_r) E_{\parallel 0}^r e^{-jk_1 (x \sin \theta_r - z \cos \theta_r)}, \end{aligned} \quad (8.65c)$$

$$\tilde{\mathbf{H}}_{\parallel}^r = -\hat{\mathbf{y}} \frac{E_{\parallel 0}^r}{\eta_1} e^{-jk_1 x_r} = -\hat{\mathbf{y}} \frac{E_{\parallel 0}^r}{\eta_1} e^{-jk_1 (x \sin \theta_r - z \cos \theta_r)}, \quad (8.65d)$$

传输波

$$\begin{aligned} \tilde{\mathbf{E}}_{\parallel}^t &= \hat{\mathbf{y}}_t E_{\parallel 0}^t e^{-jk_2 x_t} \\ &= (\hat{\mathbf{x}} \cos \theta_t - \hat{\mathbf{z}} \sin \theta_t) E_{\parallel 0}^t e^{-jk_2 (x \sin \theta_t + z \cos \theta_t)}, \end{aligned} \quad (8.65e)$$

$$\tilde{\mathbf{H}}_{\parallel}^t = \hat{\mathbf{y}} \frac{E_{\parallel 0}^t}{\eta_2} e^{-jk_2 x_t} = \hat{\mathbf{y}} \frac{E_{\parallel 0}^t}{\eta_2} e^{-jk_2 (x \sin \theta_t + z \cos \theta_t)}. \quad (8.65f)$$

通过匹配 $z = 0$ 处两种介质中 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 的切向分量，我们再次获得定义斯涅尔定律的关系，以及平行偏振菲涅尔反射和透射系数的以下表达式：

$$\Gamma_{\parallel} = \frac{E_{\parallel 0}^r}{E_{\parallel 0}^i} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}, \quad (8.66a)$$

$$\tau_{\parallel} = \frac{E_{\parallel 0}^t}{E_{\parallel 0}^i} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}. \quad (8.66b)$$

前面的表达式可以得出以下关系

$$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}. \quad (8.67)$$

我们之前在垂直偏振情况下注意到，当第二介质是 $\eta_2 = 0$ 的理想导体时，入射波在边界处发生全反射。对于平行偏振的情况也是如此；在等式中设置 $\eta_2 = 0$ (8.66a 和 b) 给出 $\Gamma_{\parallel} = -1$ 和 $\tau_{\parallel} = 0$ 。

对于非磁性材料，方程 (8.66a) 变为

$$\Gamma_{\parallel} = \frac{-(\epsilon_2/\epsilon_1) \cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{(\epsilon_2/\epsilon_1) \cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}} \quad (8.68)$$

(for $\mu_1 = \mu_2$).

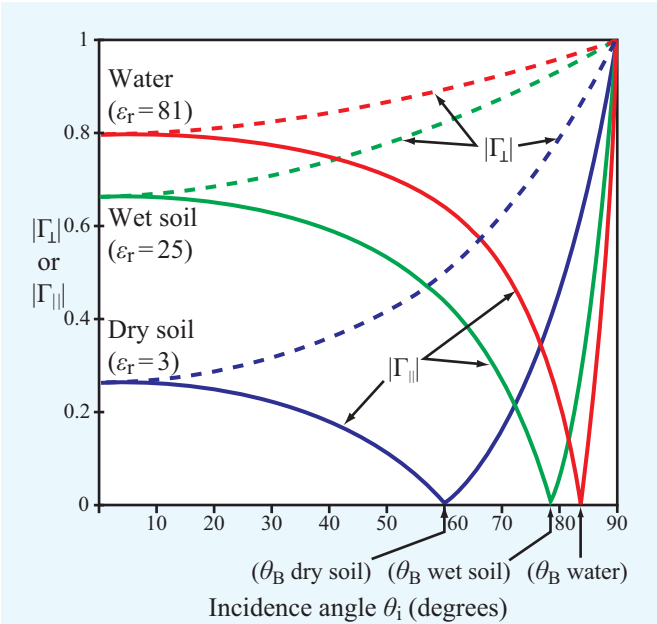


Figure 8-17 Plots for $|\Gamma_{\perp}|$ and $|\Gamma_{\parallel}|$ as a function of θ_i for a dry soil surface, a wet soil surface, and a water surface. For each surface, $|\Gamma_{\parallel}| = 0$ at the Brewster angle.

Perpendicular Polarization

For perpendicular polarization, the Brewster angle $\theta_{B\perp}$ can be obtained by setting the numerator of the expression for $\Gamma_{\perp}$, given by Eq. (8.58a), equal to zero. This happens when

$$\eta_2 \cos \theta_i = \eta_1 \cos \theta_t. \quad (8.69)$$

By (1) squaring both sides of Eq. (8.69), (2) using Eq. (8.56), (3) solving for θ_i , and then denoting θ_i as $\theta_{B\perp}$, we obtain

$$\sin \theta_{B\perp} = \sqrt{\frac{1 - (\mu_1 \epsilon_2 / \mu_2 \epsilon_1)}{1 - (\mu_1 / \mu_2)^2}}. \quad (8.70)$$

Because the denominator of Eq. (8.70) goes to zero when $\mu_1 = \mu_2$, $\theta_{B\perp}$ *does not exist for nonmagnetic materials*.

Parallel Polarization

For parallel polarization, the Brewster angle $\theta_{B\parallel}$ at which $\Gamma_{\parallel} = 0$ can be found by setting the numerator of $\Gamma_{\parallel}$ in Eq. (8.66a) equal to zero. The result is identical to Eq. (8.70), but with μ and ϵ interchanged. That is,

$$\sin \theta_{B\parallel} = \sqrt{\frac{1 - (\epsilon_1 \mu_2 / \epsilon_2 \mu_1)}{1 - (\epsilon_1 / \epsilon_2)^2}}. \quad (8.71)$$

For nonmagnetic materials,

$$\begin{aligned} \theta_{B\parallel} &= \sin^{-1} \sqrt{\frac{1}{1 + (\epsilon_1 / \epsilon_2)}} \\ &= \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}} \quad (\text{for } \mu_1 = \mu_2). \end{aligned} \quad (8.72)$$

The Brewster angle is also called the *polarizing angle*. This is because, if a wave composed of both perpendicular and parallel polarization components is incident upon a nonmagnetic surface at the Brewster angle $\theta_{B\parallel}$, the parallel polarized component is totally transmitted into the second medium, and only the perpendicularly polarized component is reflected by the surface. Natural light, including sunlight and light generated by most manufactured sources, is *unpolarized* because the direction of the electric field of the light waves varies randomly in angle over the plane perpendicular to the direction of propagation. Thus, on average, half of the intensity of natural light is perpendicularly polarized and the other half is parallel polarized. When unpolarized light is incident upon a surface at the Brewster angle, the reflected wave is strictly perpendicularly polarized. Hence, the surface acts as a polarizer.

► At the Brewster angle, the parallel-polarized component of the incident wave is totally transmitted into medium 2. ◄

8-4.3 Brewster Angle

The Brewster angle θ_B is defined as the incidence angle θ_i at which the Fresnel reflection coefficient $\Gamma = 0$.

垂直偏振

对于垂直偏振，布儒斯特角 $\theta_{B\perp}$ 可以通过设置 $\Gamma_{\perp}$ 表达式的分子来获得，如式 (1) 所示。(8.58a)，等于零。发生这种情况时

$$\eta_2 \cos \theta_i = \eta_1 \cos \theta_t. \quad (8.69)$$

通过 (1) 对方程两边求平方 (8.69)，(2) 使用方程。(8.56), (3) 求解 θ_i ，然后将 θ_i 表示为 $\theta_{B\perp}$ ，我们得到

$$\sin \theta_{B\perp} = \sqrt{\frac{1 - (\mu_1 \epsilon_2 / \mu_2 \epsilon_1)}{1 - (\mu_1 / \mu_2)^2}}. \quad (8.70)$$

因为方程的分母。当非磁性材料不存在 $\mu_1 = \mu_2$ 、 $\theta_{B\perp}$ 时，(8.70) 为零。

平行偏振

对于平行偏振，可以通过设置式 (1) 中 $\Gamma_{\parallel}$ 的分子来找到 $\Gamma_{\parallel} = 0$ 时的布儒斯特角 $\theta_{B\parallel}$ 。(8.66a) 等于零。结果与等式相同。(8.70)，但 μ 和 ϵ 互换。那是，

$$\sin \theta_{B\parallel} = \sqrt{\frac{1 - (\epsilon_1 \mu_2 / \epsilon_2 \mu_1)}{1 - (\epsilon_1 / \epsilon_2)^2}}. \quad (8.71)$$

对于非磁性材料，

$$\begin{aligned} \theta_{B\parallel} &= \sin^{-1} \sqrt{\frac{1}{1 + (\epsilon_1 / \epsilon_2)}} \\ &= \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}} \quad (\text{for } \mu_1 = \mu_2). \end{aligned} \quad (8.72)$$

布鲁斯特角也称为偏振角。这是因为，如果由垂直和平行偏振分量组成的波以布儒斯特角 $\theta_{B\parallel}$ 入射到非磁性表面，则平行偏振分量将完全透射到第二介质中，而只有垂直偏振分量被表面反射。自然光，包括太阳光和大多数人造光源产生的光，都是非偏振的，因为光波的电场方向在垂直于传播方向的平面上的角度随机变化。因此，平均而言，自然光强度的一半是垂直偏振的，另一半是平行偏振的。当非偏振光以布儒斯特角入射到表面时，反射波是严格垂直偏振的。因此，该表面充当偏振器。

为了说明 $\Gamma_{\perp}$ 和 $\Gamma_{\parallel}$ 幅度的角度变化，图 8-17 显示了空气中入射到三种不同类型介电表面的波的绘图：干燥土壤 ($\epsilon_r = 3$)、湿土壤 ($\epsilon_r = 25$) 和水 ($\epsilon_r = 81$)。对于每个表面，(1) 垂直入射时的 $\Gamma_{\perp} = \Gamma_{\parallel}$ ($\theta_i = 0$)，如预期；(2) 掠射入射角处的 $|\Gamma_{\perp}| = |\Gamma_{\parallel}| = 1$ ($\theta_i = 90^\circ$)；(3) $\Gamma_{\parallel}$ 在图 8-17 中称为布儒斯特角的角度处为零。如果这些材料也具有磁性 ($\mu_1 \neq \mu_2$)，那么 $\Gamma_{\perp}$ 也有可能某个角度消失。然而，对于非磁性材料，布儒斯特角仅存在于平行极化时，其值取决于比率 (ϵ_2 / ϵ_1)，正如我们稍后看到的。

► 在布儒斯特角处，入射波的平行偏振分量完全传输到介质 2 中。◄

8-4.3 布鲁斯特角

布鲁斯特角 θ_B 定义为菲涅耳反射系数 $\Gamma = 0$ 时的入射角 θ_i 。

Concept Question 8-4: Can total internal reflection take place for a wave incident from medium 1 (with n_1) onto medium 2 (with n_2) when $n_2 > n_1$?

Concept Question 8-5: What is the difference between the boundary conditions applied in Section 8-1.1 for normal incidence and those applied in Section 8-4.1 for oblique incidence with perpendicular polarization?

Concept Question 8-6: Why is the Brewster angle also called the polarizing angle?

Concept Question 8-7: At the boundary, the vector sum of the tangential components of the incident and reflected electric fields has to equal the tangential component of the transmitted electric field. For $\epsilon_{r1} = 1$ and $\epsilon_{r2} = 16$, determine the Brewster angle and then verify the validity of the preceding statement by sketching to scale the tangential components of the three electric fields at the Brewster angle.

Exercise 8-7: A wave in air is incident upon a soil surface at $\theta_i = 50^\circ$. If soil has $\epsilon_r = 4$ and $\mu_r = 1$, determine $\Gamma_\perp$, $\tau_\perp$, $\Gamma_\parallel$, and $\tau_\parallel$.

Answer: $\Gamma_\perp = -0.48$, $\tau_\perp = 0.52$, $\Gamma_\parallel = -0.16$, $\tau_\parallel = 0.58$. (See [EM](#).)

Exercise 8-8: Determine the Brewster angle for the boundary of Exercise 8.7.

Answer: $\theta_B = 63.4^\circ$. (See [EM](#).)

Exercise 8-9: Show that the incident, reflected, and transmitted electric and magnetic fields given by Eqs. (8.65a) through (8.65f) all have the same exponential phase function along the x direction.

Answer: With the help of Eqs. (8.55) and (8.56), all six fields are shown to vary as $e^{-jk_1 x \sin \theta_i}$. (See [EM](#).)

8-5 Reflectivity and Transmissivity

The reflection and transmission coefficients derived earlier are ratios of the reflected and transmitted electric field amplitudes to the amplitude of the incident electric field. We now examine power ratios, starting with the perpendicular polarization case. **Figure 8-18** shows a circular beam of electromagnetic

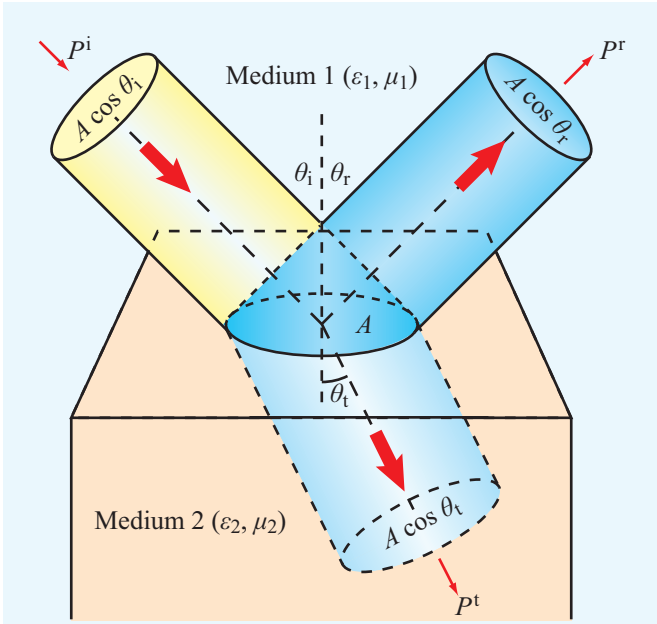


Figure 8-18 Reflection and transmission of an incident circular beam illuminating a spot of size A on the interface.

energy incident upon the boundary between two contiguous, lossless media. The area of the spot illuminated by the beam is A , and the incident, reflected, and transmitted beams have electric-field amplitudes $E_{\perp 0}^i$, $E_{\perp 0}^r$, and $E_{\perp 0}^t$, respectively. The average power densities carried by the incident, reflected, and transmitted beams are

$$S_{\perp}^i = \frac{|E_{\perp 0}^i|^2}{2\eta_1}, \quad (8.73a)$$

$$S_{\perp}^r = \frac{|E_{\perp 0}^r|^2}{2\eta_1}, \quad (8.73b)$$

$$S_{\perp}^t = \frac{|E_{\perp 0}^t|^2}{2\eta_2}, \quad (8.73c)$$

where η_1 and η_2 are the intrinsic impedances of media 1 and 2, respectively. The cross-sectional areas of the incident, reflected, and transmitted beams are

$$A_i = A \cos \theta_i, \quad (8.74a)$$

$$A_r = A \cos \theta_r, \quad (8.74b)$$

$$A_t = A \cos \theta_t, \quad (8.74c)$$

概念题 8-4: 波由介质1 (折射率 n_1) 入射到介质2 (折射率 n_2) , 当 $n_2 > n_1$ 时, 能否发生全内反射?

概念题 8-5: 第 8-1.1 节垂直入射所用的边界条件, 与第 8-4.1 节垂直极化波斜入射所用的边界条件有何区别?

概念题 8-6: 为什么布儒斯特角又称为起偏角?

概念题 8-7: 在界面处, 入射电场与反射电场的切向分量之矢量和, 必须等于透射电场的切向分量。已知 $\epsilon_{r1} = 1$ 、 $\epsilon_{r2} = 16$, 求布儒斯特角, 并按比例画出该入射角下三个电场的切向分量, 以验证上述关系。

练习 8-7: 空气中的波以 $\theta_i = 50^\circ$ 入射到土壤表面。若土壤 $\epsilon_r = 4$ 、 $\mu_r = 1$, 求 $\Gamma_\perp$ 、 $\tau_\perp$ 、 $\Gamma_\parallel$ 、 $\tau_\parallel$ 。

答案: $\Gamma_\perp = -0.48$, $\tau_\perp = 0.52$, $\Gamma_\parallel = -0.16$, $\tau_\parallel = 0.58$ 。(参见配套 EM 资源。)

练习 8-8: 求练习 8-7 中界面的布儒斯特角。

答案: $\theta_B = 63.4^\circ$ 。(参见配套 EM 资源。)

练习 8-9: 证明式 (8.65a) 至 (8.65f) 中的入射、反射和透射电场及磁场, 沿 x 方向具有相同的指数相位因子。

答案: 利用式 (8.55)、(8.56), 可得六个场均含因子 $e^{-jk_1 x \sin \theta_i}$ 。(参见配套 EM 资源。)

8-5 反射率与透射率

前面得到的反射系数与透射系数, 分别为反射、透射电场振幅与入射电场振幅之比。下面从垂直极化的情况入手, 研究功率之比。图 8-18 表示一束圆形截面的电磁波束

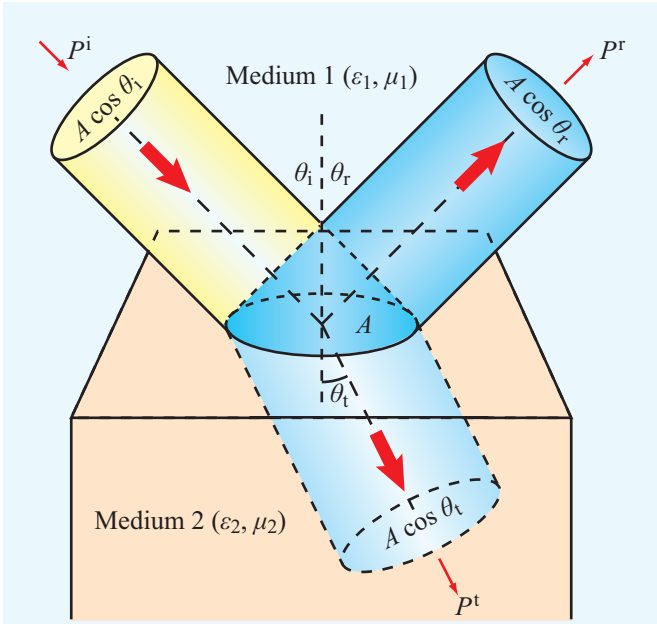


图 8-18 圆形入射波束的反射与透射; 它在界面上的照射区域面积为 A 。

图例: Medium 1、Medium 2 分别为介质1、介质2。

入射到两个相邻无损介质的界面。波束照射区域的面积为 A ; 入射、反射和透射波束的电场振幅分别为 $E_{\perp 0}^i$ 、 $E_{\perp 0}^r$ 、 $E_{\perp 0}^t$ 。三个波束携带的平均功率密度分别为

$$S_{\perp}^i = \frac{|E_{\perp 0}^i|^2}{2\eta_1}, \quad (8.73a)$$

$$S_{\perp}^r = \frac{|E_{\perp 0}^r|^2}{2\eta_1}, \quad (8.73b)$$

$$S_{\perp}^t = \frac{|E_{\perp 0}^t|^2}{2\eta_2}, \quad (8.73c)$$

其中, η_1 和 η_2 分别为介质1、介质2的本征阻抗。入射、反射和透射波束的横截面积分别为

$$A_i = A \cos \theta_i, \quad (8.74a)$$

$$A_r = A \cos \theta_r, \quad (8.74b)$$

$$A_t = A \cos \theta_t, \quad (8.74c)$$

and the corresponding average powers carried by the beams are

$$P_{\perp}^i = S_{\perp}^i A_i = \frac{|E_{\perp 0}^i|^2}{2\eta_1} A \cos \theta_i, \tag{8.75a}$$

$$P_{\perp}^r = S_{\perp}^r A_r = \frac{|E_{\perp 0}^r|^2}{2\eta_1} A \cos \theta_r, \tag{8.75b}$$

$$P_{\perp}^t = S_{\perp}^t A_t = \frac{|E_{\perp 0}^t|^2}{2\eta_2} A \cos \theta_t. \tag{8.75c}$$

The **reflectivity** R (also called **reflectance** in optics) is defined as the ratio of the reflected to the incident power. The reflectivity for perpendicular polarization is then

$$R_{\perp} = \frac{P_{\perp}^r}{P_{\perp}^i} = \frac{|E_{\perp 0}^r|^2 \cos \theta_r}{|E_{\perp 0}^i|^2 \cos \theta_i} = \left| \frac{E_{\perp 0}^r}{E_{\perp 0}^i} \right|^2, \tag{8.76}$$

where we used the fact that $\theta_r = \theta_i$ in accordance with Snell’s law of reflection. The ratio of the reflected to incident electric field amplitudes, $|E_{\perp 0}^r/E_{\perp 0}^i|$, is equal to the magnitude of the reflection coefficient $\Gamma_{\perp}$. Hence,

$$R_{\perp} = |\Gamma_{\perp}|^2, \tag{8.77}$$

and similarly, for parallel polarization

$$R_{\parallel} = \frac{P_{\parallel}^r}{P_{\parallel}^i} = |\Gamma_{\parallel}|^2. \tag{8.78}$$

The **transmissivity** T (or **transmittance** in optics) is defined as the ratio of the transmitted power to incident power:

$$T_{\perp} = \frac{P_{\perp}^t}{P_{\perp}^i} = \frac{|E_{\perp 0}^t|^2}{|E_{\perp 0}^i|^2} \frac{\eta_1}{\eta_2} \frac{A \cos \theta_t}{A \cos \theta_i} = |\tau_{\perp}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right), \tag{8.79a}$$

$$T_{\parallel} = \frac{P_{\parallel}^t}{P_{\parallel}^i} = |\tau_{\parallel}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right). \tag{8.79b}$$

► The incident, reflected, and transmitted waves do not have to obey any such laws as conservation of electric field, conservation of magnetic field, or conservation of power density, but they do have to obey the law of conservation of power. ◀

In fact, in many cases, the transmitted electric field is larger than the incident electric field. Conservation of power requires that the incident power equals the sum of the reflected and transmitted powers. That is, for perpendicular polarization,

$$P_{\perp}^i = P_{\perp}^r + P_{\perp}^t, \tag{8.80}$$

or

$$\frac{|E_{\perp 0}^i|^2}{2\eta_1} A \cos \theta_i = \frac{|E_{\perp 0}^r|^2}{2\eta_1} A \cos \theta_r + \frac{|E_{\perp 0}^t|^2}{2\eta_2} A \cos \theta_t. \tag{8.81}$$

Use of Eqs. (8.76), (8.79a), and (8.79b) leads to

$$R_{\perp} + T_{\perp} = 1, \tag{8.82a}$$

$$R_{\parallel} + T_{\parallel} = 1, \tag{8.82b}$$

or

$$|\Gamma_{\perp}|^2 + |\tau_{\perp}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right) = 1, \tag{8.83a}$$

$$|\Gamma_{\parallel}|^2 + |\tau_{\parallel}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right) = 1. \tag{8.83b}$$

Figure 8-19 shows plots for $(R_{\parallel}, T_{\parallel})$ as a function of θ_i for an air–glass interface. Note that the sum of $R_{\parallel}$ and $T_{\parallel}$ is always equal to 1, as mandated by Eq. (8.82b). We also note that, at the Brewster angle θ_B , $R_{\parallel} = 0$ and $T_{\parallel} = 1$.

Table 8-2 provides a summary of the general expressions for Γ , τ , R , and T for both normal and oblique incidence.

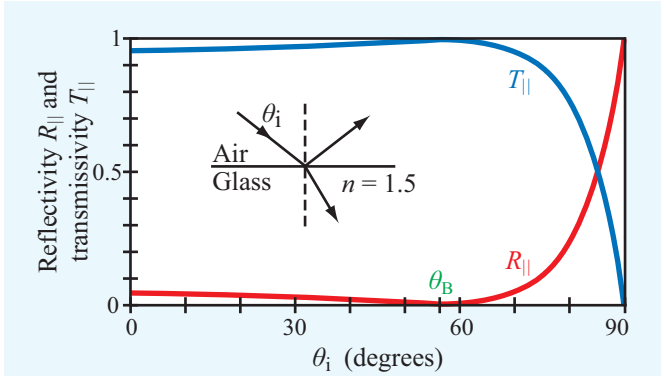


Figure 8-19 Angular plots for $(R_{\parallel}, T_{\parallel})$ for an air–glass interface.

光束所携带的相应平均功率为

$$P_{\perp}^i = S_{\perp}^i A_i = \frac{|E_{\perp 0}^i|^2}{2\eta_1} A \cos \theta_i, \tag{8.75a}$$

$$P_{\perp}^r = S_{\perp}^r A_r = \frac{|E_{\perp 0}^r|^2}{2\eta_1} A \cos \theta_r, \tag{8.75b}$$

$$P_{\perp}^t = S_{\perp}^t A_t = \frac{|E_{\perp 0}^t|^2}{2\eta_2} A \cos \theta_t. \tag{8.75c}$$

反射率 R (在光学中也称为反射率) 定义为反射功率与入射功率之比。垂直偏振的反射率为

$$R_{\perp} = \frac{P_{\perp}^r}{P_{\perp}^i} = \frac{|E_{\perp 0}^r|^2 \cos \theta_r}{|E_{\perp 0}^i|^2 \cos \theta_i} = \left| \frac{E_{\perp 0}^r}{E_{\perp 0}^i} \right|^2, \tag{8.76}$$

我们根据斯涅尔反射定律使用了 $\theta_r = \theta_i$ 这一事实。反射电场幅度与入射电场幅度之比 $|E_{\perp 0}^r/E_{\perp 0}^i|$ 等于反射系数 $\Gamma_{\perp}$ 的大小。因此,

$$R_{\perp} = |\Gamma_{\perp}|^2, \tag{8.77}$$

类似地, 对于平行偏振

$$R_{\parallel} = \frac{P_{\parallel}^r}{P_{\parallel}^i} = |\Gamma_{\parallel}|^2. \tag{8.78}$$

透射率 T (或光学中的透射率) 定义为透射功率与入射功率之比:

$$T_{\perp} = \frac{P_{\perp}^t}{P_{\perp}^i} = \frac{|E_{\perp 0}^t|^2}{|E_{\perp 0}^i|^2} \frac{\eta_1}{\eta_2} \frac{A \cos \theta_t}{A \cos \theta_i} = |\tau_{\perp}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right), \tag{8.79a}$$

$$T_{\parallel} = \frac{P_{\parallel}^t}{P_{\parallel}^i} = |\tau_{\parallel}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right). \tag{8.79b}$$

► 入射波、反射波和透射波不必遵守电场守恒定律、磁场守恒定律或功率密度守恒定律, 但它们必须遵守功率守恒定律。 ◀

事实上, 在许多情况下, 传输电场大于入射电场。功率守恒要求入射功率等于反射功率和传输功率之和。也就是说, 对于垂直偏振,

$$P_{\perp}^i = P_{\perp}^r + P_{\perp}^t, \tag{8.80}$$

或者

$$\frac{|E_{\perp 0}^i|^2}{2\eta_1} A \cos \theta_i = \frac{|E_{\perp 0}^r|^2}{2\eta_1} A \cos \theta_r + \frac{|E_{\perp 0}^t|^2}{2\eta_2} A \cos \theta_t. \tag{8.81}$$

使用等式. (8.76)、(8.79a) 和 (8.79b) 导致

$$R_{\perp} + T_{\perp} = 1, \tag{8.82a}$$

$$R_{\parallel} + T_{\parallel} = 1, \tag{8.82b}$$

或者

$$|\Gamma_{\perp}|^2 + |\tau_{\perp}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right) = 1, \tag{8.83a}$$

$$|\Gamma_{\parallel}|^2 + |\tau_{\parallel}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right) = 1. \tag{8.83b}$$

图 8-19 显示了空气-玻璃界面中 $(R_{\parallel}, T_{\parallel})$ 与 θ_i 的函数关系图。请注意, $R_{\parallel}$ 和 $T_{\parallel}$ 之和始终等于 1, 如方程式 1 所要求的那样. (8.82b)。我们还注意到, 在布儒斯特角 θ_B 处, $R_{\parallel} = 0$ 且 $T_{\parallel} = 1$ 。表 8-2 总结了正入射和斜入射的 Γ 、 τ 、 R 和 T 的通用表达式。

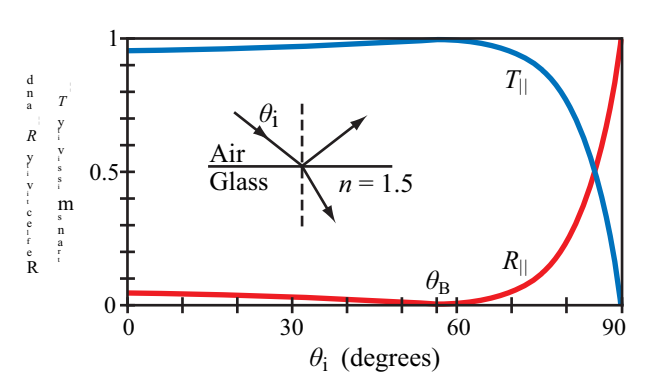


图 8-19 空气-玻璃界面的 $(R_{\parallel}, T_{\parallel})$ 角度图。

Table 8-2 Expressions for Γ , τ , R , and T for wave incidence from a medium with intrinsic impedance η_1 onto a medium with intrinsic impedance η_2 . Angles θ_i and θ_t are the angles of incidence and transmission, respectively.

Property	Normal Incidence $\theta_i = \theta_t = 0$	Perpendicular Polarization	Parallel Polarization
Reflection coefficient	$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Transmission coefficient	$\tau = \frac{2\eta_2}{\eta_2 + \eta_1}$	$\tau_{\perp} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\tau_{\parallel} = \frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Relation of Γ to τ	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$
Reflectivity	$R = \Gamma ^2$	$R_{\perp} = \Gamma_{\perp} ^2$	$R_{\parallel} = \Gamma_{\parallel} ^2$
Transmissivity	$T = \tau ^2 \left(\frac{\eta_1}{\eta_2} \right)$	$T_{\perp} = \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$T_{\parallel} = \tau_{\parallel} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
Relation of R to T	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{\parallel} = 1 - R_{\parallel}$

Notes: (1) $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; (2) $\eta_1 = \sqrt{\mu_1 / \epsilon_1}$; (3) $\eta_2 = \sqrt{\mu_2 / \epsilon_2}$; (4) for nonmagnetic media, $\eta_2 / \eta_1 = n_1 / n_2$.

Module 8.3 Oblique Incidence Upon specifying the frequency, polarization and incidence angle of a plane wave incident upon a planar boundary between two lossless media, this module displays vector information and plots of the reflection and transmission coefficients as a function of the incidence angle.

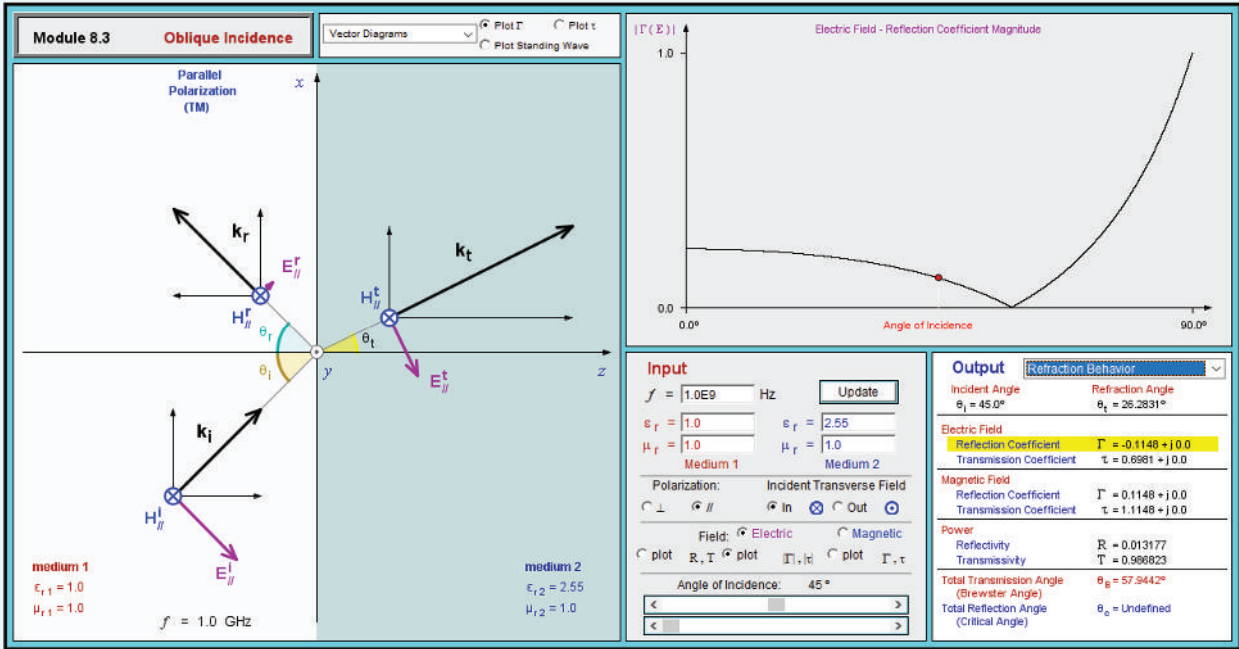
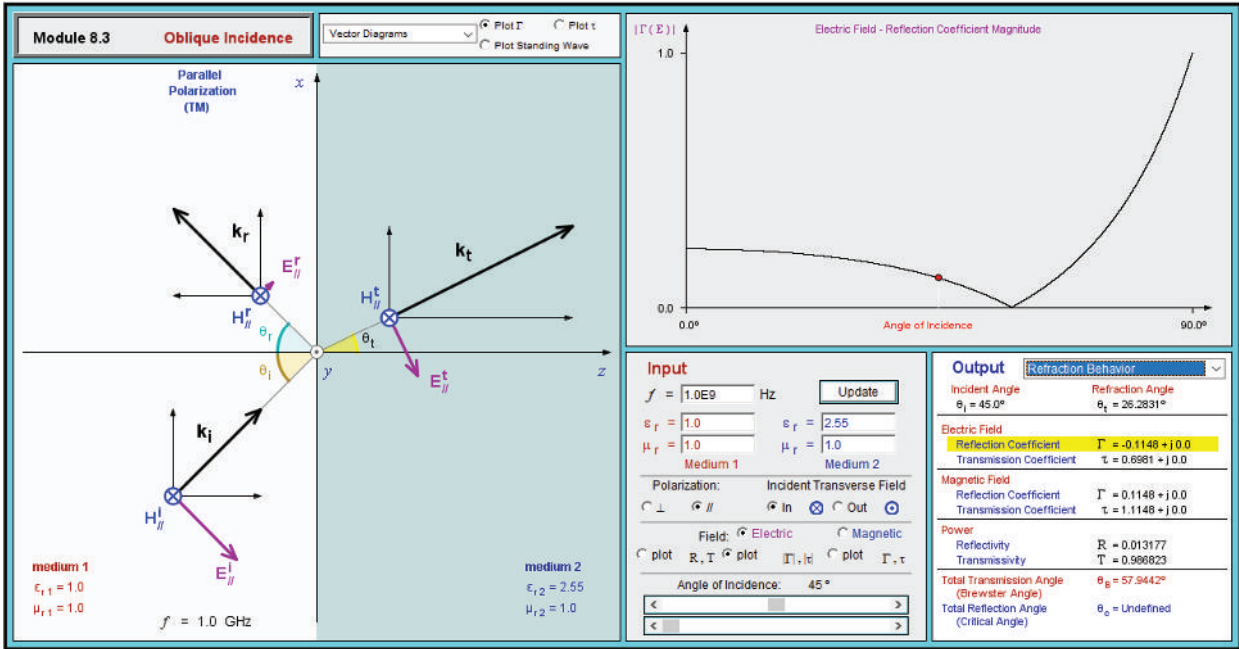


表 8-2 波从具有固有阻抗 η_1 的介质入射到具有固有阻抗 η_2 的介质时的 Γ 、 τ 、 R 和 T 表达式。角度 θ_i 和 θ_t 分别是入射角和透射角。

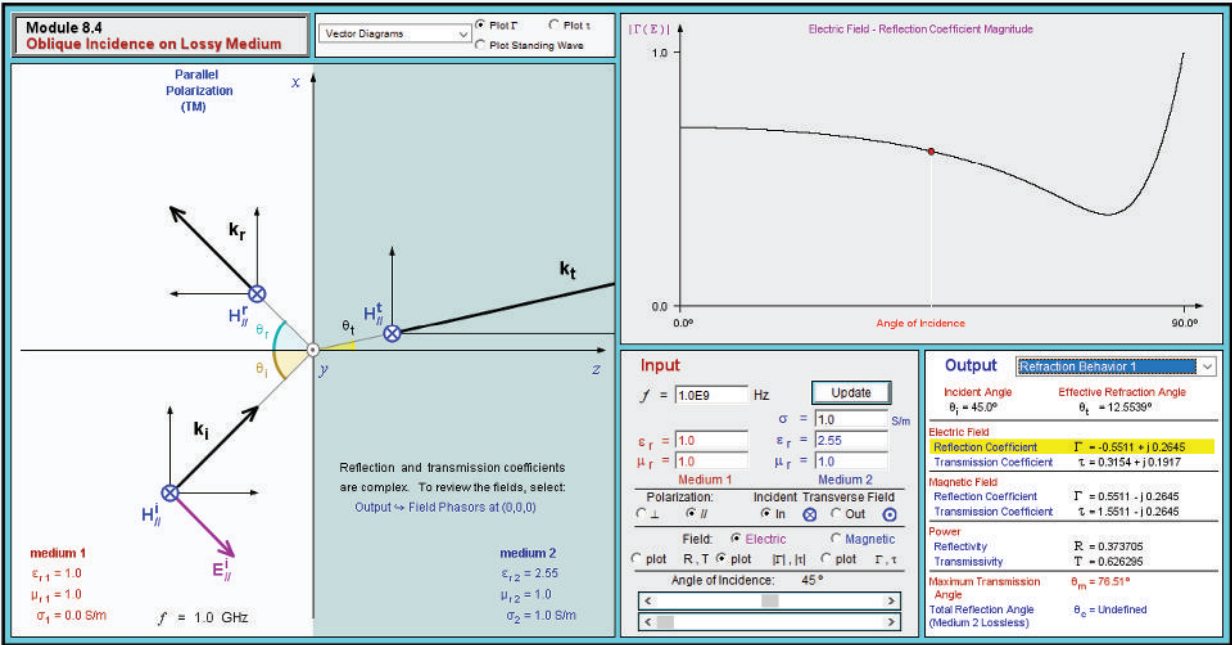
Property	Normal Incidence $\theta_i = \theta_t = 0$	Perpendicular Polarization	Parallel Polarization
Reflection coefficient	$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Transmission coefficient	$\tau = \frac{2\eta_2}{\eta_2 + \eta_1}$	$\tau_{\perp} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\tau_{\parallel} = \frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Relation of Γ to τ	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$
Reflectivity	$R = \Gamma ^2$	$R_{\perp} = \Gamma_{\perp} ^2$	$R_{\parallel} = \Gamma_{\parallel} ^2$
Transmissivity	$T = \tau ^2 \left(\frac{\eta_1}{\eta_2} \right)$	$T_{\perp} = \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$T_{\parallel} = \tau_{\parallel} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
Relation of R to T	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{\parallel} = 1 - R_{\parallel}$

Notes: (1) $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; (2) $\eta_1 = \sqrt{\mu_1 / \epsilon_1}$; (3) $\eta_2 = \sqrt{\mu_2 / \epsilon_2}$; (4) for nonmagnetic media, $\eta_2 / \eta_1 = n_1 / n_2$.

模块 8.3 倾斜入射 在指定入射到两个无损介质之间的平面边界上的平面波的频率、偏振和入射角后，该模块显示矢量信息以及作为入射角函数的反射和透射系数图。



Module 8.4 Oblique Incidence in Lossy Medium This module extends the capabilities of Module 8.1 to situations in which medium 2 is lossy.



Example 8-7: Beam of Light

A 5 W beam of light with circular cross section is incident in air upon the plane boundary of a dielectric medium with index of refraction of 5. If the angle of incidence is 60° and the incident wave is parallel polarized, determine the transmission angle and the powers contained in the reflected and transmitted beams.

Solution: From Eq. (8.56),

$$\sin \theta_t = \frac{n_1}{n_2} \sin \theta_i = \frac{1}{5} \sin 60^\circ = 0.17$$

or

$$\theta_t = 10^\circ.$$

With $\epsilon_2/\epsilon_1 = n_2^2/n_1^2 = (5)^2 = 25$, the reflection coefficient for parallel polarization follows from Eq. (8.68) as

$$\begin{aligned} \Gamma_{\parallel} &= \frac{-(\epsilon_2/\epsilon_1) \cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{(\epsilon_2/\epsilon_1) \cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}} \\ &= \frac{-25 \cos 60^\circ + \sqrt{25 - \sin^2 60^\circ}}{25 \cos 60^\circ + \sqrt{25 - \sin^2 60^\circ}} = -0.435. \end{aligned}$$

The reflected and transmitted powers therefore are

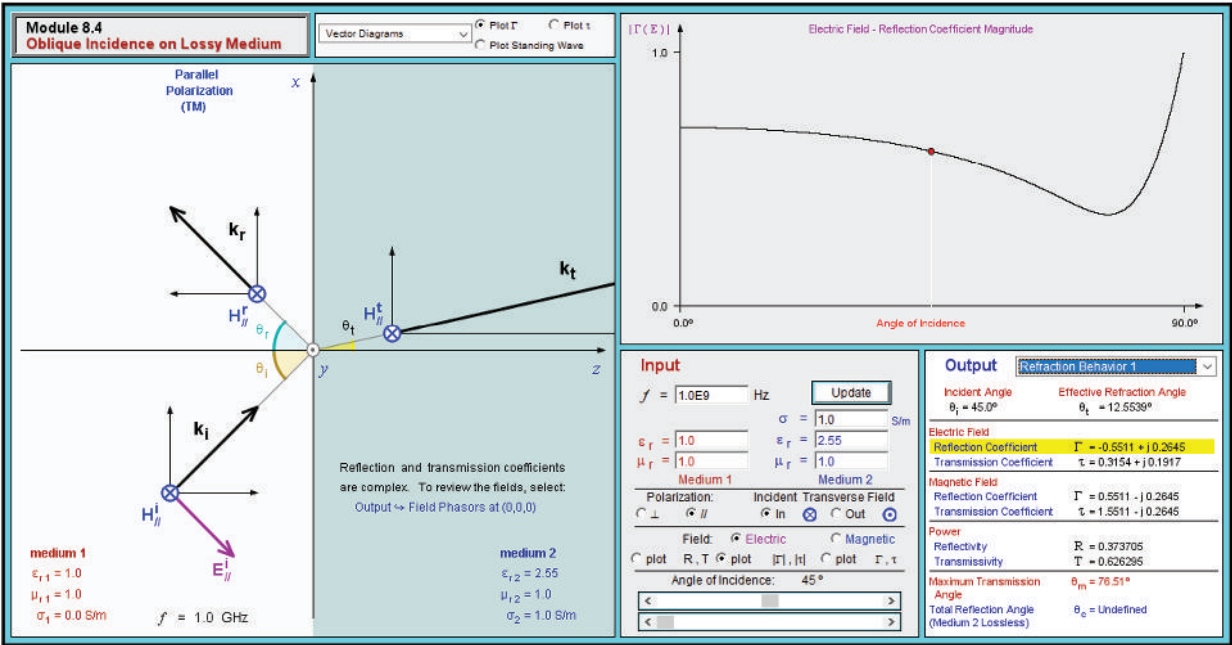
$$\begin{aligned} P_{\parallel}^r &= P_{\parallel}^i |\Gamma_{\parallel}|^2 = 5(0.435)^2 = 0.95 \text{ W}, \\ P_{\parallel}^t &= P_{\parallel}^i - P_{\parallel}^r = 5 - 0.95 = 4.05 \text{ W}. \end{aligned}$$

8-6 Waveguides

Earlier in Chapter 2, we considered two families of transmission lines, namely those that support **transverse-electromagnetic** (TEM) modes, and those that do not. Transmission lines belonging to the TEM family (Fig. 2-4), including coaxial, two-wire, and parallel-plate lines, support **E** and **H** fields that are orthogonal to the direction of propagation. Fields supported by lines in the other group—often called **higher-order transmission lines**—may have **E** or **H** orthogonal to the direction of propagation $\hat{\mathbf{k}}$, but not both simultaneously. Thus, *at least one component of **E** or **H** is along $\hat{\mathbf{k}}$.*

► If **E** is transverse to $\hat{\mathbf{k}}$ but **H** is not, we call it a **transverse electric** (TE) mode, and if **H** is transverse to $\hat{\mathbf{k}}$ but **E** is not, we call it a **transverse magnetic** (TM) mode. ◀

模块 8.4 有损介质中的斜入射 该模块将模块 8.1 的功能扩展到介质 2 有损的情况。



例8-7：光束

具有圆形横截面的 5 W 光束在空气中入射到折射率为 5 的电介质的平面边界上。如果入射角为 60° 并且入射波为平行偏振，请确定透射角以及反射和透射光束中包含的功率。

解决方案：从方程。 (8.56) ,

$$\sin \theta_t = \frac{n_1}{n_2} \sin \theta_i = \frac{1}{5} \sin 60^\circ = 0.17$$

或者

$$\theta_t = 10^\circ.$$

当 $\epsilon_2/\epsilon_1 = n_2^2/n_1^2 = (5)^2 = 25$ 时，平行偏振的反射系数可从等式得出： (8.68) 作为

$$\begin{aligned} \Gamma_{\parallel} &= \frac{-(\epsilon_2/\epsilon_1) \cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{(\epsilon_2/\epsilon_1) \cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}} \\ &= \frac{-25 \cos 60^\circ + \sqrt{25 - \sin^2 60^\circ}}{25 \cos 60^\circ + \sqrt{25 - \sin^2 60^\circ}} = -0.435. \end{aligned}$$

因此，反射功率和传输功率为

$$\begin{aligned} P_{\parallel}^r &= P_{\parallel}^i |\Gamma_{\parallel}|^2 = 5(0.435)^2 = 0.95 \text{ W}, \\ P_{\parallel}^t &= P_{\parallel}^i - P_{\parallel}^r = 5 - 0.95 = 4.05 \text{ W}. \end{aligned}$$

8-6 波导

在第 2 章前面，我们考虑了两类传输线，即支持横向电磁 (TEM) 模式的传输线和不支持横向电磁模式的传输线。属于 TEM 系列的传输线 (图 2-4)，包括同轴、两线和平行板线，支持与传播方向正交的 **E** 和 **H** 场。另一组中的线 (通常称为高阶传输线) 支持的场可能具有与传播方向 $\hat{\mathbf{k}}$ 正交的 **E** 或 **H**，但不能同时两者。因此，**E** 或 **H** 的至少一个分量沿着 $\hat{\mathbf{k}}$ 。

► 如果 **E** 横向于 $\hat{\mathbf{k}}$ 但 **H** 不横向，我们将其称为横向电 (TE) 模式，如果 **H** 横向于 $\hat{\mathbf{k}}$ 但 **E** 不横向，我们将其称为横向磁 (TM) 模式。 ◀

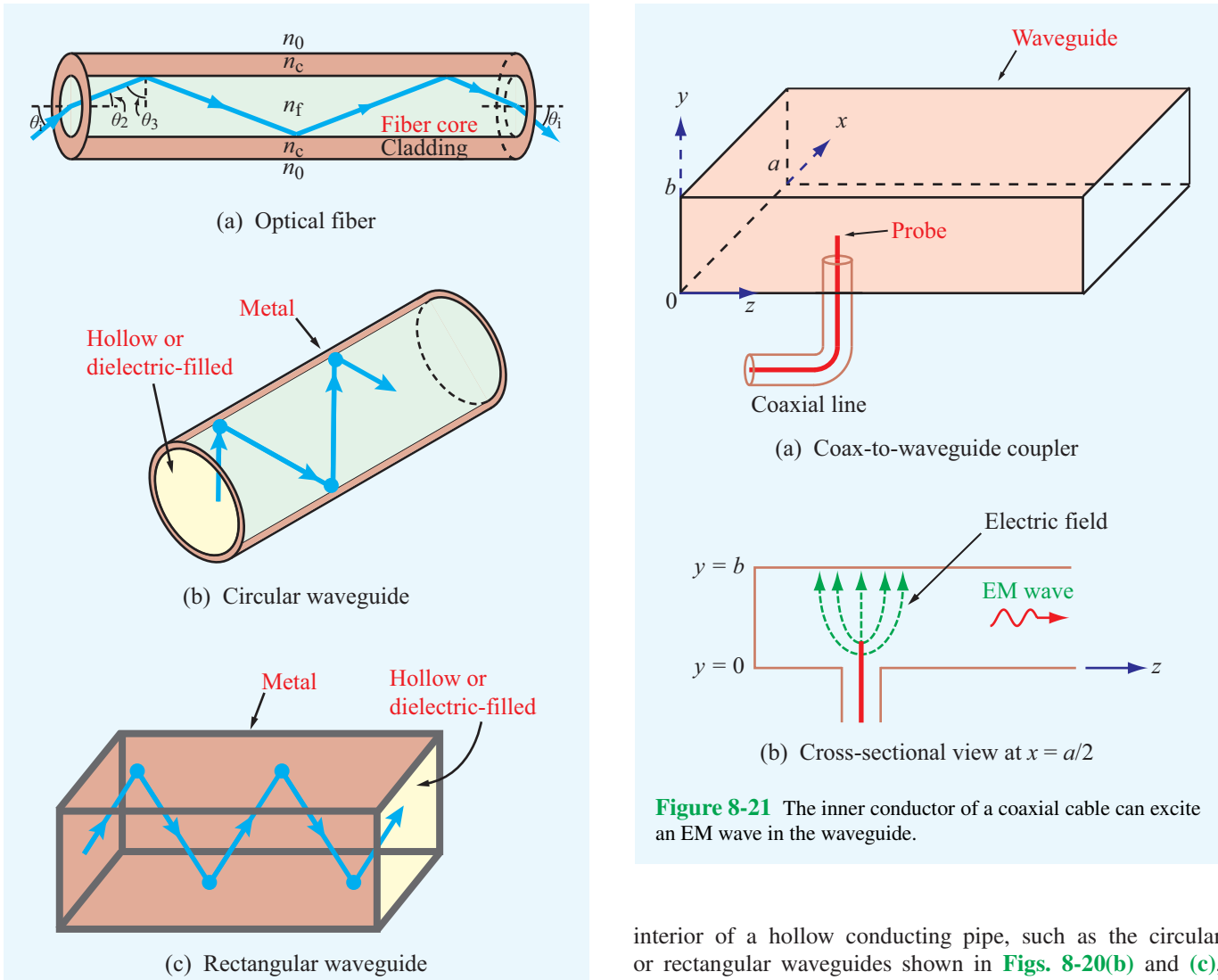


Figure 8-20 Wave travel by successive reflections in (a) an optical fiber, (b) a circular metal waveguide, and (c) a rectangular metal waveguide.

Among all higher-order transmission lines, the two most commonly used are optical fiber and the metal waveguide. As noted in Section 8-3, a wave is guided along an optical fiber through successive zigzags by taking advantage of the total internal reflection at the boundary between the (inner) core and the (outer) cladding (Fig. 8-20(a)). Another way to achieve internal reflection at the core's boundary is to have its surface coated by a conducting material. Under the proper conditions (we shall elaborate later) a wave excited in the

interior of a hollow conducting pipe, such as the circular or rectangular waveguides shown in Figs. 8-20(b) and (c), undergoes a process similar to that of successive internal reflection in an optical fiber, resulting in propagation down the pipe. Most waveguide applications call for air-filled guides, but in some cases, the waveguide may be filled with a dielectric material used to alter its propagation velocity or impedance, or it may be vacuum-pumped to eliminate air molecules in order to prevent voltage breakdown, thereby increasing its power-handling capabilities.

Figure 8-21 illustrates how a coaxial cable can be connected to a rectangular waveguide. With its outer conductor connected to the metallic waveguide enclosure, the coaxial cable's inner conductor protrudes through a tiny hole into the waveguide's interior (without touching the conducting surface). Time-varying electric field lines extending between the protruding inner conductor and the inside surface of the guide provide the excitation necessary to transfer a signal from the coaxial

Figure 8-21 The inner conductor of a coaxial cable can excite an EM wave in the waveguide.

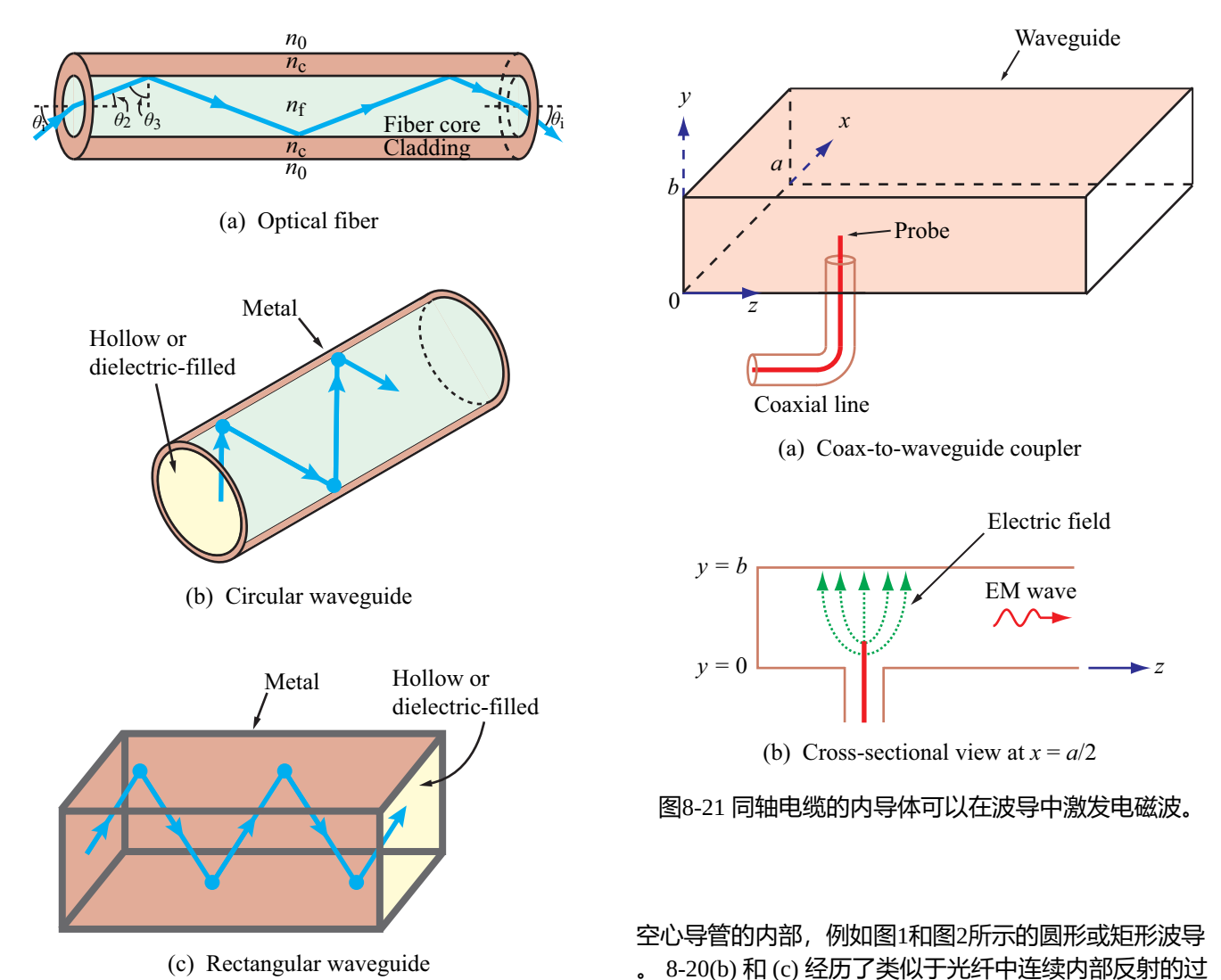


图8-21 同轴电缆的内导体可以在波导中激发电磁波。

空心导管的内部，例如如图1和图2所示的圆形或矩形波导。8-20(b)和(c)经历了类似于光纤中连续内部反射的过程，导致沿着管道传播。大多数波导应用需要充气波导，但在某些情况下，波导可能填充有介电材料，用于改变其传播速度或阻抗，或者可以对其进行真空泵浦以消除空气分子，以防止电压击穿，从而提高其功率处理能力。

图8-21说明了如何将同轴电缆连接到矩形波导。同轴电缆的外导体连接到金属波导外壳，内导体通过一个小孔伸入波导内部（不接触导电表面）。在突出的内导体和导管内表面之间延伸的时变电场线提供了从同轴电缆传输信号所需的激励。

图 8-20 波在 (a) 光纤、(b) 圆形金属波导和 (c) 矩形金属波导中通过连续反射传播。

在所有高阶传输线中，最常用的两种是光纤和金属波导。如第 8-3 节所述，利用（内）纤芯和（外）包层之间边界处的全内反射，波被引导通过连续的锯齿形光纤（图 8-20(a)）。在核心边界实现内部反射的另一种方法是在其表面涂上导电材料。在适当的条件下（我们稍后将详细说明），在

line to the guide. Conversely, the center conductor can act like a probe, coupling a signal from the waveguide to the coaxial cable.

For guided transmission at frequencies below 30 GHz, the coaxial cable is by far the most widely used transmission line. At higher frequencies, however, the coaxial cable has a number of limitations: (a) in order for it to propagate only TEM modes, the cable’s inner and outer conductors have to be reduced in size to satisfy a certain size-to-wavelength requirement, making it more difficult to fabricate, (b) the smaller cross section reduces the cable’s power-handling capacity (limited by dielectric breakdown), and (c) the attenuation due to dielectric losses increases with frequency. For all of these reasons, metal waveguides have been used as an alternative to coaxial lines for many radar and communication applications that operate at frequencies in the 5–100 GHz range, particularly those requiring the transmission of high levels of radio-frequency (RF) power. Even though waveguides with circular and elliptical cross sections have been used in some microwave systems, the rectangular shape has been the more prevalent geometry.

8-7 General Relations for **E** and **H**

The purpose of the next two sections is to derive expressions for **E** and **H** for the TE and TM modes in a rectangular waveguide and to examine their wave properties. We choose the coordinate system shown in Fig. 8-22, where the propagation occurs along **z**. For TE modes, the electric field is transverse to the direction of propagation. Hence, **E** may have components along **x̂** and **ŷ** but not along **ẑ**. In contrast, **H** has a **ẑ**-directed component and may have components along either **x̂** or **ŷ** or along both. The converse is true for TM modes.

Our solution procedure consists of four steps:

- (1) Maxwell’s equations are manipulated to develop general expressions for the phasor-domain transverse field components $\tilde{E}_x$, $\tilde{E}_y$, $\tilde{H}_x$, and $\tilde{H}_y$ in terms of $\tilde{E}_z$ and $\tilde{H}_z$. When

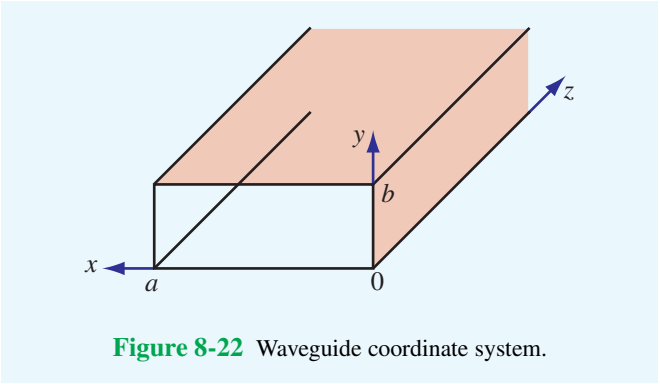


Figure 8-22 Waveguide coordinate system.

specialized to the TE case, these expressions become functions of $\tilde{H}_z$ only, and the converse is true for the TM case.

- (2) The homogeneous wave equations given by Eqs. (7.15) and (7.16) are solved to obtain valid solutions for $\tilde{E}_z$ (TM case) and $\tilde{H}_z$ (TE case) in a waveguide.
- (3) The expressions derived in step 1 are then used to find $\tilde{E}_x$, $\tilde{E}_y$, $\tilde{H}_x$, and $\tilde{H}_y$.
- (4) The solutions obtained in step 3 are analyzed to determine the phase velocity and other properties of the TE and TM waves.

The intent of the present section is to realize the stated goals of step 1. We begin with a general form for the **E** and **H** fields in the phasor domain:

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}}\tilde{E}_x + \hat{\mathbf{y}}\tilde{E}_y + \hat{\mathbf{z}}\tilde{E}_z, \tag{8.84a}$$

$$\tilde{\mathbf{H}} = \hat{\mathbf{x}}\tilde{H}_x + \hat{\mathbf{y}}\tilde{H}_y + \hat{\mathbf{z}}\tilde{H}_z. \tag{8.84b}$$

In general, all six components of $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ may depend on (x,y,z) , and while we do not yet know how they functionally depend on (x,y) , our prior experience suggests that $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ of a wave traveling along the $+z$ direction should exhibit a dependence on z of the form $e^{-j\beta z}$, where β is a yet-to-be determined *phase constant*. Hence, we adopt the form

$$\tilde{E}_x(x,y,z) = \tilde{e}_x(x,y) e^{-j\beta z}, \tag{8.85}$$

where $\tilde{e}_x(x,y)$ describes the dependence of $\tilde{E}_x(x,y,z)$ on (x,y) only. The form of Eq. (8.85) can be used for all other components of $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ as well. Thus,

$$\tilde{\mathbf{E}} = (\hat{\mathbf{x}}\tilde{e}_x + \hat{\mathbf{y}}\tilde{e}_y + \hat{\mathbf{z}}\tilde{e}_z)e^{-j\beta z}, \tag{8.86a}$$

$$\tilde{\mathbf{H}} = (\hat{\mathbf{x}}\tilde{h}_x + \hat{\mathbf{y}}\tilde{h}_y + \hat{\mathbf{z}}\tilde{h}_z)e^{-j\beta z}. \tag{8.86b}$$

The notation is intended to clarify that, in contrast to $\tilde{E}$ and $\tilde{H}$, which vary with (x,y,z) , the lower case $\tilde{e}$ and $\tilde{h}$ vary with (x,y) only.

In a lossless, source-free medium (such as the inside of a waveguide) characterized by permittivity ϵ and permeability μ (and conductivity $\sigma = 0$), Maxwell’s curl equations are given by Eqs. (7.2b and d) with $\mathbf{J} = 0$,

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}, \tag{8.87a}$$

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon\tilde{\mathbf{E}}. \tag{8.87b}$$

Upon inserting Eqs. (8.86a and b) into Eqs. (8.87a and b) and recalling that each of the curl equations actually consists of

线到指南。相反，中心导体可以像探针一样，将信号从波导耦合到同轴电缆。

对于 30 GHz 以下频率的引导传输，同轴电缆是迄今为止使用最广泛的传输线。然而，在较高频率下，同轴电缆有许多限制：(a) 为了使其仅传播 TEM 模式，必须减小电缆的内导体和外导体的尺寸以满足一定的尺寸与波长要求，从而使其更难以制造，(b) 较小的横截面会降低电缆的功率处理能力（受到介电击穿的限制），以及 (c) 由于介电损耗导致的衰减随着频率的增加而增加。由于所有这些原因，金属波导已被用作许多雷达和通信应用中同轴线的替代品，这些雷达和通信应用的工作频率在 5-100 GHz 范围内，特别是那些需要传输高水平射频 (RF) 功率的应用。尽管具有圆形和椭圆形横截面的波导已经在一些微波系统中使用，但矩形形状是更普遍的几何形状。

8-7 E和H的一般关系

接下来两节的目的是推导矩形波导中 TE 和 TM 模式的 **E** 和 **H** 表达式，并检查它们的波特特性。我们选择图 8-22 所示的坐标系，其中传播沿 **z** 发生。对于 TE 模式，电场垂直于传播方向。因此，**E** 可能具有沿 **x̂** 和 **ŷ** 的分量，但不沿 **ẑ** 的分量。相反，**H** 具有 **ẑ** 方向的分量，并且可以具有沿 **x̂** 或 **ŷ** 或沿两者的分量。对于 TM 模式则相反。

我们的解决方案过程包括四个步骤：

- (1) 操纵麦克斯韦方程，以 $\tilde{E}_z$ 和 $\tilde{H}_z$ 的形式，建立相量域横向场分量 $\tilde{E}_x$ 、 $\tilde{E}_y$ 、 $\tilde{H}_x$ 和 $\tilde{H}_y$ 的一般表达式。什么时候

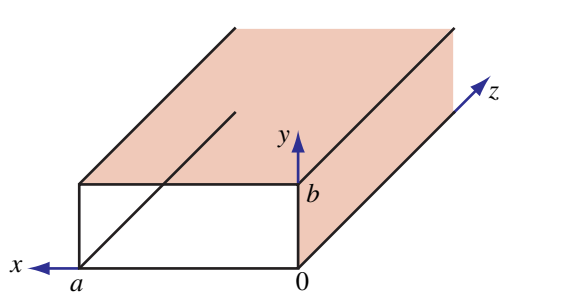


图 8-22 波导坐标系。

专门针对 TE 情况，这些表达式仅成为 $\tilde{H}_z$ 的函数，而对于 TM 情况则相反。

- (2) 齐次波动方程由方程给出。求解 (7.15) 和 (7.16) 以获得波导中 $\tilde{E}_z$ (TM情况)和 $\tilde{H}_z$ (TE情况)的有效解。

- (3) 然后使用步骤 1 中导出的表达式来查找 $\tilde{E}_x$ 、 $\tilde{E}_y$ 、 $\tilde{H}_x$ 和 $\tilde{H}_y$ 。

- (4) 对步骤3中得到的解进行分析，以确定TE波和TM波的相速度和其他特性。

本节的目的是实现步骤 1 的既定目标。我们从相量域中 **E** 和 **H** 场的一般形式开始：

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}}\tilde{E}_x + \hat{\mathbf{y}}\tilde{E}_y + \hat{\mathbf{z}}\tilde{E}_z, \tag{8.84a}$$

$$\tilde{\mathbf{H}} = \hat{\mathbf{x}}\tilde{H}_x + \hat{\mathbf{y}}\tilde{H}_y + \hat{\mathbf{z}}\tilde{H}_z. \tag{8.84b}$$

一般来说， $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 的所有六个分量都可能依赖于 (x,y,z) ，虽然我们还不知道它们在功能上如何依赖于 (x,y) ，但我们之前的经验表明，沿着 $+z$ 方向传播的波的 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 应该表现出对 z 形式 $e^{-j\beta z}$ 的依赖性，其中 β 是尚未确定的相位常数。因此，我们采用以下形式

$$\tilde{E}_x(x,y,z) = \tilde{e}_x(x,y) e^{-j\beta z}, \tag{8.85}$$

其中 $\tilde{e}_x(x,y)$ 描述了 $\tilde{E}_x(x,y,z)$ 仅对 (x,y) 的依赖性。方程的形式(8.85) 也可用于 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 的所有其他分量。因此，

$$\tilde{\mathbf{E}} = (\hat{\mathbf{x}}\tilde{e}_x + \hat{\mathbf{y}}\tilde{e}_y + \hat{\mathbf{z}}\tilde{e}_z)e^{-j\beta z}, \tag{8.86a}$$

$$\tilde{\mathbf{H}} = (\hat{\mathbf{x}}\tilde{h}_x + \hat{\mathbf{y}}\tilde{h}_y + \hat{\mathbf{z}}\tilde{h}_z)e^{-j\beta z}. \tag{8.86b}$$

该符号旨在澄清，与随 (x,y,z) 变化的 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 不同，小写的 $\tilde{e}$ 和 $\tilde{h}$ 仅随 (x,y) 变化。

在以介电常数 ϵ 和磁导率 μ (以及电导率 $\sigma = 0$) 为特征的无损、无源介质 (例如波导内部) 中，麦克斯韦旋度方程由等式 1 给出。(7.2b 和 d) 其中 $\mathbf{J} = 0$,

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}, \tag{8.87a}$$

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon\tilde{\mathbf{E}}. \tag{8.87b}$$

插入方程式后。(8.86a 和 b) 代入等式。(8.87a 和 b) 并回顾每个旋度方程实际上由

three separate equations—one for each of the unit vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$, we obtain the following relationships:

$$\frac{\partial \tilde{e}_z}{\partial y} + j\beta \tilde{e}_y = -j\omega\mu \tilde{h}_x, \quad (8.88a)$$

$$-j\beta \tilde{e}_x - \frac{\partial \tilde{e}_z}{\partial x} = -j\omega\mu \tilde{h}_y, \quad (8.88b)$$

$$\frac{\partial \tilde{e}_y}{\partial x} - \frac{\partial \tilde{e}_x}{\partial y} = -j\omega\mu \tilde{h}_z, \quad (8.88c)$$

$$\frac{\partial \tilde{h}_z}{\partial y} + j\beta \tilde{h}_y = j\omega\varepsilon \tilde{e}_x, \quad (8.88d)$$

$$-j\beta \tilde{h}_x - \frac{\partial \tilde{h}_z}{\partial x} = j\omega\varepsilon \tilde{e}_y, \quad (8.88e)$$

$$\frac{\partial \tilde{h}_y}{\partial x} - \frac{\partial \tilde{h}_x}{\partial y} = j\omega\varepsilon \tilde{e}_z. \quad (8.88f)$$

Equations (8.88a-f) incorporate the fact that differentiation with respect to z is equivalent to multiplication by $-j\beta$. By manipulating these equations algebraically, we can obtain expressions for the x and y components of $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ in terms of their z components, namely

$$\tilde{E}_x = \frac{-j}{k_c^2} \left(\beta \frac{\partial \tilde{E}_z}{\partial x} + \omega\mu \frac{\partial \tilde{H}_z}{\partial y} \right), \quad (8.89a)$$

$$\tilde{E}_y = \frac{j}{k_c^2} \left(-\beta \frac{\partial \tilde{E}_z}{\partial y} + \omega\mu \frac{\partial \tilde{H}_z}{\partial x} \right), \quad (8.89b)$$

$$\tilde{H}_x = \frac{j}{k_c^2} \left(\omega\varepsilon \frac{\partial \tilde{E}_z}{\partial y} - \beta \frac{\partial \tilde{H}_z}{\partial x} \right), \quad (8.89c)$$

$$\tilde{H}_y = \frac{-j}{k_c^2} \left(\omega\varepsilon \frac{\partial \tilde{E}_z}{\partial x} + \beta \frac{\partial \tilde{H}_z}{\partial y} \right). \quad (8.89d)$$

Here

$$k_c^2 = k^2 - \beta^2 = \omega^2\mu\varepsilon - \beta^2, \quad (8.90)$$

where k is the **unbounded-medium wavenumber** earlier defined as

$$k = \omega\sqrt{\mu\varepsilon}. \quad (8.91)$$

For reasons that become clear later (in Section 8-8), the constant k_c is called the **cutoff wavenumber**. In view of Eqs. (8.89a-d), the x and y components of $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ now can be found readily—so long as we have mathematical expressions for $\tilde{E}_z$ and $\tilde{H}_z$. For the TE mode, $\tilde{E}_z = 0$, so all we need to know is $\tilde{H}_z$, and the converse is true for the TM case.

8-8 TM Modes in Rectangular Waveguide

In the preceding section, we developed expressions for $\tilde{E}_x$, $\tilde{E}_y$, $\tilde{H}_x$, and $\tilde{H}_y$ in terms of $\tilde{E}_z$ and $\tilde{H}_z$. Since **$\tilde{H}_z = 0$ for the TM mode**, our task reduces to obtaining a valid solution for $\tilde{E}_z$. Our starting point is the **homogeneous wave equation** for $\tilde{\mathbf{E}}$. For a lossless medium characterized by an unbounded-medium wavenumber k , the wave equation is given by Eq. (7.19) as

$$\nabla^2 \tilde{\mathbf{E}} + k^2 \tilde{\mathbf{E}} = 0. \quad (8.92)$$

To satisfy Eq. (8.92), each of its $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ components has to be satisfied independently. Its $\hat{\mathbf{z}}$ component is given by

$$\frac{\partial^2 \tilde{E}_z}{\partial x^2} + \frac{\partial^2 \tilde{E}_z}{\partial y^2} + \frac{\partial^2 \tilde{E}_z}{\partial z^2} + k^2 \tilde{E}_z = 0. \quad (8.93)$$

By adopting the mathematical form given by Eq. (8.85), namely,

$$\tilde{E}_z(x, y, z) = \tilde{e}_z(x, y) e^{-j\beta z}, \quad (8.94)$$

Eq. (8.93) reduces to

$$\frac{\partial^2 \tilde{e}_z}{\partial x^2} + \frac{\partial^2 \tilde{e}_z}{\partial y^2} + k_c^2 \tilde{e}_z = 0, \quad (8.95)$$

where k_c^2 is as defined by Eq. (8.90).

The form of the partial differential equation (separate, uncoupled derivatives with respect to x and y) allows us to assume a product solution of the form

$$\tilde{e}_z(x, y) = X(x) Y(y). \quad (8.96)$$

Substituting Eq. (8.96) into Eq. (8.95) and then dividing all terms by $X(x) Y(y)$ leads to:

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + k_c^2 = 0. \quad (8.97)$$

To satisfy Eq. (8.97), each of the first two terms has to equal a constant. Hence, we define separation constants k_x and k_y such that

$$\frac{d^2 X}{dx^2} + k_x^2 X = 0, \quad (8.98a)$$

$$\frac{d^2 Y}{dy^2} + k_y^2 Y = 0, \quad (8.98b)$$

and

$$k_c^2 = k_x^2 + k_y^2. \quad (8.99)$$

三个独立的方程 - 每个单位向量 $\hat{\mathbf{x}}$ 、 $\hat{\mathbf{y}}$ 和 $\hat{\mathbf{z}}$, 我们得到以下关系:

$$\frac{\partial \tilde{e}_z}{\partial y} + j\beta \tilde{e}_y = -j\omega\mu \tilde{h}_x, \quad (8.88a)$$

$$-j\beta \tilde{e}_x - \frac{\partial \tilde{e}_z}{\partial x} = -j\omega\mu \tilde{h}_y, \quad (8.88b)$$

$$\frac{\partial \tilde{e}_y}{\partial x} - \frac{\partial \tilde{e}_x}{\partial y} = -j\omega\mu \tilde{h}_z, \quad (8.88c)$$

$$\frac{\partial \tilde{h}_z}{\partial y} + j\beta \tilde{h}_y = j\omega\varepsilon \tilde{e}_x, \quad (8.88d)$$

$$-j\beta \tilde{h}_x - \frac{\partial \tilde{h}_z}{\partial x} = j\omega\varepsilon \tilde{e}_y, \quad (8.88e)$$

$$\frac{\partial \tilde{h}_y}{\partial x} - \frac{\partial \tilde{h}_x}{\partial y} = j\omega\varepsilon \tilde{e}_z. \quad (8.88f)$$

方程 (8.88a-f) 包含了这样一个事实: 对 z 的微分相当于乘以 $-j\beta$ 。通过对这些方程进行代数运算, 我们可以得到 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 的 x 和 y 分量以其 z 分量的表达式, 即

$$\tilde{E}_x = \frac{-j}{k_c^2} \left(\beta \frac{\partial \tilde{E}_z}{\partial x} + \omega\mu \frac{\partial \tilde{H}_z}{\partial y} \right), \quad (8.89a)$$

$$\tilde{E}_y = \frac{j}{k_c^2} \left(-\beta \frac{\partial \tilde{E}_z}{\partial y} + \omega\mu \frac{\partial \tilde{H}_z}{\partial x} \right), \quad (8.89b)$$

$$\tilde{H}_x = \frac{j}{k_c^2} \left(\omega\varepsilon \frac{\partial \tilde{E}_z}{\partial y} - \beta \frac{\partial \tilde{H}_z}{\partial x} \right), \quad (8.89c)$$

$$\tilde{H}_y = \frac{-j}{k_c^2} \left(\omega\varepsilon \frac{\partial \tilde{E}_z}{\partial x} + \beta \frac{\partial \tilde{H}_z}{\partial y} \right). \quad (8.89d)$$

这里

$$k_c^2 = k^2 - \beta^2 = \omega^2\mu\varepsilon - \beta^2, \quad (8.90)$$

其中 k 是无界介质波数, 之前定义为

$$k = \omega\sqrt{\mu\varepsilon}. \quad (8.91)$$

由于稍后会变得清楚的原因 (在第 8-8 节中), 常数 k_c 称为截止波数。鉴于方程式. (8.89a-d), 现在可以很容易地找到 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 的 x 和 y 分量——只要我们有 $\tilde{E}_z$ 和 $\tilde{H}_z$ 的数学表达式。对于TE模式, $\tilde{E}_z = 0$, 所以我们只需要知道 $\tilde{H}_z$, 对于TM情况则相反。

矩形波导中的 8-8 TM 模式

在上一节中, 我们根据 $\tilde{E}_z$ 和 $\tilde{H}_z$ 开发了 $\tilde{E}_x$ 、 $\tilde{E}_y$ 、 $\tilde{H}_x$ 和 $\tilde{H}_y$ 的表达式。由于 TM 模式的 $\tilde{H}_z = 0$, 我们的任务简化为获取 $\tilde{E}_z$ 的有效解。我们的起点是 $\tilde{\mathbf{E}}$ 的齐次波动方程。对于以无界介质波数 k 为特征的无损介质, 波动方程由式 (1) 给出。(7.19) 作为

$$\nabla^2 \tilde{\mathbf{E}} + k^2 \tilde{\mathbf{E}} = 0. \quad (8.92)$$

满足方程. (8.92) 中, 其每个 $\hat{\mathbf{x}}$ 、 $\hat{\mathbf{y}}$ 和 $\hat{\mathbf{z}}$ 分量都必须独立满足。它的 $\hat{\mathbf{z}}$ 分量由下式给出

$$\frac{\partial^2 \tilde{E}_z}{\partial x^2} + \frac{\partial^2 \tilde{E}_z}{\partial y^2} + \frac{\partial^2 \tilde{E}_z}{\partial z^2} + k^2 \tilde{E}_z = 0. \quad (8.93)$$

采用方程给出的数学形式。(8.85), 即

$$\tilde{E}_z(x, y, z) = \tilde{e}_z(x, y) e^{-j\beta z}, \quad (8.94)$$

等式. (8.93) 减少为

$$\frac{\partial^2 \tilde{e}_z}{\partial x^2} + \frac{\partial^2 \tilde{e}_z}{\partial y^2} + k_c^2 \tilde{e}_z = 0, \quad (8.95)$$

其中 k_c^2 由方程式定义。(8.90)。

偏微分方程的形式 (关于 x 和 y 的独立、非耦合导数) 允许我们假设以下形式的乘积解

$$\tilde{e}_z(x, y) = X(x) Y(y). \quad (8.96)$$

代入方程式(8.96) 代入方程. (8.95) 然后将所有项除以 $X(x) Y(y)$ 得出:

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + k_c^2 = 0. \quad (8.97)$$

满足方程. (8.97), 前两项必须等于一个常数。因此, 我们定义分离常数 k_x 和 k_y 使得

$$\frac{d^2 X}{dx^2} + k_x^2 X = 0, \quad (8.98a)$$

$$\frac{d^2 Y}{dy^2} + k_y^2 Y = 0, \quad (8.98b)$$

和

$$k_c^2 = k_x^2 + k_y^2. \quad (8.99)$$

Before proposing solutions for Eqs. (8.98a and b), we should consider the constraints that the solutions must meet. The electric field $\tilde{E}_z$ is parallel to all four walls of the waveguide. Since $\mathbf{E} = 0$ in the conducting walls, *the boundary conditions require $\tilde{E}_z$ in the waveguide cavity to go to zero as x approaches 0 and a and as y approaches 0 and b (Fig. 8-22).* To satisfy these boundary conditions, sinusoidal solutions are chosen for $X(x)$ and $Y(y)$ as:

$$\tilde{e}_z = X(x)Y(y) = (A \cos k_x x + B \sin k_x x)(C \cos k_y y + D \sin k_y y). \quad (8.100)$$

These forms for $X(x)$ and $Y(y)$ definitely satisfy the differential equations given by Eqs. (8.98a and b). The boundary conditions for $\tilde{e}_z$ are

$$\tilde{e}_z = 0 \quad \text{at } x = 0 \text{ and } a, \quad (8.101a)$$

$$\tilde{e}_z = 0 \quad \text{at } y = 0 \text{ and } b. \quad (8.101b)$$

Satisfying $\tilde{e}_z = 0$ at $x = 0$ requires that we set $A = 0$, and similarly, satisfying $\tilde{e}_z = 0$ at $y = 0$ requires $C = 0$. Satisfying $\tilde{e}_z = 0$ at $x = a$ requires

$$k_x = \frac{m\pi}{a}, \quad m = 1, 2, 3, \dots \quad (8.102a)$$

and similarly, satisfying $\tilde{e}_z = 0$ at $y = b$ requires

$$k_y = \frac{n\pi}{b}, \quad n = 1, 2, 3, \dots \quad (8.102b)$$

Consequently,

$$\tilde{E}_z = \tilde{e}_z e^{-j\beta z} = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.103)$$

where $E_0 = BD$ is the amplitude of the wave in the guide. Keeping in mind that $\tilde{H}_z = 0$ for the TM mode, the transverse components of $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ now can be obtained by applying Eq. (8.103) to Eqs. (8.89a-d),

$$\tilde{E}_x = \frac{-j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.104a)$$

$$\tilde{E}_y = \frac{-j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.104b)$$

$$\tilde{H}_x = \frac{j\omega\epsilon}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.104c)$$

$$\tilde{H}_y = \frac{-j\omega\epsilon}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}. \quad (8.104d)$$

Each combination of the integers m and n represents a viable solution—or a mode—denoted TM_{mn} . Associated with each mn mode are specific field distributions for the region inside the guide. **Figure 8-23** depicts the $\mathbf{E}$ and $\mathbf{H}$ field lines for the TM_{11} mode across two different cross sections of the guide.

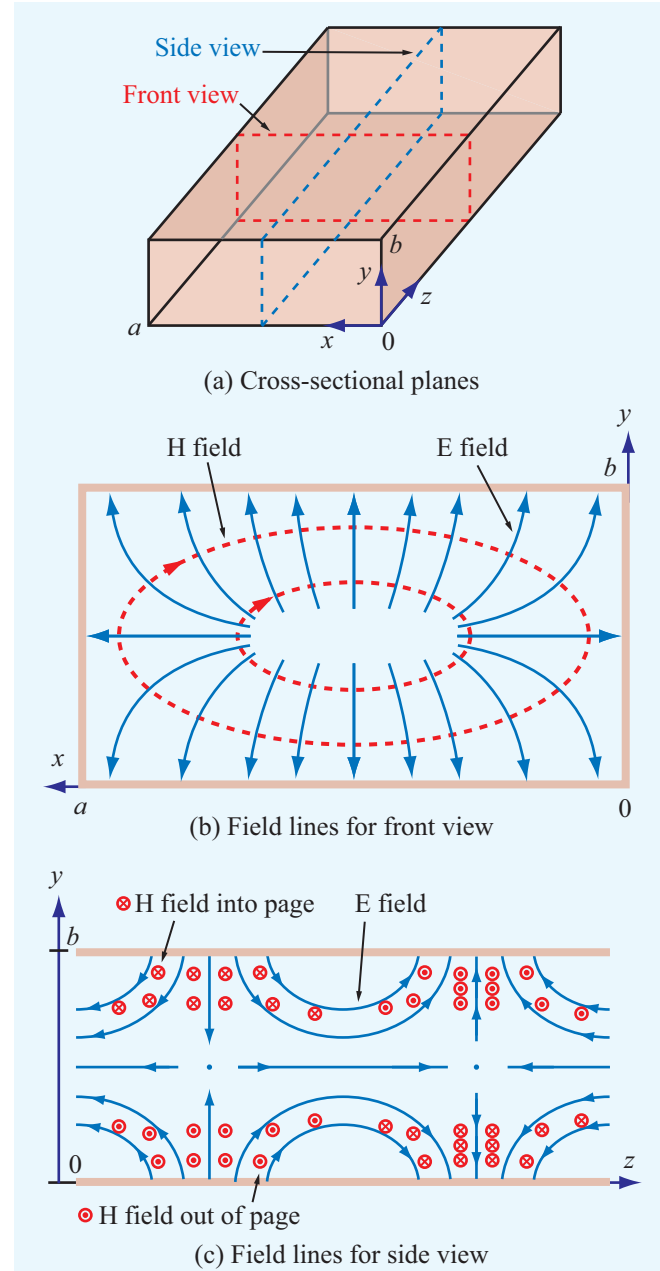


Figure 8-23 TM_{11} electric and magnetic field lines across two cross-sectional planes.

According to Eqs. (8.103) and (8.104e), a rectangular waveguide with cross section $(a \times b)$ can support the propagation of waves with many different, but discrete, field configurations specified by the integers m and n . The only quantity in the

在提出方程式的解决方案之前。(8.98a 和 b)，我们应该考虑解必须满足的约束。电场 $\tilde{E}_z$ 平行于波导的所有四个壁。由于导电壁中的 $\mathbf{E} = 0$ ，边界条件要求当 x 接近 0 和 a 以及 y 接近 0 和 b 时，波导腔中的 $\tilde{E}_z$ 为零 (图 8-22)。为了满足这些边界条件，为 $X(x)$ 和 $Y(y)$ 选择正弦解，如下所示：

$$\tilde{e}_z = X(x)Y(y) = (A \cos k_x x + B \sin k_x x)(C \cos k_y y + D \sin k_y y). \quad (8.100)$$

$X(x)$ 和 $Y(y)$ 的这些形式绝对满足方程 1 给出的微分方程。(8.98a 和 b)。 $\tilde{e}_z$ 的边界条件是

$$\tilde{e}_z = 0 \quad \text{at } x = 0 \text{ and } a, \quad (8.101a)$$

$$\tilde{e}_z = 0 \quad \text{at } y = 0 \text{ and } b. \quad (8.101b)$$

在 $x = 0$ 处满足 $\tilde{e}_z = 0$ 需要我们设置 $A = 0$ ，同样，在 $y = 0$ 处满足 $\tilde{e}_z = 0$ 需要 $C = 0$ 。在 $x = a$ 处满足 $\tilde{e}_z = 0$ 需要

$$k_x = \frac{m\pi}{a}, \quad m = 1, 2, 3, \dots \quad (8.102a)$$

同样，在 $y = b$ 处满足 $\tilde{e}_z = 0$ 需要

$$k_y = \frac{n\pi}{b}, \quad n = 1, 2, 3, \dots \quad (8.102b)$$

最后，

$$\tilde{E}_z = \tilde{e}_z e^{-j\beta z} = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.103)$$

其中 $E_0 = BD$ 是波导中波的振幅。请记住，对于 TM 模式， $\tilde{H}_z = 0$ ， $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 的横向分量现在可以通过应用等式获得。(8.103) 至等式。(8.89a-d)，

$$\tilde{E}_x = \frac{-j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.104a)$$

$$\tilde{E}_y = \frac{-j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.104b)$$

$$\tilde{H}_x = \frac{j\omega\epsilon}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.104c)$$

$$\tilde{H}_y = \frac{-j\omega\epsilon}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}. \quad (8.104d)$$

整数 m 和 n 的每个组合都代表一个可行的解决方案或模式，表示为 TM_{mn} 。与每个 mn 模式相关的是指南内区域的特定场分布。图 8-23 描绘了 TM_{11} 模式在波导两个不同横截面上的 $\mathbf{E}$ 和 $\mathbf{H}$ 场线。

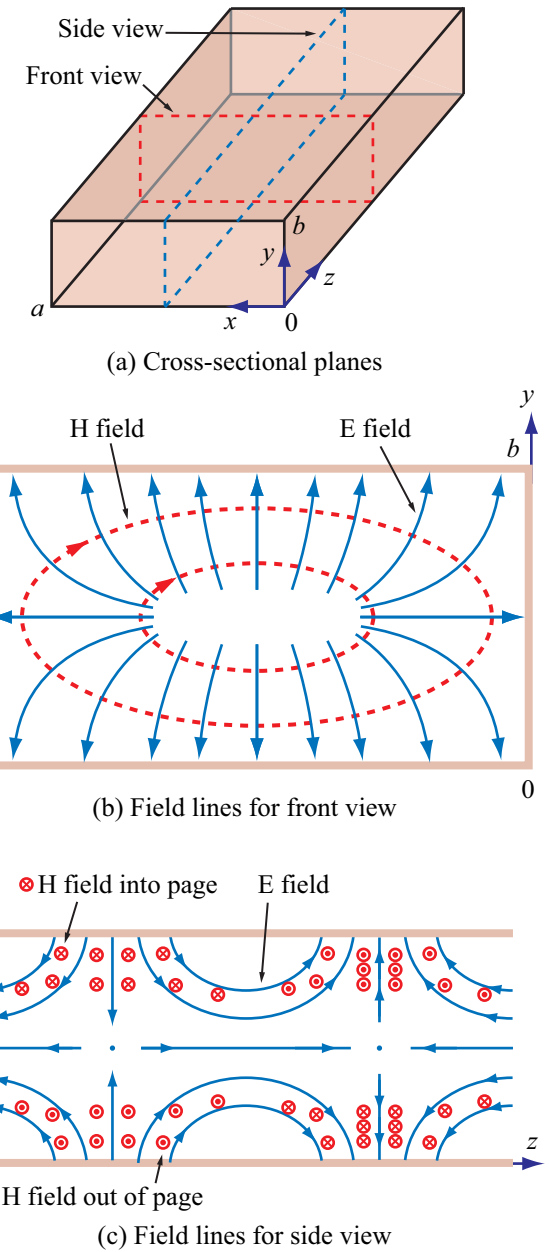


图 8-23 穿过两个横截面的 TM_{11} 电场线和磁场线。

根据方程式。根据 (8.103) 和 (8.104e)，具有横截面 $(a \times b)$ 的矩形波导可以支持具有由整数 m 和 n 指定的许多不同但离散的场配置的波的传播。中唯一的数量

fields’ expressions that we have yet to determine is the propagation constant β , which is contained in the exponential $e^{-j\beta z}$. By combining Eqs. (8.90), (8.99), and (8.102), we obtain the following expression for β :

$$\beta = \sqrt{k^2 - k_c^2} = \sqrt{\omega^2 \mu \epsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}. \quad (8.105)$$

(TE and TM)

Even though the expression for β was derived for TM modes, it is equally applicable to TE modes.

The exponential $e^{-j\beta z}$ describes a wave traveling in the $+z$ direction—provided that β is real—which corresponds to $k > k_c$. If $k < k_c$, β becomes imaginary: $\beta = -j\alpha$ with α real, in which case $e^{-j\beta z} = e^{-\alpha z}$, yielding **evanescent waves** characterized by amplitudes that decay rapidly with z due to the attenuation function $e^{-\alpha z}$. Corresponding to each mode (m, n) , there is a **cutoff frequency** f_{mn} at which $\beta = 0$. By setting $\beta = 0$ in Eq. (8.105) and then solving for f , we have

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}, \quad (8.106)$$

(TE and TM)

where $u_{p0} = 1/\sqrt{\mu\epsilon}$ is the phase velocity of a TEM wave in an unbounded medium with constitutive parameters ϵ and μ .

► A wave in a given mode can propagate through the guide only if its frequency $f > f_{mn}$, as only then $\beta = \text{real}$. ◀

The mode with the lowest cutoff frequency is known as the **dominant mode**. *The dominant mode is TM_{11} among TM modes and TE_{10} among TE modes. Whereas a value of zero for m or n is allowed for TE modes, it is not for TM modes* [because if either m or n is zero, $\tilde{E}_z$ in Eq. (8.103) becomes zero and all other field components vanish as well].

By combining Eqs. (8.105) and (8.106), we can express β in terms of f_{mn} ,

$$\beta = \frac{\omega}{u_{p0}} \sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}. \quad \textbf{(TE and TM)} \quad (8.107)$$

The phase velocity of a TE or TM wave in a waveguide is

$$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}}. \quad \textbf{(TE and TM)} \quad (8.108)$$

The transverse electric field consists of components $\tilde{E}_x$ and $\tilde{E}_y$ given by Eqs. (8.104a and b). For a wave traveling in the $+z$ direction, the magnetic field associated with $\tilde{E}_x$ is $\tilde{H}_y$ [according to the right hand rule given by Eq. (7.39a)]. Similarly, the magnetic field associated with $\tilde{E}_y$ is $-\tilde{H}_x$. The ratios, obtained by employing Eq. (8.104e), constitute the wave impedance in the guide,

$$Z_{\text{TM}} = \frac{\tilde{E}_x}{\tilde{H}_y} = -\frac{\tilde{E}_y}{\tilde{H}_x} = \frac{\beta \eta}{k} = \eta \sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}, \quad (8.109)$$

where $\eta = \sqrt{\mu/\epsilon}$ is the intrinsic impedance of the dielectric material filling the guide.

Example 8-8: Mode Properties

A TM wave propagating in a dielectric-filled waveguide of unknown permittivity has a magnetic field with the y component given by

$$H_y = 6 \cos(25\pi x) \sin(100\pi y) \\ \times \sin(1.5\pi \times 10^{10} t - 109\pi z) \quad (\text{mA/m}).$$

If the guide dimensions are $a = 2b = 4$ cm, determine (a) the mode numbers, (b) the relative permittivity of the material in the guide, and (c) the phase velocity, and (d) obtain an expression for E_x .

Solution: (a) By comparison with the expression for $\tilde{H}_y$ given by Eq. (8.104d), we deduce that the argument of x is $(m\pi/a)$ and the argument of y is $(n\pi/b)$. Hence,

$$25\pi = \frac{m\pi}{4 \times 10^{-2}}, \quad 100\pi = \frac{n\pi}{2 \times 10^{-2}},$$

which yield $m = 1$ and $n = 2$. Therefore, the mode is TM_{12} .

(b) The second sine function in the expression for H_y represents $\sin(\omega t - \beta z)$, which means that

$$\omega = 1.5\pi \times 10^{10} \text{ (rad/s) or } f = 7.5 \text{ GHz,} \\ \beta = 109\pi \text{ (rad/m).}$$

By rewriting Eq. (8.105) to obtain an expression for $\epsilon_r = \epsilon/\epsilon_0$ in terms of the other quantities, we have

$$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right],$$

场表达式中尚待确定的是传播常数 β ，它出现在指数因子 $e^{-j\beta z}$ 中。联立式 (8.90)、(8.99) 和 (8.102)，可得 β 的表达式：

$$\beta = \sqrt{k^2 - k_c^2} = \sqrt{\omega^2 \mu \epsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}. \quad (8.105)$$

(TE 和 TM)

尽管 β 的表达式是针对 TM 模式导出的，但它同样适用于 TE 模式。

指数 $e^{-j\beta z}$ 描述了沿 $+z$ 方向传播的波（假设 β 为实数），对应于 $k > k_c$ 。如果 $k < k_c$ ， β 变为虚数： $\beta = -j\alpha$ 且 α 为实数，在这种情况下 $e^{-j\beta z} = e^{-\alpha z}$ ，产生倏逝波，其特征是由于衰减函数 $e^{-\alpha z}$ ，其振幅随 z 快速衰减。对应于每个模态 (m, n) ，存在一个截止频率 f_{mn} ，其中 $\beta = 0$ 。通过将式(1)中的 $\beta = \text{设置为} 0$ ，可以得到：(8.105) 然后求解 f ，我们有

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}, \quad (8.106)$$

(TE and TM)

其中 $u_{p0} = 1/\sqrt{\mu\epsilon}$ 是具有本构参数 ϵ 和 μ 的无界介质中 TEM 波的相速度。

► 给定模式下的波只有其频率为 $f > f_{mn}$ 时才能通过波导传播，只有 $\beta = \text{才是真实的}$ 。 ◀

具有最低截止频率的模式称为主模式。TM模式中的主导模式是 TM_{11} ，TE模式中的主导模式是 TE_{10} 。虽然 TE 模式允许 m 或 n 为零，但 TM 模式则不允许 [因为如果 m 或 n 为零，则方程 1 中的 $\tilde{E}_z$ 为 0。(8.103) 变为零，所有其他场分量也消失]。

通过结合等式。(8.105) 和 (8.106)，我们可以用 f_{mn} 来表示 β ，

$$\beta = \frac{\omega}{u_{p0}} \sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}. \quad \textbf{(TE and TM)} \quad (8.107)$$

波导中 TE 或 TM 波的相速度为

$$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}}. \quad \textbf{(TE and TM)} \quad (8.108)$$

横向电场由方程 1 给出的分量 $\tilde{E}_x$ 和 $\tilde{E}_y$ 组成。(8.104a 和 b)。对于沿 $+z$ 方向传播的波，与 $\tilde{E}_x$ 相关的磁场为 $\tilde{H}_y$ [根据等式 1 给出的右手法则] (7.39a)]。类似地，与 $\tilde{E}_y$ 相关的磁场是 $-\tilde{H}_x$ 。通过使用等式获得的比率。(8.104e)，构成波导中的波阻抗，

$$Z_{\text{TM}} = \frac{\tilde{E}_x}{\tilde{H}_y} = -\frac{\tilde{E}_y}{\tilde{H}_x} = \frac{\beta \eta}{k} = \eta \sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}, \quad (8.109)$$

其中 $\eta = \sqrt{\mu/\epsilon}$ 是填充波导的介电材料的固有阻抗。

例 8-8：模式属性

TM 波在介电常数未知的电介质填充波导中传播，其磁场的 y 分量为

$$H_y = 6 \cos(25\pi x) \sin(100\pi y) \\ \times \sin(1.5\pi \times 10^{10} t - 109\pi z) \quad (\text{mA/m}).$$

如果波导尺寸为 $a = 2b = 4$ cm，则确定 (a) 模数、(b) 波导中材料的相对介电常数、(c) 相速度，以及 (d) 获得 E_x 的表达式。

解：(a) 通过与式(1)给出的 $\tilde{H}_y$ 的表达式进行比较。(8.104d)，我们推导出 x 的参数是 $(m\pi/a)$ ， y 的参数是 $(n\pi/b)$ 。因此，

$$25\pi = \frac{m\pi}{4 \times 10^{-2}}, \quad 100\pi = \frac{n\pi}{2 \times 10^{-2}},$$

产生 $m = 1$ 和 $n = 2$ 。因此，众数为 TM_{12} 。

(b) H_y 表达式中的第二个正弦函数表示 $\sin(\omega t - \beta z)$ ，这意味着

$$\omega = 1.5\pi \times 10^{10} \text{ (rad/s) or } f = 7.5 \text{ GHz,} \\ \beta = 109\pi \text{ (rad/m).}$$

通过重写方程。(8.105) 为了获得 $\epsilon_r = \epsilon/\epsilon_0$ 用其他量表示的表达式，我们有

$$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right],$$

where c is the speed of light. Inserting the available values, we obtain

$$\epsilon_r = \frac{(3 \times 10^8)^2}{(1.5\pi \times 10^{10})^2} \cdot \left[(109\pi)^2 + \left(\frac{\pi}{4 \times 10^{-2}} \right)^2 + \left(\frac{2\pi}{2 \times 10^{-2}} \right)^2 \right] = 9.$$

(c)

$$u_p = \frac{\omega}{\beta} = \frac{1.5\pi \times 10^{10}}{109\pi} = 1.38 \times 10^8 \text{ m/s},$$

which is slower than the speed of light. However, as explained later in Section 8-10, the phase velocity in a waveguide may exceed c , but the group velocity with which energy is carried down the guide is the group velocity u_g , which is never greater than c .

(d) From Eq. (8.109),

$$Z_{\text{TM}} = \eta \sqrt{1 - (f_{12}/f)^2}.$$

Application of Eq. (8.106) yields $f_{12} = 5.15$ GHz for the TM_{12} mode. Using that in the expression for Z_{TM} , in addition to $f = 7.5$ GHz and

$$\eta = \sqrt{\mu/\epsilon} = \frac{\sqrt{\mu_0/\epsilon_0}}{\sqrt{\epsilon_r}} = \frac{377}{\sqrt{9}} = 125.67 \, \Omega,$$

gives

$$Z_{\text{TM}} = 91.3 \, \Omega.$$

Hence,

$$\begin{aligned} E_x &= Z_{\text{TM}} H_y = 91.3 \times 6 \cos(25\pi x) \sin(100\pi y) \\ &\quad \times \sin(1.5\pi \times 10^{10} t - 109\pi z) \quad (\text{mV/m}) \\ &= 0.55 \cos(25\pi x) \sin(100\pi y) \\ &\quad \times \sin(1.5\pi \times 10^{10} t - 109\pi z) \quad (\text{V/m}). \end{aligned}$$

Concept Question 8-8: What are the primary limitations of coaxial cables at frequencies higher than 30 GHz?

Concept Question 8-9: Can a TE mode have a zero magnetic field along the direction of propagation?

Concept Question 8-10: What is the rationale for choosing a solution for $\tilde{e}_z$ that involves sine and cosine functions?

Concept Question 8-11: What is an evanescent wave?

Exercise 8-10: For a square waveguide with $a = b$, what is the value of the ratio $\tilde{E}_x/\tilde{E}_y$ for the TM_{11} mode?

Answer: $\tan(\pi y/a)/\tan(\pi x/a)$.

Exercise 8-11: What is the cutoff frequency for the dominant TM mode in a waveguide filled with a material with $\epsilon_r = 4$? The waveguide dimensions are $a = 2b = 5$ cm.

Answer: For TM_{11} , $f_{11} = 3.35$ GHz.

Exercise 8-12: What is the magnitude of the phase velocity of a TE or TM mode at $f = f_{mn}$?

Answer: $u_p = \infty$! [See explanation in Section 8-10.]

8-9 TE Modes in Rectangular Waveguide

In the TM case, where the wave has no magnetic field component along the z direction (i.e., $\tilde{H}_z = 0$), we started our treatment in the preceding section by obtaining a solution for $\tilde{E}_z$, and then we used it to derive expressions for the tangential components of $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$. For the TE case, the same basic procedure can be applied, except we reversed the roles of $\tilde{E}_z$ and $\tilde{H}_z$. Such a process leads to:

$$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b} \right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.110a)$$

$$\tilde{E}_y = \frac{-j\omega\mu}{k_c^2} \left(\frac{m\pi}{a} \right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.110b)$$

$$\tilde{H}_x = \frac{j\beta}{k_c^2} \left(\frac{m\pi}{a} \right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.110c)$$

$$\tilde{H}_y = \frac{j\beta}{k_c^2} \left(\frac{n\pi}{b} \right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.110d)$$

$$\tilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.110e)$$

and of course, $\tilde{E}_z = 0$. The expressions for f_{mn} , β , and u_p given earlier by Eqs. (8.106), (8.107), and (8.108) remain unchanged.

► Because not all of the fields vanish if m or n assume a value of zero, the lowest order TE mode is TE_{10} if $a > b$ or TE_{01} if $a < b$. It is customary to assign a to be the longer dimension, in which case the TE_{10} mode is the de facto dominant mode. ◀

其中c是光速。插入可用值，我们得到

$$\epsilon_r = \frac{(3 \times 10^8)^2}{(1.5\pi \times 10^{10})^2} \cdot \left[(109\pi)^2 + \left(\frac{\pi}{4 \times 10^{-2}} \right)^2 + \left(\frac{2\pi}{2 \times 10^{-2}} \right)^2 \right] = 9.$$

(c)

$$u_p = \frac{\omega}{\beta} = \frac{1.5\pi \times 10^{10}}{109\pi} = 1.38 \times 10^8 \text{ m/s},$$

它比光速慢。然而，正如稍后第 8-10 节中所解释的，波导中的相速度可能超过 c ，但能量沿着波导传递的速度是群速度 u_g ，它永远不会大于 c 。

(d) 从方程式(8.109),

$$Z_{\text{TM}} = \eta \sqrt{1 - (f_{12}/f)^2}.$$

方程的应用(8.106) 为 TM_{12} 模式产生 $f_{12} = 5.15$ GHz。在 Z_{TM} 的表达式中使用该值，此外还有 $f = 7.5$ GHz 和

$$\eta = \sqrt{\mu/\epsilon} = \frac{\sqrt{\mu_0/\epsilon_0}}{\sqrt{\epsilon_r}} = \frac{377}{\sqrt{9}} = 125.67 \, \Omega,$$

给出

$$Z_{\text{TM}} = 91.3 \, \Omega.$$

因此,

$$\begin{aligned} E_x &= Z_{\text{TM}} H_y = 91.3 \times 6 \cos(25\pi x) \sin(100\pi y) \\ &\quad \times \sin(1.5\pi \times 10^{10} t - 109\pi z) \quad (\text{mV/m}) \\ &= 0.55 \cos(25\pi x) \sin(100\pi y) \\ &\quad \times \sin(1.5\pi \times 10^{10} t - 109\pi z) \quad (\text{V/m}). \end{aligned}$$

概念问题 8-8：频率高于 30 GHz 时同轴电缆的主要限制是什么？

概念问题 8-9：TE 模式沿传播方向的磁场可以为零吗？

概念问题 8-10：选择涉及正弦和余弦函数的 $\tilde{e}_z$ 解的基本原理是什么？

概念问题 8-11：什么是倏逝波？

Exercise 8-10: For a square waveguide with $a = b$, what is the value of the ratio $\tilde{E}_x/\tilde{E}_y$ for the TM_{11} mode?

Answer: $\tan(\pi y/a)/\tan(\pi x/a)$.

Exercise 8-11: What is the cutoff frequency for the dominant TM mode in a waveguide filled with a material with $\epsilon_r = 4$? The waveguide dimensions are $a = 2b = 5$ cm.

Answer: For TM_{11} , $f_{11} = 3.35$ GHz.

Exercise 8-12: What is the magnitude of the phase velocity of a TE or TM mode at $f = f_{mn}$?

Answer: $u_p = \infty$! [See explanation in Section 8-10.]

8-9 矩形波导中的TE模式

在 TM 情况下，波沿 z 方向没有磁场分量（即 $\tilde{H}_z = 0$ ），我们在上一节中通过获得 $\tilde{E}_z$ 的解开始处理，然后用它推导 $\tilde{E}$ 和 $\tilde{H}$ 的切向分量的表达式。对于 TE 情况，可以应用相同的基本过程，只是我们颠倒了 $\tilde{E}_z$ 和 $\tilde{H}_z$ 的角色。这样的过程导致：

$$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b} \right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.110a)$$

$$\tilde{E}_y = \frac{-j\omega\mu}{k_c^2} \left(\frac{m\pi}{a} \right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.110b)$$

$$\tilde{H}_x = \frac{j\beta}{k_c^2} \left(\frac{m\pi}{a} \right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.110c)$$

$$\tilde{H}_y = \frac{j\beta}{k_c^2} \left(\frac{n\pi}{b} \right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.110d)$$

$$\tilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.110e)$$

当然， $\tilde{E}_z = 0$ 。 f_{mn} 、 β 和 u_p 的表达式前面由等式给出。(8.106)、(8.107)和(8.108)保持不变。

► 因为如果 m 或 n 的值为零，则并非所有场都会消失，因此最低阶 TE 模式是 TE_{10} （如果 $a > b$ ）或 TE_{01} （如果 $a < b$ ）。通常将 a 指定为较长的维度，在这种情况下， TE_{10} 模式是事实上的主导模式。 ◀

Table 8-3 Wave properties for TE and TM modes in a rectangular waveguide with dimensions $a \times b$ that are filled with a dielectric material with constitutive parameters ϵ and μ . The TEM case—shown here for reference—pertains to plane-wave propagation in an unbounded medium.

Rectangular Waveguides		Plane Wave
TE Modes	TM Modes	TEM Mode
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_y = \frac{-j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_z = 0$ $\tilde{H}_x = -\tilde{E}_y/Z_{TE}$ $\tilde{H}_y = \tilde{E}_x/Z_{TE}$ $\tilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $Z_{TE} = \eta/\sqrt{1-(f_c/f)^2}$	$\tilde{E}_x = \frac{-j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_y = \frac{-j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_z = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{H}_x = -\tilde{E}_y/Z_{TM}$ $\tilde{H}_y = \tilde{E}_x/Z_{TM}$ $\tilde{H}_z = 0$ $Z_{TM} = \eta\sqrt{1-(f_c/f)^2}$	$\tilde{E}_x = E_{x0}e^{-j\beta z}$ $\tilde{E}_y = E_{y0}e^{-j\beta z}$ $\tilde{E}_z = 0$ $\tilde{H}_x = -\tilde{E}_y/\eta$ $\tilde{H}_y = \tilde{E}_x/\eta$ $\tilde{H}_z = 0$ $\eta = \sqrt{\mu/\epsilon}$
<i>Properties Common to TE and TM Modes</i> $f_c = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$ $\beta = k\sqrt{1-(f_c/f)^2}$ $u_p = \frac{\omega}{\beta} = u_{p0}/\sqrt{1-(f_c/f)^2}$		$f_c = \text{not applicable}$ $k = \omega\sqrt{\mu\epsilon}$ $u_{p0} = 1/\sqrt{\mu\epsilon}$

Another difference between the TE and TM modes relates to the expression for the wave impedance. For TE,

$$Z_{TE} = \frac{\tilde{E}_x}{\tilde{H}_y} = -\frac{\tilde{E}_y}{\tilde{H}_x} = \frac{\eta}{\sqrt{1-(f_{mn}/f)^2}}. \quad (8.111)$$

A summary of the expressions for the various wave attributes of TE and TM modes is given in **Table 8-3**. As a reference, corresponding expressions for the TEM mode on a coaxial transmission line are included as well.

Example 8-9: Cutoff Frequencies

For a hollow rectangular waveguide with dimensions $a = 3$ cm and $b = 2$ cm, determine the cutoff frequencies for all modes up to 20 GHz. Over what frequency range will the guide support the propagation of a single dominant mode?

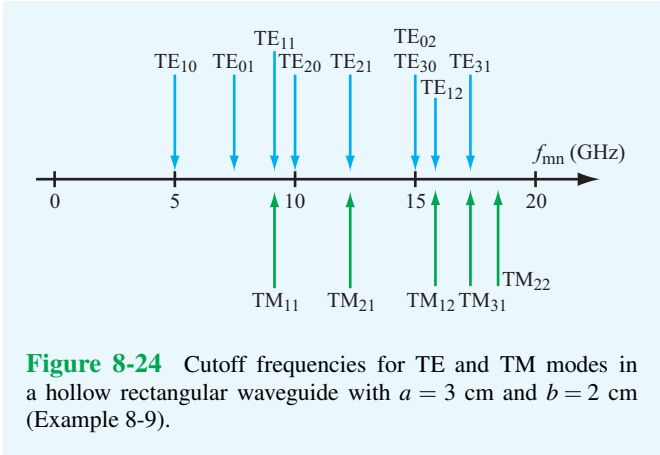


Figure 8-24 Cutoff frequencies for TE and TM modes in a hollow rectangular waveguide with $a = 3$ cm and $b = 2$ cm (Example 8-9).

Solution: A hollow guide has $\mu = \mu_0$ and $\epsilon = \epsilon_0$. Hence, $u_{p0} = 1/\sqrt{\mu_0\epsilon_0} = c$. Application of Eq. (8.106) gives the cutoff frequencies shown in **Fig. 8-24**, which start at 5 GHz for the TE₁₀ mode. To avoid all other modes, the frequency of operation should be restricted to the 5–7.5 GHz range.

表 8-3 尺寸为 a × b 的矩形波导中 TE 和 TM 模式的波特特性，该波导填充有本构参数为 ε 和 μ 的介电材料。 TEM 案例（此处显示仅供参考）涉及无界介质中的平面波传播。

Rectangular Waveguides		Plane Wave
TE Modes	TM Modes	TEM Mode
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_y = \frac{-j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_z = 0$ $\tilde{H}_x = -\tilde{E}_y/Z_{TE}$ $\tilde{H}_y = \tilde{E}_x/Z_{TE}$ $\tilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $Z_{TE} = \eta/\sqrt{1-(f_c/f)^2}$	$\tilde{E}_x = \frac{-j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_y = \frac{-j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{E}_z = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ $\tilde{H}_x = -\tilde{E}_y/Z_{TM}$ $\tilde{H}_y = \tilde{E}_x/Z_{TM}$ $\tilde{H}_z = 0$ $Z_{TM} = \eta\sqrt{1-(f_c/f)^2}$	$\tilde{E}_x = E_{x0}e^{-j\beta z}$ $\tilde{E}_y = E_{y0}e^{-j\beta z}$ $\tilde{E}_z = 0$ $\tilde{H}_x = -\tilde{E}_y/\eta$ $\tilde{H}_y = \tilde{E}_x/\eta$ $\tilde{H}_z = 0$ $\eta = \sqrt{\mu/\epsilon}$
<i>Properties Common to TE and TM Modes</i> $f_c = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$ $\beta = k\sqrt{1-(f_c/f)^2}$ $u_p = \frac{\omega}{\beta} = u_{p0}/\sqrt{1-(f_c/f)^2}$		$f_c = \text{not applicable}$ $k = \omega\sqrt{\mu\epsilon}$ $u_{p0} = 1/\sqrt{\mu\epsilon}$

TE 和 TM 模式之间的另一个区别与波阻抗的表达式有关。对于TE来说，

$$Z_{TE} = \frac{\tilde{E}_x}{\tilde{H}_y} = -\frac{\tilde{E}_y}{\tilde{H}_x} = \frac{\eta}{\sqrt{1-(f_{mn}/f)^2}}. \quad (8.111)$$

表 8-3 总结了 TE 和 TM 模式的各种波属性的表达式。作为参考，还包括同轴传输线上 TEM 模式的相应表达式。

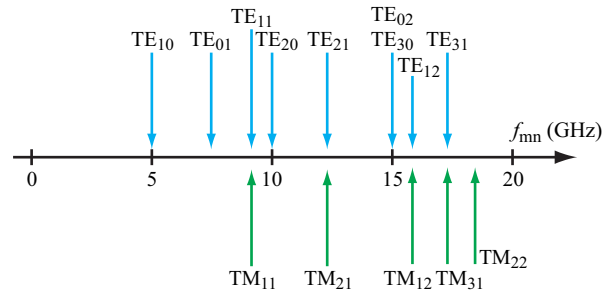


图 8-24 a = 3 cm 和 b = 2 cm 的空心矩形波导中 TE 和 TM 模式的截止频率（示例 8-9）。

例 8-9： 截止频率

对于尺寸 a = 3 cm 和 b = 2 cm 的空心矩形波导，确定高达 20 GHz 的所有模式的截止频率。波导将在什么频率范围内支持单一主模的传播？

解：空心导轨有μ = μ₀和ε = ε₀。因此， u_{p0} = 1/√μ₀ε₀ = c。方程的应用(8.106) 给出了如图 8-24 所示的截止频率，TE₁₀ 模式的起始频率为 5 GHz。为了避免所有其他模式，工作频率应限制在 5–7.5 GHz 范围内。

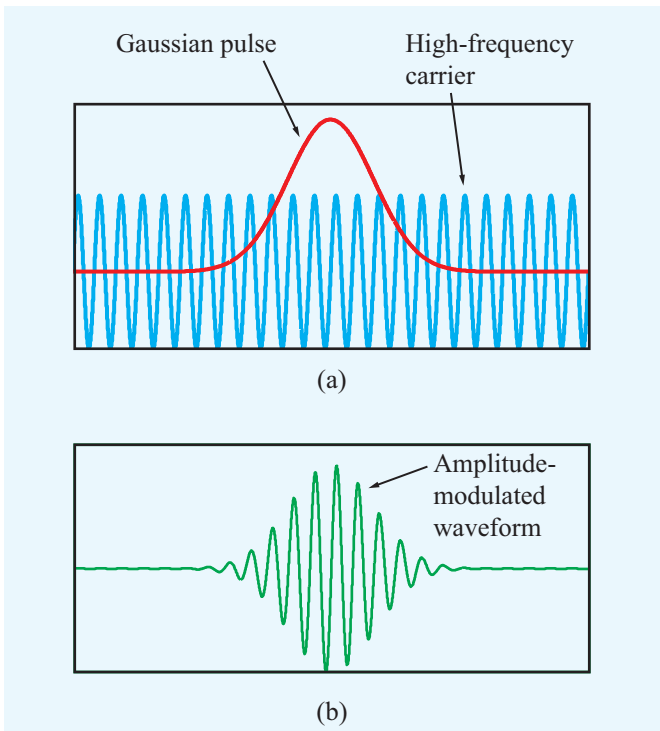


Figure 8-25 The amplitude-modulated high-frequency waveform in (b) is the product of the Gaussian-shaped pulse with the sinusoidal high-frequency carrier in (a).

8-10 Propagation Velocities

When a wave is used to carry a message through a medium or along a transmission line, information is encoded into the wave's amplitude, frequency, or phase. A simple example is shown in **Fig. 8-25**, where a high-frequency sinusoidal wave of frequency f is amplitude-modulated by a low-frequency Gaussian pulse. The waveform in **Fig. 8-25(b)** is the result of multiplying the Gaussian pulse shape in **Fig. 8-25(a)** by the carrier waveform.

By Fourier analysis, the waveform in **Fig. 8-25(b)** is equivalent to the superposition of a *group* of sinusoidal waves with specific amplitudes and frequencies. Exact equivalence may require a large (or infinite) number of frequency components, but in practice, it is often possible to represent the modulated waveform to a fairly high degree of fidelity with a wave group that extends over a relatively narrow bandwidth surrounding the high-frequency carrier f . The velocity that the envelope—or equivalently the wave group—travels through the medium is called the **group velocity** u_g . As such, u_g is the velocity of the energy carried by the wave group and of the information

encoded in it. Depending on whether or not the propagation medium is dispersive, u_g may or may not be equal to the phase velocity u_p . In Section 2-1.1, we described a dispersive transmission line as one “on which the phase velocity is not a constant as a function of frequency,” and as a result the shape of a pulse transmitted through it gets progressively distorted as it moves down the line. A rectangular waveguide constitutes a dispersive transmission line because the phase velocity of a TE or TM mode propagating through it is a strong function of frequency [per Eq. (8.108)], particularly at frequencies close to the cutoff frequency f_{mn} . As we see shortly, if $f \gg f_{mn}$, the TE and TM modes become approximately TEM in character—not only in terms of the directional arrangement of the electric and magnetic fields but also in terms of the frequency dependence of the phase velocity.

We now examine u_p and u_g in more detail. The phase velocity defined as the velocity of the sinusoidal pattern of the wave is given by

$$u_p = \frac{\omega}{\beta}, \quad (8.112)$$

while the group velocity u_g is given by

$$u_g = \frac{1}{d\beta/d\omega}. \quad (8.113)$$

Even though we will not derive Eq. (8.113) in this book, it is nevertheless important that we understand its properties for TE and TM modes in a metal waveguide. Using the expression for β given by Eq. (8.107),

$$u_g = \frac{1}{d\beta/d\omega} = u_{p0} \sqrt{1 - (f_{mn}/f)^2}, \quad (8.114)$$

where—as before— u_{p0} is the phase velocity in an unbounded dielectric medium. In view of Eq. (8.108) for the phase velocity u_p ,

$$u_p u_g = u_{p0}^2. \quad (8.115)$$

Above cutoff ($f > f_{mn}$), $u_p \geq u_{p0}$ and $u_g \leq u_{p0}$. As $f \rightarrow \infty$, or more precisely as $(f_{mn}/f) \rightarrow 0$, TE and TM modes approach the TEM case for which $u_p = u_g = u_{p0}$.

A useful graphical tool for describing the propagation properties of a medium or transmission line is the **ω - β diagram**. In **Fig. 8-26**, the straight line starting at the origin represents the ω - β relationship for a TEM wave propagating in an unbounded medium (or on a TEM transmission line). The TEM line provides a reference to which the ω - β curves of the TE/TM modes can be compared. At a given location on the ω - β line or curve, the ratio of the value of ω to that of

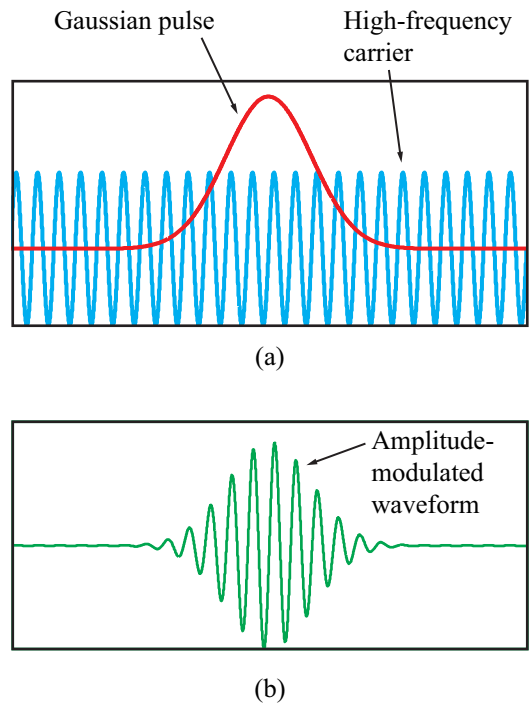


图 8-25 (b) 中的调幅高频波形是 (a) 中的高斯形脉冲与正弦高频载波的乘积。

8-10 传播速度

当波用于通过介质或沿着传输线携带消息时，信息被编码为波的幅度、频率或相位。一个简单的例子如图8-25所示，其中频率为 f 的高频正弦波由低频高斯脉冲进行幅度调制。图8-25(b)中的波形是将图8-25(a)中的高斯脉冲形状乘以载波波形的结果。

通过傅里叶分析，图8-25 (b) 中的波形相当于一组具有特定幅度和频率的正弦波的叠加。精确等效可能需要大量（或无限）的频率分量，但在实践中，通常可以使用在高频载波 f 周围相对较窄的带宽上延伸的波群来表示相当高保真度的调制波形。包络线（或等效的波群）穿过介质的速度称为群速度 u_g 。因此， u_g 是波群携带的能量和信息的速度

编码在其中。根据传播介质是否色散， u_g 可能等于也可能不等于相速度 u_p 。在第 2-1.1 节中，我们将色散传输线描述为“相速度不是频率函数的常数”，因此通过它传输的脉冲的形状随着沿线移动而逐渐失真。矩形波导构成色散传输线，因为通过它传播的 TE 或 TM 模式的相速度是频率的强函数（根据方程 1]。 (8.108)]，特别是在接近截止频率 f_{mn} 的频率下。正如我们很快看到的，如果 $f \gg f_{mn}$ ，TE 和 TM 模式在特性上变得近似 TEM——不仅在电场和磁场的方向排列方面，而且在相速度的频率依赖性方面。

我们现在更详细地研究 u_p 和 u_g 。相速度定义为波的正弦模式的速度，由下式给出

$$u_p = \frac{\omega}{\beta}, \quad (8.112)$$

而群速度 u_g 则由下式给出

$$u_g = \frac{1}{d\beta/d\omega}. \quad (8.113)$$

即使我们不会推导出方程。(8.113) 在本书中，我们了解金属波导中 TE 和 TM 模式的特性仍然很重要。使用等式给出的 β 表达式。(8.107),

$$u_g = \frac{1}{d\beta/d\omega} = u_{p0} \sqrt{1 - (f_{mn}/f)^2}, \quad (8.114)$$

其中，与之前一样， u_{p0} 是无界电介质中的相速度。鉴于方程。(8.108) 对于相速度 u_p ,

$$u_p u_g = u_{p0}^2. \quad (8.115)$$

高于截止值 ($f > f_{mn}$)、 $u_p \geq u_{p0}$ 和 $u_g \leq u_{p0}$ 。当 $f \rightarrow \infty$ 时，或者更精确地作为 $(f_{mn}/f) \rightarrow 0$ 时，TE 和 TM 模式接近 TEM 情况，其中 $u_p = u_g = u_{p0}$ 。 ω - β 图是描述介质或传输线的传播特性的有用图形工具。在图8-26中，从原点开始的直线表示TEM波在无界介质（或TEM传输线上）传播的 ω - β 关系。TEM线提供了可以比较 TE/TM 模式的 ω - β 曲线的参考。在 ω - β 直线或曲线上的给定位置， ω 的值与

8-10 PROPAGATION VELOCITIES

373

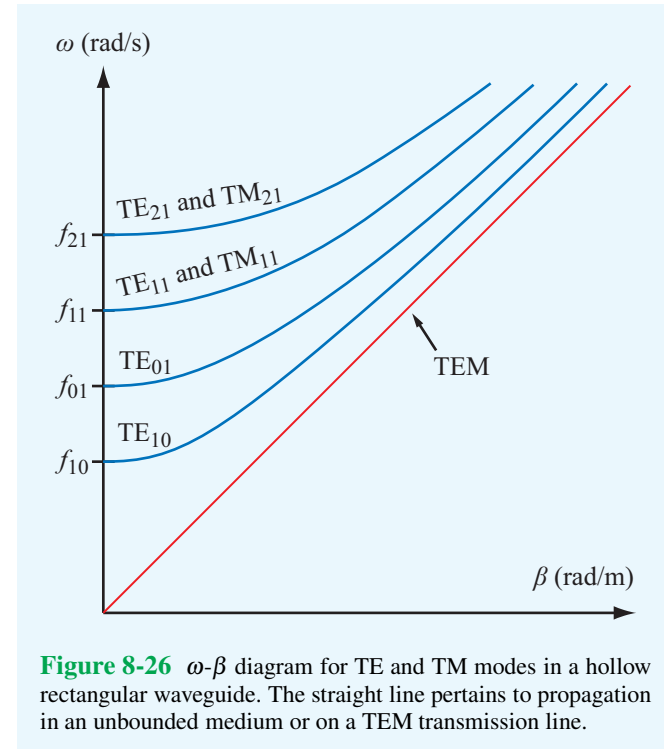


Figure 8-26 ω - β diagram for TE and TM modes in a hollow rectangular waveguide. The straight line pertains to propagation in an unbounded medium or on a TEM transmission line.

β defines $u_p = \omega/\beta$, whereas it is the slope $d\omega/d\beta$ of the curve at that point that defines the group velocity u_g . For the TEM line, the ratio and the slope have identical values (hence, $u_p = u_g$), and the line starts at $\omega = 0$. In contrast, the curve for each of the indicated TE/TM modes starts at a cutoff frequency specific to that mode below which the waveguide cannot support the propagation of a wave in that mode. At frequencies close to cutoff, u_p and u_g assume very different values; in fact, at cutoff $u_p = \infty$ and $u_g = 0$. On the other end of the frequency spectrum—at frequencies much higher than f_{mn} —the ω - β curves of the TE/TM modes approach the TEM line. We should note that for TE and TM modes, u_p may easily exceed the speed of light, but u_g will not. Since it is u_g that represents the actual transport of energy, Einstein's assertion that there is an upper bound on the speed of physical phenomena is not violated.

So far, we have described the fields in the guide, but we have yet to interpret them in terms of plane waves that zigzag along the guide through successive reflections. To do just that, consider the simple case of a TE₁₀ mode. For $m = 1$ and $n = 0$, the only nonzero component of the electric field given by Eq. (8.110) is $\tilde{E}_y$, so

$$\tilde{E}_y = -j \frac{\omega \mu}{k_c^2} \left(\frac{\pi}{a} \right) H_0 \sin \left(\frac{\pi x}{a} \right) e^{-j\beta z}. \quad (8.116)$$

Using the identity $\sin \theta = (e^{j\theta} - e^{-j\theta})/2j$ for any argument θ , we obtain

$$\begin{aligned} \tilde{E}_y &= \left(\frac{\omega \mu \pi H_0}{2k_c^2 a} \right) (e^{-j\pi x/a} - e^{j\pi x/a}) e^{-j\beta z} \\ &= E'_0 (e^{-j\beta(z+\pi x/\beta a)} - e^{-j\beta(z-\pi x/\beta a)}) \\ &= E'_0 (e^{-j\beta z'} - e^{-j\beta z''}), \end{aligned} \quad (8.117)$$

where we have consolidated the quantities multiplying the two exponential terms into the constant E'_0 . The first exponential term represents a wave with propagation constant β traveling in the z' direction, where

$$z' = z + \frac{\pi x}{\beta a}, \quad (8.118a)$$

and the second term represents a wave traveling in the z'' direction with

$$z'' = z - \frac{\pi x}{\beta a}. \quad (8.118b)$$

From the diagram shown in **Fig. 8-27(a)**, it is evident that the z' direction is at an angle θ' relative to z , and the z'' direction is at an angle $\theta'' = -\theta'$. This means that the electric field $\tilde{E}_y$ (and its associated magnetic field $\tilde{\mathbf{H}}$) of the TE₁₀ mode is composed of two TEM waves shown in **Fig. 8-27(b)** with both traveling in the $+z$ direction by zigzagging between the opposite walls of the waveguide. Along the zigzag directions (z' and z''), the phase velocity of the individual wave components is u_{p0} , but the phase velocity of the combination of the two waves along z is u_p .

Example 8-10: Zigzag Angle

For the TE₁₀ mode, express the zigzag angle θ' in terms of the ratio (f/f_{10}) , and then evaluate it at $f = f_{10}$ and for $f \gg f_{10}$.

Solution: From **Fig. 8-27**,

$$\theta'_{10} = \tan^{-1} \left(\frac{\pi}{\beta_{10} a} \right),$$

where the subscript 10 has been added as a reminder that the expression applies to the TE₁₀ mode specifically. For $m = 1$ and $n = 0$, Eq. (8.106) reduces to $f_{10} = u_{p0}/2a$. After replacing β with the expression given by Eq. (8.107) and replacing a with $u_{p0}/2f_{10}$, we obtain

$$\theta' = \tan^{-1} \left[\frac{1}{\sqrt{(f/f_{10})^2 - 1}} \right].$$

8-10 PROPAGATION VELOCITIES

373

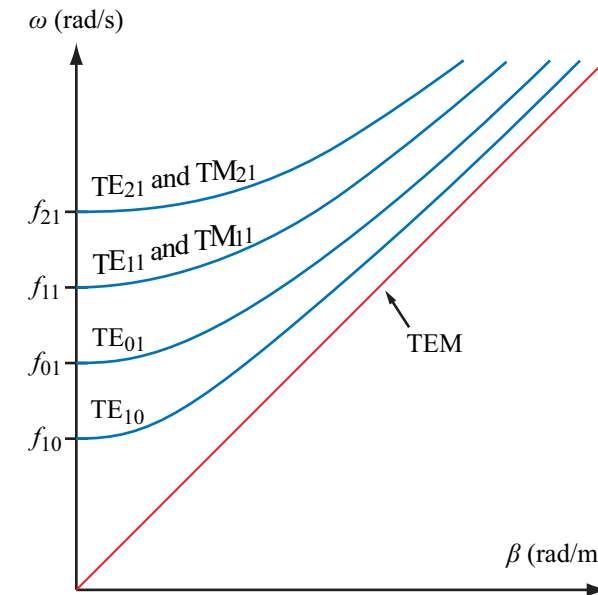


图 8-26 空心矩形波导中 TE 和 TM 模式的 ω - β 图。该直线适用于无界介质或 TEM 传输线上的传播。

β 定义了 $u_p = \omega/\beta$ ，而该点处曲线的斜率 $d\omega/d\beta$ 则定义了群速度 u_g 。对于 TEM 线，比率和斜率具有相同的值（因此， $u_p = u_g$ ），并且线从 $\omega = 0$ 开始。相反，每个指示的 TE/TM 模式的曲线从特定于该模式的截止频率开始，低于该频率波导无法支持该模式下波的传播。在接近截止的频率处， u_p 和 u_g 呈现非常不同的值；事实上，在截止 $u_p = \infty$ 和 $u_g = 0$ 处。在频谱的另一端（在远高于 f_{mn} 的频率处），TE/TM 模式的 ω - β 曲线接近 TEM 线。我们应该注意到，对于 TE 和 TM 模式， u_p 可能很容易超过光速，但 u_g 不会。由于 u_g 代表了实际的能量传输，因此并不违反爱因斯坦关于物理现象速度存在上限的断言。

到目前为止，我们已经描述了波导中的场，但我们还没有用平面波来解释它们，平面波通过连续的反射沿着波导蜿蜒。为此，请考虑 TE₁₀ 模式的简单情况。对于 $m = 1$ 和 $n = 0$ ，由方程式给出的电场的唯一非零分量。(8.110) 是 $\tilde{E}_y$ ，所以

$$\tilde{E}_y = -j \frac{\omega \mu}{k_c^2} \left(\frac{\pi}{a} \right) H_0 \sin \left(\frac{\pi x}{a} \right) e^{-j\beta z}. \quad (8.116)$$

使用恒等式 $\sin \theta = (e^{j\theta} - e^{-j\theta})/2j$ 对于任何参数 θ ，我们得到

$$\begin{aligned} \tilde{E}_y &= \left(\frac{\omega \mu \pi H_0}{2k_c^2 a} \right) (e^{-j\pi x/a} - e^{j\pi x/a}) e^{-j\beta z} \\ &= E'_0 (e^{-j\beta(z+\pi x/\beta a)} - e^{-j\beta(z-\pi x/\beta a)}) \\ &= E'_0 (e^{-j\beta z'} - e^{-j\beta z''}), \end{aligned} \quad (8.117)$$

我们将两个指数项相乘的量合并为常数 E'_0 。第一个指数项表示传播常数为 β 的波，沿 z' 方向传播，其中

$$z' = z + \frac{\pi x}{\beta a}, \quad (8.118a)$$

第二项表示沿 z'' 方向传播的波

$$z'' = z - \frac{\pi x}{\beta a}. \quad (8.118b)$$

从图8-27(a)所示的图中可以看出， z' 方向相对于 z 成角度 θ' ，并且 z'' 方向相对于 z 成角度 $\theta'' = -\theta'$ 。这意味着 TE₁₀ 模式的电场 $\tilde{E}_y$ (及其相关磁场 $\tilde{\mathbf{H}}$) 由图 8-27(b) 中所示的两个 TEM 波组成，两个 TEM 波均在波导的相对壁之间锯齿形沿 $+z$ 方向传播。沿之字形方向 (z' 和 z'')，各个波分量的相速度为 u_{p0} ，但沿 z 的两个波组合的相速度为 u_p 。

示例 8-10: Z 字形角度

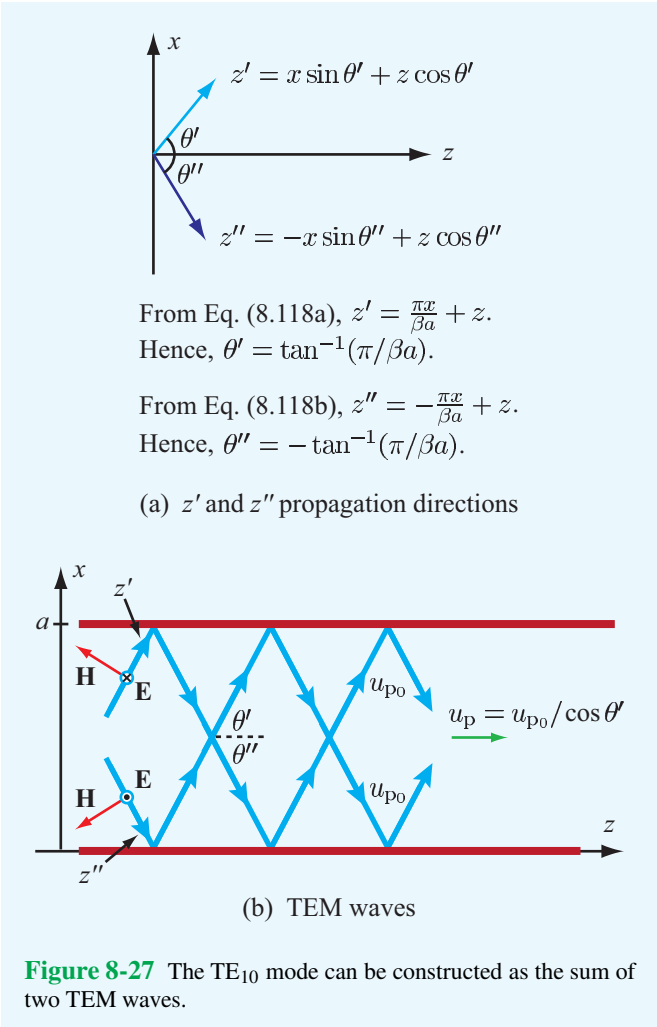
对于 TE₁₀ 模式，用比率 (f/f_{10}) 表示锯齿角 θ' ，然后在 $f = f_{10}$ 和 $f \gg f_{10}$ 处对其进行评估。

解：由图8-27可知，

$$\theta'_{10} = \tan^{-1} \left(\frac{\pi}{\beta_{10} a} \right),$$

其中添加下标 10 是为了提醒该表达式特别适用于 TE₁₀ 模式。对于 $m = 1$ 和 $n = 0$ ，等式：(8.106) 简化为 $f_{10} = u_{p0}/2a$ 。将 β 替换为等式给出的表达式后：(8.107) 并将 a 替换为 $u_{p0}/2f_{10}$ ，我们得到

$$\theta' = \tan^{-1} \left[\frac{1}{\sqrt{(f/f_{10})^2 - 1}} \right].$$



At $f = f_{10}$, $\theta' = 90^\circ$, which means that the wave bounces back and forth at normal incidence between the two side walls of the waveguide, making no progress in the z direction. At the other end of the frequency spectrum, when $f \gg f_{10}$, θ' approaches 0 and the wave becomes TEM-like as it travels straight down the guide.

Concept Question 8-12: For TE waves, the dominant mode is TE_{10} , but for TM the dominant mode is TM_{11} . Why is it not TM_{10} ?

Concept Question 8-13: Why is it acceptable for u_p to exceed the speed of light c , but not so for u_g ?

Exercise 8-13: What do the wave impedances for TE and TM look like as f approaches f_{mn} ?

Answer: At $f = f_{mn}$, Z_{TE} looks like an open circuit, and Z_{TM} looks like a short circuit.

Exercise 8-14: What are the values for (a) u_p , (b) u_g , and (c) the zigzag angle θ' at $f = 2f_{10}$ for a TE_{10} mode in a hollow waveguide?

Answer: (a) $u_p = 1.15c$, (b) $u_g = 0.87c$, (c) $\theta' = 30^\circ$.

8-11 Cavity Resonators

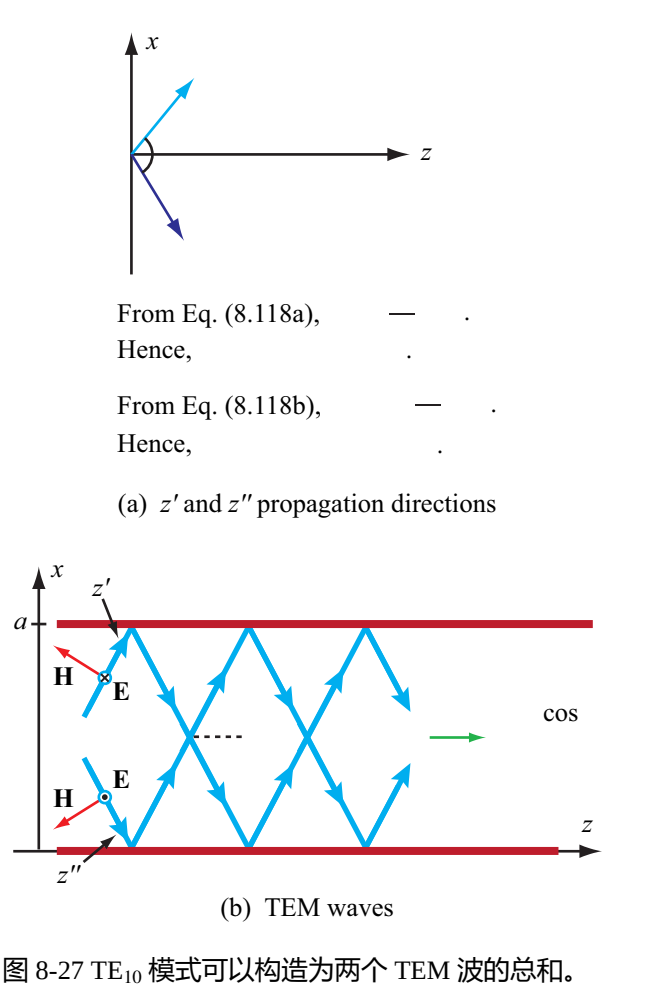
A rectangular waveguide has metal walls on four sides. When the two remaining sides are terminated with conducting walls, the waveguide becomes a cavity. By designing cavities to *resonate* at specific frequencies, they can be used as circuit elements in *microwave oscillators*, *amplifiers*, and *bandpass filters*.

The rectangular cavity shown in **Fig. 8-28(a)** with dimensions $(a \times b \times d)$ is connected to two coaxial cables that feed and extract signals into and from the cavity via input and output probes. As a bandpass filter, the function of a resonant cavity is to block all spectral components of the input signal except for those with frequencies that fall within a narrow band surrounding a specific center frequency f_0 , which is the cavity's *resonant frequency*. Comparison of the spectrum in **Fig. 8-28(b)**, which describes the range of frequencies that might be contained in a typical input signal, with the narrow output spectrum in **Fig. 8-28(c)** demonstrates the filtering action imparted by the cavity.

In a rectangular waveguide, the fields constitute standing waves along the x and y directions and a propagating wave along $\hat{z}$. The terms TE and TM were defined relative to the propagation direction; TE meant that $\mathbf{E}$ was entirely transverse to $\hat{z}$, and TM meant that $\mathbf{H}$ had no component along $\hat{z}$. In a cavity, there is no unique propagation direction, as *no* fields propagate. Instead, standing waves exist along all three directions. Hence, the terms TE and TM need to be modified by defining the fields relative to one of the three rectangular axes. For the sake of consistency, *we will continue to define the transverse direction to be any direction contained in the plane whose normal is $\hat{z}$* .

The TE mode in the rectangular waveguide consists of a single propagating wave whose $\tilde{H}_z$ component is given by Eq. (8.110e) as

$$\tilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.119)$$



在 $f = f_{10}$ 、 $\theta' = 90^\circ$ 处，这意味着波在波导的两个侧壁之间以法向入射来回反射，在 z 方向上没有前进。在频谱的另一端，当 $f \gg f_{10}$ 、 θ' 接近 0 时，波在沿着波导直接传播时变得类似 TEM。

概念问题8-12：对于TE波，主模是 TE_{10} ，但对于TM波，主模是 TM_{11} 。为什么不是 TM_{10} ？

概念问题 8-13：为什么 u_p 超过光速 c 可以接受，而 u_g 则不能？

练习8-13：当接近 f_{mn} 时，TE和TM的波阻抗是什么样的？

答案：在 $f = f_{mn}$ 处， Z_{TE} 看起来像开路，而 Z_{TM} 看起来像短路。

练习 8-14：空心波导中 TE_{10} 模式的 (a) u_p 、(b) u_g 和 (c) $f = 2 f_{10}$ 处的锯齿角 θ' 的值是多少？

答案：(a) $u_p = 1.15c$, (b) $u_g = 0.87c$, (c) $\theta' = 30^\circ$ 。

8-11 空腔谐振器

矩形波导的四个侧面都有金属壁。当其余两个侧面以导电壁终止时，波导就变成了空腔。通过设计在特定频率下谐振的空腔，它们可以用作微波振荡器、放大器和带通滤波器中的电路元件。

矩形腔体如图8-28(a)所示，尺寸为 $(a \times b \times d)$ 连接到两根同轴电缆，通过输入和输出探头将信号馈入腔体或从腔体提取信号。作为带通滤波器，谐振腔的功能是阻挡输入信号的所有频谱分量，除了那些频率落在围绕特定中心频率 f 的窄带内的频谱分量。 f_0 这是腔体的谐振频率。图 8-28(b) 中的频谱描述了典型输入信号中可能包含的频率范围，与图 8-28(c) 中的窄输出频谱的比较表明了腔体所产生的滤波作用。

在矩形波导中，场沿 x 和 y 方向构成驻波，并沿 z 构成传播波。术语 TE 和 TM 是相对于传播方向定义的；TE 表示 $\mathbf{E}$ 完全横向于 z ，TM 表示 $\mathbf{H}$ 没有沿 z 的分量。在空腔中，没有唯一的传播方向，因为没有场传播。相反，驻波沿着所有三个方向都存在。因此，需要通过定义相对于三个矩形轴之一的场来修改术语 TE 和 TM。为了保持一致性，我们将继续将横向方向定义为法线为 $\hat{z}$ 的平面中包含的任何方向。

矩形波导中的 TE 模式由单个传播波组成，其 $\tilde{H}_z$ 分量由方程 1 给出。(8.110e) 为

$$\tilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \quad (8.119)$$

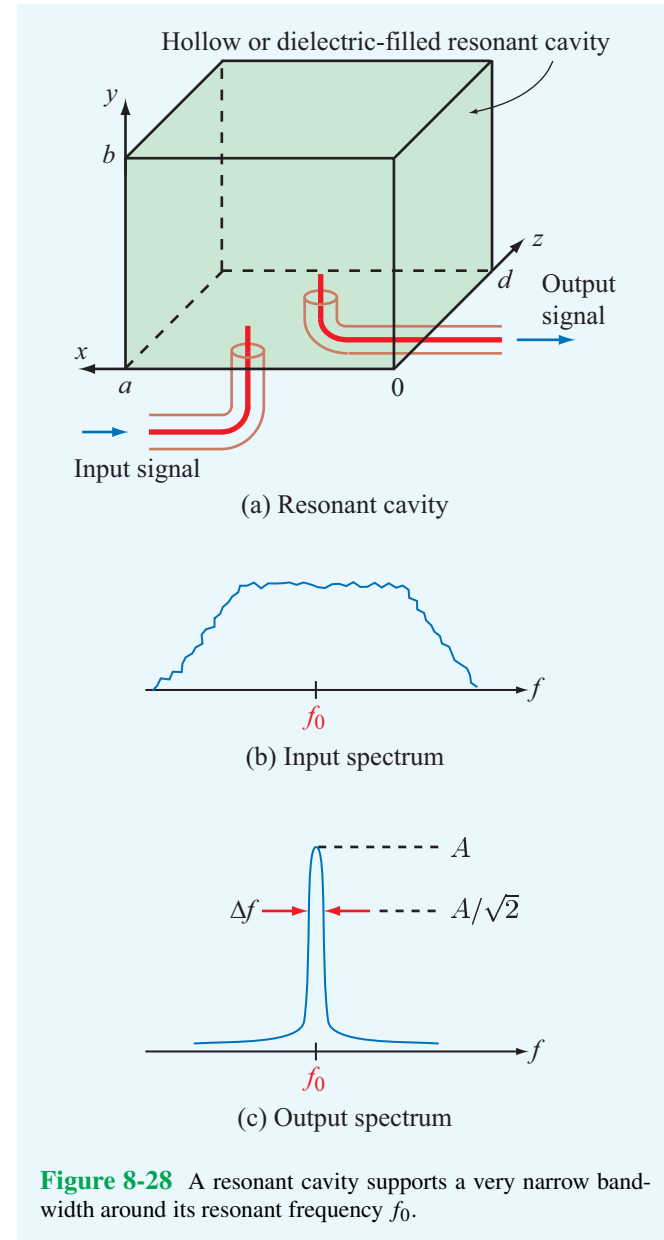


Figure 8-28 A resonant cavity supports a very narrow bandwidth around its resonant frequency f_0 .

where the phase factor $e^{-j\beta z}$ signifies propagation along $+\hat{z}$. Because the cavity has conducting walls at both $z = 0$ and $z = d$, it will contain two such waves: one with amplitude H_0 traveling along $+\hat{z}$ and another with amplitude H_0^- traveling along $-\hat{z}$. Hence,

$$\tilde{H}_z = (H_0 e^{-j\beta z} + H_0^- e^{j\beta z}) \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right). \quad (8.120)$$

Boundary conditions require the normal component of $\tilde{\mathbf{H}}$ to be

zero at a conducting boundary. Consequently, $\tilde{H}_z$ must be zero at $z = 0$ and $z = d$. To satisfy these conditions, it is necessary that $H_0^- = -H_0$ and $\beta d = p\pi$, with $p = 1, 2, 3, \dots$. In this case, Eq. (8.120) becomes

$$\tilde{H}_z = -2jH_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) \sin\left(\frac{p\pi z}{d}\right). \quad (8.121)$$

Given that $\tilde{E}_z = 0$ for the TE modes, all of the other components of $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ can be derived readily through the application of the relationships given by Eq. (8.89). A similar procedure also can be used to characterize cavity modes for the TM case.

8-11.1 Resonant Frequency

The consequence of the quantization condition imposed on β , namely $\beta = p\pi/d$ with p assuming only integer values, is that for any specific set of integer values of (m, n, p) the wave inside the cavity can exist at only a single **resonant frequency**, f_{mnp} , whose value has to satisfy Eq. (8.105). The resulting expression for f_{mnp} is

$$f_{mnp} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}. \quad (8.122)$$

For TE, the indices m and n start at 0, but p starts at 1. The exact opposite applies to TM. By way of an example, the resonant frequency for a TE_{101} mode in a hollow cavity with dimensions $a = 2$ cm, $b = 3$ cm, and $d = 4$ cm is $f_{101} = 8.38$ GHz.

8-11.2 Quality Factor

In the ideal case, if a group of frequencies is introduced into the cavity to excite a certain TE or TM mode, only the frequency component at exactly f_{mnp} of that mode will survive, and all others will attenuate. If a probe is used to couple a sample of the resonant wave out of the cavity, the output signal will be a monochromatic sinusoidal wave at f_{mnp} . In practice, the cavity exhibits a frequency response similar to that shown in **Fig. 8-28(c)**, which is very narrow, but it is not a perfect spike. The bandwidth Δf of the cavity is defined as the frequency range between the two frequencies (on either side of f_{mnp}) at which the amplitude is $1/\sqrt{2}$ of the maximum amplitude (at f_{mnp}). The **normalized bandwidth**, defined as $\Delta f/f_{mnp}$, is approximately equal to the reciprocal of the **quality factor** Q of the cavity:

$$Q \approx \frac{f_{mnp}}{\Delta f}. \quad (8.123)$$

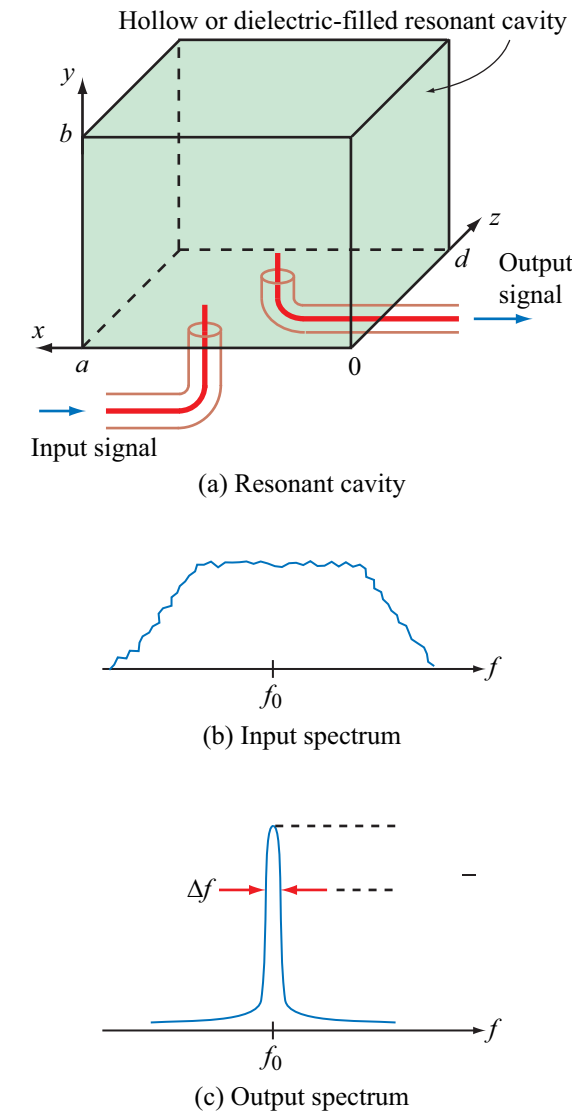


图 8-28 谐振腔在其谐振频率 f_0 附近支持非常窄的带宽。

其中相位因子 $e^{-j\beta z}$ 表示沿 $+\hat{z}$ 的传播。由于空腔在 $z = 0$ 和 $z = d$ 处均具有传导壁，因此它将包含两个这样的波：一个波幅为 H_0 ，沿 $+\hat{z}$ 传播，另一个波幅为 H_0^- ，沿 $-\hat{z}$ 传播。因此，

$$\tilde{H}_z = (H_0 e^{-j\beta z} + H_0^- e^{j\beta z}) \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right). \quad (8.120)$$

边界条件要求 $\tilde{H}$ 的法向分量为

导电边界为零。因此，在 $z = 0$ 和 $z = d$ 处， $\tilde{H}_z$ 必须为零。为了满足这些条件，需要 $H_0^- = -H_0$ 和 $\beta d = p\pi$ ，其中 $p = 1, 2, 3, \dots$ 。在这种情况下，等式 (8.120) 变为

$$\tilde{H}_z = -2jH_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) \sin\left(\frac{p\pi z}{d}\right). \quad (8.121)$$

给定 TE 模式的 $\tilde{E}_z = 0$ ， $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 的所有其他分量可以通过应用方程 1 给出的关系轻松导出。(8.89)。类似的过程也可用于表征 TM 情况的腔模式。

8-11.1 谐振频率

对 β 施加量化条件（即 $\beta = p\pi/d$ ，其中 p 仅假设整数值）的结果是，对于 (m, n, p) 的任何特定整数值集，腔内的波只能以单个谐振频率 f_{mnp} 存在，其值必须满足方程 1。(8.105)。 f_{mnp} 的结果表达式为

$$f_{mnp} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}. \quad (8.122)$$

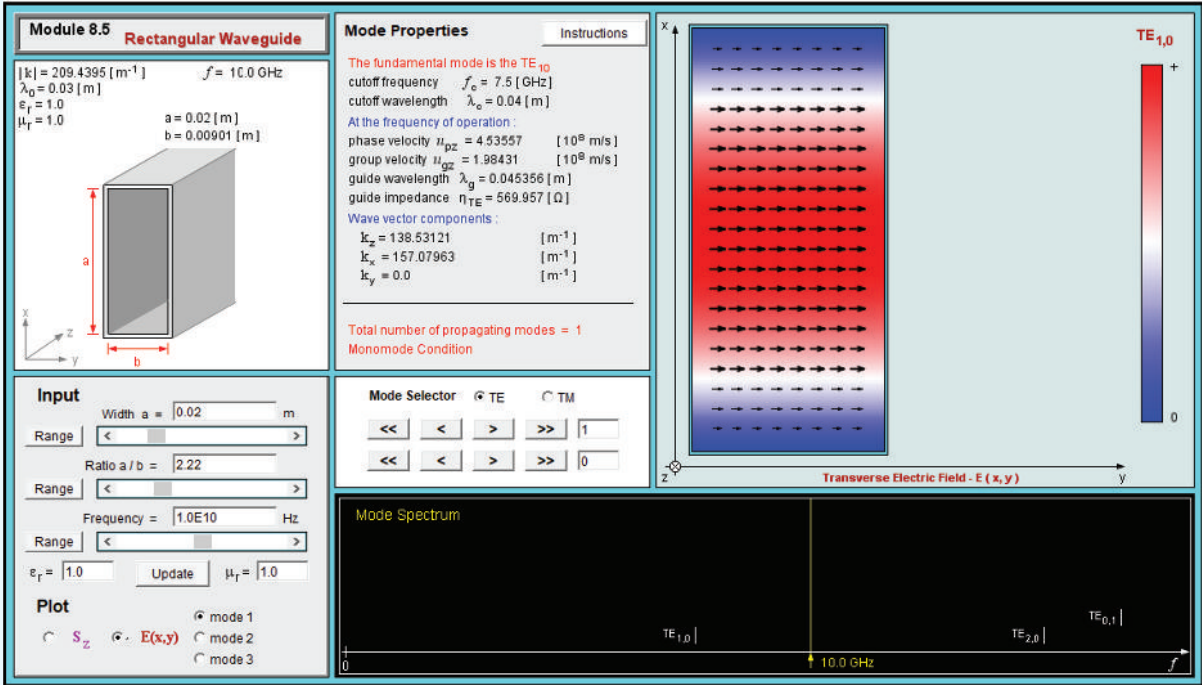
对于 TE，索引 m 和 n 从 0 开始，但 p 从 1 开始。TM 的情况正好相反。举例来说，尺寸为 $a=2$ cm、 $b=3$ cm 和 $d=4$ cm 的空腔中 TE_{101} 模式的谐振频率为 $f_{101} = 8.38$ GHz。

8-11.2 品质因数

在理想情况下，如果将一组频率引入腔体以激发某种 TE 或 TM 模式，则只有该模式的 f_{mnp} 处的频率分量会保留下来，所有其他频率分量都会衰减。如果使用探头将谐振波样本耦合到腔外，则输出信号将是 f_{mnp} 处的单色正弦波。实际上，空腔表现出类似于图 8-28(c) 所示的频率响应，该频率响应非常窄，但它并不是完美的尖峰。腔体的带宽 Δf 定义为两个频率（ f_{mnp} 两侧）之间的频率范围，在此范围内，幅度为最大幅度（在 f_{mnp} 处）的 $1/\sqrt{2}$ 。归一化带宽定义为 $\Delta f/f_{mnp}$ ，约等于腔体品质因数 Q 的倒数：

$$Q \approx \frac{f_{mnp}}{\Delta f}. \quad (8.123)$$

Module 8.5 Rectangular Waveguide Upon specifying the waveguide dimensions, the frequency f , and the mode type (TE or TM) and number, this module provides information about the wave impedance, cutoff frequency, and other wave attributes. It also displays the electric and magnetic field distributions inside the guide.



► The quality factor is defined in terms of the ratio of the energy stored in the cavity volume to the energy dissipated in the cavity walls through conduction. ◀

For an ideal cavity with perfectly conducting walls, no energy loss is incurred, as a result of which Q is infinite and $\Delta f \approx 0$. Metals have very high (but not infinite) conductivities, so a real cavity with metal walls stores most of the energy coupled into it in its volume. However, it also loses some energy to heat conduction. A typical value for Q is on the order of 10,000, which is much higher than can be realized with lumped RLC circuits.

Example 8-11: Q of a Resonant Cavity

The quality factor for a hollow resonant cavity operating in the TE_{101} mode is

$$Q = \frac{1}{\delta_s} \frac{abd(a^2 + d^2)}{[a^3(d + 2b) + d^3(a + 2b)]}, \quad (8.124)$$

where $\delta_s = 1/\sqrt{\pi f_{mnp} \mu_0 \sigma_c}$ is the skin depth and σ_c is the conductivity of the conducting walls. Design a cubic cavity with a TE_{101} resonant frequency of 12.6 GHz and evaluate its bandwidth. The cavity walls are made of copper.

Solution: For $a = b = d$, $m = 1$, $n = 0$, $p = 1$, and $u_{p0} = c = 3 \times 10^8$ m/s, Eq. (8.122) simplifies to

$$f_{101} = \frac{3\sqrt{2} \times 10^8}{2a} \quad (\text{Hz}).$$

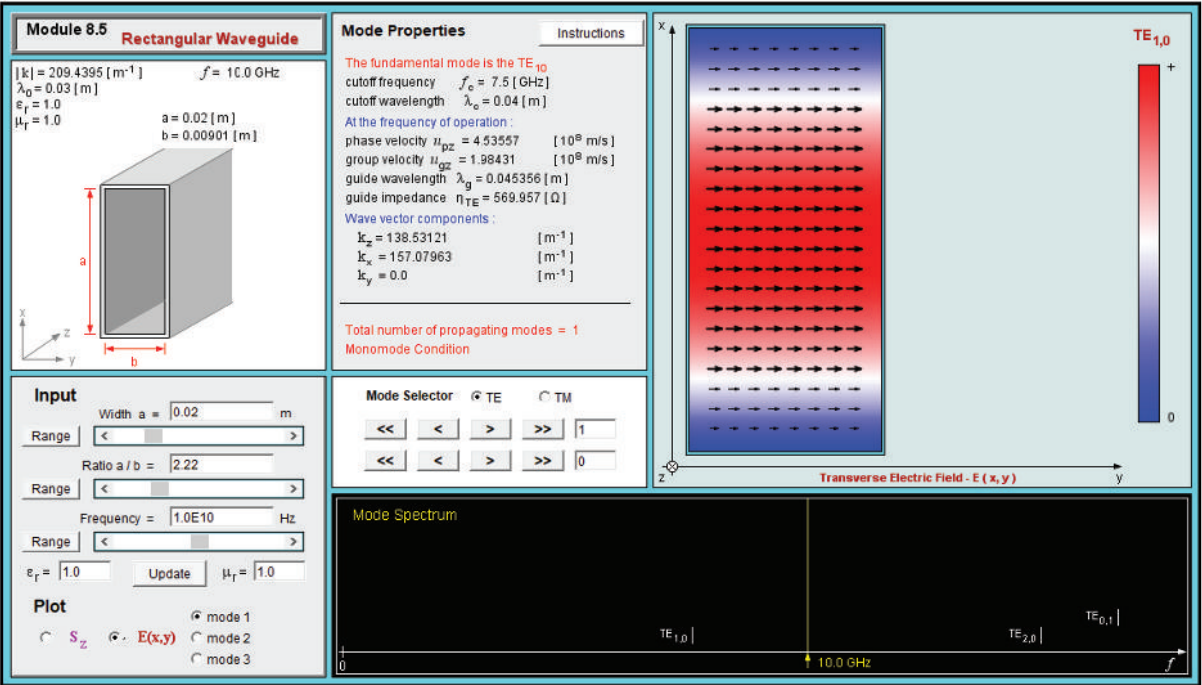
For $f_{101} = 12.6$ GHz, this gives

$$a = 1.68 \text{ cm}.$$

At $f_{101} = 12.6$ GHz, the skin depth for copper (with $\sigma_c = 5.8 \times 10^7$ S/m) is

$$\begin{aligned} \delta_s &= \frac{1}{[\pi f_{101} \mu_0 \sigma_c]^{1/2}} \\ &= \frac{1}{[\pi \times 12.6 \times 10^9 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7]^{1/2}} \\ &= 5.89 \times 10^{-7} \text{ m}. \end{aligned}$$

模块 8.5 矩形波导 在指定波导尺寸、频率 f 以及模式类型 (TE 或 TM) 和数量后, 此模块提供有关波阻抗、截止频率和其他波属性的信息。它还显示导轨内的电场和磁场分布。



► 品质因数是根据腔体积中存储的能量与通过传导在腔壁中耗散的能量的比率来定义的。◀

对于具有理想导电壁的理想空腔, 不会产生能量损失, 因此 Q 无穷大且 $\Delta f \approx 0$ 。金属具有非常高 (但不是无限) 的电导率, 因此具有金属壁的真实空腔将耦合到其体积中的大部分能量存储起来。然而, 它也会因焦耳热而损失一些能量。 Q 的典型值约为 10,000, 该值远高于集总 RLC 电路所能实现的值。

例 8-11: 谐振腔的 Q 值

在 TE_{101} 模式下工作的空心谐振腔的品质因数为

$$Q = \frac{1}{\delta_s} \frac{abd(a^2 + d^2)}{[a^3(d + 2b) + d^3(a + 2b)]}, \quad (8.124)$$

其中 $\delta_s = 1/\sqrt{\pi f_{mnp} \mu_0 \sigma_c}$ 是趋肤深度, σ_c 是导电壁的电导率。设计一个 TE_{101} 谐振频率为 12.6 GHz 的立方腔并评估其带宽。空腔壁由铜制成。

解: 对于 $a = b = d$ 、 $m = 1$ 、 $n = 0$ 、 $p = 1$ 和 $u_{p0} = c = 3 \times 10^8$ m/s, 方程: (8.122) 简化为

$$f_{101} = \frac{3\sqrt{2} \times 10^8}{2a} \quad (\text{Hz}).$$

对于 $f_{101} = 12.6$ GHz, 这给出

$$a = 1.68 \text{ cm}.$$

在 $f_{101} = 12.6$ GHz 时, 铜的趋肤深度 ($\sigma_c = 5.8 \times 10^7$ S/m) 为

$$\begin{aligned} \delta_s &= \frac{1}{[\pi f_{101} \mu_0 \sigma_c]^{1/2}} \\ &= \frac{1}{[\pi \times 12.6 \times 10^9 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7]^{1/2}} \\ &= 5.89 \times 10^{-7} \text{ m}. \end{aligned}$$

Upon setting $a = b = d$ in Eq. (8.124), the expression for Q of a cubic cavity becomes

$$Q = \frac{a}{3\delta_s} = \frac{1.68 \times 10^{-2}}{3 \times 5.89 \times 10^{-7}} \approx 9,500.$$

Hence, the cavity bandwidth is

$$\Delta f \approx \frac{f_{101}}{Q} \approx \frac{12.6 \times 10^9}{9,500} \approx 1.3 \text{ MHz}.$$

Chapter 8 Summary

Concepts

- The relations describing the reflection and transmission behavior of a plane EM wave at the boundary between two different media are the consequence of satisfying the conditions of continuity of the tangential components of **E** and **H** across the boundary.
- Snell’s laws state that $\theta_i = \theta_t$ and

$$\sin \theta_t = (n_1/n_2) \sin \theta_i.$$

For media such that $n_2 < n_1$, the incident wave is reflected totally by the boundary when $\theta_i \geq \theta_c$, where θ_c is the critical angle given by $\theta_c = \sin^{-1}(n_2/n_1)$.

- By successive multiple reflections, light can be guided through optical fibers. The maximum data rate of digital pulses that can be transmitted along optical fibers is dictated by modal dispersion.

- At the Brewster angle for a given polarization, the incident wave is transmitted totally across the boundary. For nonmagnetic materials, the Brewster angle exists for parallel polarization only.
- Any plane wave incident on a plane boundary can be synthesized as the sum of a perpendicularly polarized wave and a parallel polarized wave.
- Transmission-line equivalent models can be used to characterize wave propagation, reflection by, and transmission through boundaries between different media.
- Waves can travel through a metal waveguide in the form of transverse electric (TE) and transverse magnetic (TM) modes. For each mode, the waveguide has a cutoff frequency below which a wave cannot propagate.
- A cavity resonator can support standing waves at specific resonant frequencies.

Important Terms

Provide definitions or explain the meaning of the following terms:

ω - β diagram
acceptance angle θ_a
angles of incidence, reflection, and transmission
Brewster angle θ_B
cladding
critical angle θ_c
cutoff frequency f_{mn}
cutoff wavenumber k_c
dominant mode
evanescent wave
fiber core
grazing incidence
group velocity u_g

index of refraction n
modal dispersion
modes
optical fibers
parallel polarization
perpendicular polarization
phase-matching condition
plane of incidence
polarizing angle
quality factor Q
reflection coefficient Γ
reflectivity (reflectance) R
refraction angle
resonant cavity

resonant frequency
Snell’s laws
standing-wave ratio S
surface wave
total internal reflection
transmission coefficient τ
transmissivity (transmittance) T
transverse electric (TE) polarization
transverse magnetic (TM) polarization
unbounded-medium wavenumber
unpolarized
wavefront

在等式中设置 a = b = d 后： (8.124), 立方腔的 Q 表达式变为

$$Q = \frac{a}{3\delta_s} = \frac{1.68 \times 10^{-2}}{3 \times 5.89 \times 10^{-7}} \approx 9,500.$$

因此, 腔体带宽为

$$\Delta f \approx \frac{f_{101}}{Q} \approx \frac{12.6 \times 10^9}{9,500} \approx 1.3 \text{ MHz}.$$

第8章总结

概念

- 描述平面电磁波在两种不同介质边界处的反射和传输行为的关系是满足 E 和 H 跨越边界的切向分量连续性条件的结果。
- 斯涅尔定律指出 $\theta_i = \theta_t$ 和

$$\sin \theta_t = (n_1/n_2) \sin \theta_i.$$

对于 $n_2 < n_1$ 这样的介质, 当 $\theta_i \geq \theta_c$ 时, 入射波被边界全反射, 其中 θ_c 是由 $\theta_c = \sin^{-1}(n_2/n_1)$ 给出的临界角。

- 通过连续的多次反射, 光可以通过光纤被引导。可以沿光纤传输的数字脉冲的最大数据速率由模态色散决定。

- 在给定偏振的布鲁斯特角下, 入射波完全穿过边界传播。对于非磁性材料, 布儒斯特角仅存在于平行极化。
- 任何入射到平面边界上的平面波都可以合成为垂直偏振波和平行偏振波之和。

- 传输线等效模型可用于表征不同介质之间边界的波传播、反射和传输。

- 波可以以横电 (TE) 和横磁 (TM) 模式的形式在金属波导中传播。对于每种模式, 波导都有一个截止频率, 低于该频率波就无法传播。
- 空腔谐振器可以支持特定谐振频率的驻波。

重要条款

提供以下术语的定义或解释其含义:

ω - β 图接收角 θ_a 入射角、
反射角和透射角 布儒斯特角
 θ_B 包层临界角 θ_c 截止频率 f_{mn}
截止波数 k_c 主模倏逝波
光纤芯掠入射群速度 u_g

折射率 n 模色散模式 光
纤 平行偏振 垂直偏振
相位匹配条件 入射平面
偏振角 品质因数 Q 反射
系数 Γ 反射率 (反射率
) R 折射角 谐振腔

谐振频率 斯涅尔定律 驻波比 S
表面波 全内反射 透射系数 τ
透射率 (透射率) T 横向电 (TE)
偏振 横向磁 (TM) 偏振 无界
介质波数 非偏振波前

Mathematical and Physical Models

Normal Incidence

$$\Gamma = \frac{E_0^r}{E_0^i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\tau = \frac{E_0^t}{E_0^i} = \frac{2\eta_2}{\eta_2 + \eta_1}$$

$$\tau = 1 + \Gamma$$

$$\Gamma = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} \quad (\text{if } \mu_1 = \mu_2)$$

Snell's Laws

$$\theta_i = \theta_r$$

$$\frac{\sin \theta_i}{\sin \theta_t} = \frac{u_{p2}}{u_{p1}} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}}$$

Oblique Incidence

Perpendicular Polarization

$$\Gamma_{\perp} = \frac{E_{\perp 0}^r}{E_{\perp 0}^i} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$$

$$\tau_{\perp} = \frac{E_{\perp 0}^t}{E_{\perp 0}^i} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$$

Parallel Polarization

$$\Gamma_{\parallel} = \frac{E_{\parallel 0}^r}{E_{\parallel 0}^i} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

$$\tau_{\parallel} = \frac{E_{\parallel 0}^t}{E_{\parallel 0}^i} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

Brewster Angle

$$\theta_{B\parallel} = \sin^{-1} \sqrt{\frac{1}{1 + (\epsilon_1/\epsilon_2)}} = \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

Waveguides

$$\beta = \sqrt{\omega^2 \mu \epsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}}$$

$$u_p u_g = u_{p0}^2$$

$$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_{mn}/f)^2}}$$

$$Z_{TM} = \eta \sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}$$

Resonant Cavity

$$f_{mnp} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

$$Q \approx \frac{f_{mnp}}{\Delta f}$$

PROBLEMS

Section 8-1: Reflection and Transmission at Normal Incidence

*8.1 A plane wave in air with an electric field amplitude of 20 V/m is incident normally upon the surface of a lossless, nonmagnetic medium with $\epsilon_r = 25$. Determine:

- (a) the reflection and transmission coefficients,

(b) the standing-wave ratio in the air medium, and

(c) the average power densities of the incident, reflected, and transmitted waves.

8.2 A plane wave traveling in medium 1 with $\epsilon_{r1} = 2.25$ is normally incident upon medium 2 with $\epsilon_{r2} = 4$. Both media are made of nonmagnetic, nonconducting materials. The electric field of the incident wave is given by

$$\mathbf{E}^i = \hat{\mathbf{y}} 8 \cos(6\pi \times 10^9 t - 30\pi x) \quad (\text{V/m}).$$

(a) Obtain time-domain expressions for the electric and magnetic fields in each of the two media.

* Answer(s) available in Appendix E.

数学与物理模型

垂直入射

$$\Gamma = \frac{E_0^r}{E_0^i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\tau = \frac{E_0^t}{E_0^i} = \frac{2\eta_2}{\eta_2 + \eta_1}$$

$$\tau = 1 + \Gamma$$

$$\Gamma = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} \quad (\text{if } \mu_1 = \mu_2)$$

斯涅尔定律

$$\theta_i = \theta_r$$

$$\frac{\sin \theta_i}{\sin \theta_t} = \frac{u_{p2}}{u_{p1}} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}}$$

斜入射

垂直极化

$$\Gamma_{\perp} = \frac{E_{\perp 0}^r}{E_{\perp 0}^i} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$$

$$\tau_{\perp} = \frac{E_{\perp 0}^t}{E_{\perp 0}^i} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$$

平行极化

$$\Gamma_{\parallel} = \frac{E_{\parallel 0}^r}{E_{\parallel 0}^i} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

$$\tau_{\parallel} = \frac{E_{\parallel 0}^t}{E_{\parallel 0}^i} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

问题

第8-1节 垂直入射时的反射与透射

*8.1 空气中电场幅度为20 V/m的平面波，垂直入射到相对介电率 $\epsilon_r = 25$ 的无损非磁性介质表面。求：

(a) 反射系数和透射系数，(b) 空气介质中的驻波比，以及

* 标记的习题在附录 E 提供答案。

布儒斯特角

$$\theta_{B\parallel} = \sin^{-1} \sqrt{\frac{1}{1 + (\epsilon_1/\epsilon_2)}} = \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

波导

$$\beta = \sqrt{\omega^2 \mu \epsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}}$$

$$u_p u_g = u_{p0}^2$$

$$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_{mn}/f)^2}}$$

$$Z_{TM} = \eta \sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}$$

谐振腔

$$f_{mnp} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

$$Q \approx \frac{f_{mnp}}{\Delta f}$$

(c) 入射波、反射波和透射波的平均功率密度。

8.2 在 $\epsilon_{r1} = 2.25$ 的介质 1 中传播的平面波垂直入射到 $\epsilon_{r2} = 4$ 的介质 2 上。两种介质均由非磁性、非导电材料制成。入射波的电场由下式给出

$$\mathbf{E}^i = \hat{\mathbf{y}} 8 \cos(6\pi \times 10^9 t - 30\pi x) \quad (\text{V/m}).$$

(a) 获得两种介质中电场和磁场的时域表达式。

(b) Determine the average power densities of the incident, reflected and transmitted waves.

8.3 A plane wave traveling in a medium with $\epsilon_{r1} = 9$ is normally incident upon a second medium with $\epsilon_{r2} = 4$. Both media are made of nonmagnetic, nonconducting materials. The magnetic field of the incident plane wave is given by

$$\mathbf{H}^i = \hat{\mathbf{z}} 2 \cos(2\pi \times 10^9 t - ky) \quad (\text{A/m}).$$

(a) Obtain time-domain expressions for the electric and magnetic fields in each of the two media.

***8.4** Determine the average power densities of the incident, reflected, and transmitted waves.

8.4 A 200 MHz, left-hand circularly polarized plane wave with an electric field modulus of 5 V/m is normally incident in air upon a dielectric medium with $\epsilon_r = 4$ and occupies the region defined by $z \geq 0$.

(a) Write an expression for the electric field phasor of the incident wave given that the field is a positive maximum at $z = 0$ and $t = 0$.

(b) Calculate the reflection and transmission coefficients.

(c) Write expressions for the electric field phasors of the reflected wave, the transmitted wave, and the total field in the region $z \leq 0$.

(d) Determine the percentages of the incident average power reflected by the boundary and transmitted into the second medium.

8.5 Repeat Problem 8.4, but replace the dielectric medium with a poor conductor characterized by $\epsilon_r = 2.25$, $\mu_r = 1$, and $\sigma = 10^{-4} \text{ S/m}$.

8.6 A 50 MHz plane wave with electric field amplitude of 50 V/m is normally incident in air onto a semi-infinite, perfect dielectric medium with $\epsilon_r = 36$. Determine:

***8.7** (a) Γ ,

(b) the average power densities of the incident and reflected waves, and

(c) the distance in the air medium from the boundary to the nearest minimum of the electric field intensity, $|\mathbf{E}|$.

***8.8** What is the maximum amplitude of the total electric field in the air medium of Problem 8.6, and at what nearest distance from the boundary does it occur?

8.8 Repeat Problem 8.6, but replace the dielectric medium with a conductor with $\epsilon_r = 1$, $\mu_r = 1$, and $\sigma = 2.78 \times 10^{-3} \text{ S/m}$.

***8.9** The three regions shown in **Fig. P8.9** contain perfect dielectrics. For a wave in medium 1, incident normally upon

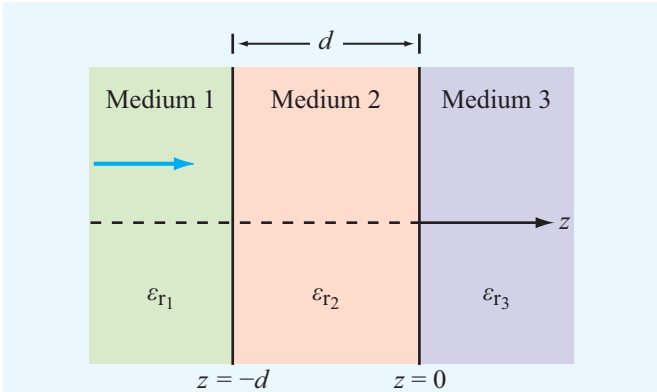


Figure P8.9 Dielectric layers for Problems 8.9 to 8.11.

the boundary at $z = -d$, what combination of ϵ_{r2} and d produces no reflection? Express your answers in terms of ϵ_{r1} , ϵ_{r3} and the oscillation frequency of the wave, f .

8.10 For the configuration shown in **Fig. P8.9**, use transmission-line equations (or the Smith chart) to calculate the input impedance at $z = -d$ for $\epsilon_{r1} = 1$, $\epsilon_{r2} = 9$, $\epsilon_{r3} = 4$, $d = 1.2 \text{ m}$, and $f = 50 \text{ MHz}$. Also determine the fraction of the incident average power density reflected by the structure. Assume all media are lossless and nonmagnetic.

***8.11** Repeat Problem 8.10, but interchange ϵ_{r1} and ϵ_{r3} .

8.12 Orange light of wavelength $0.61 \mu\text{m}$ in air enters a block of glass with $\epsilon_r = 1.44$. What color would it appear to a sensor embedded in the glass? The wavelength ranges of colors are violet (0.39 to $0.45 \mu\text{m}$), blue (0.45 to $0.49 \mu\text{m}$), green (0.49 to $0.58 \mu\text{m}$), yellow (0.58 to $0.60 \mu\text{m}$), orange (0.60 to $0.62 \mu\text{m}$), and red (0.62 to $0.78 \mu\text{m}$).

***8.13** A plane wave of unknown frequency is normally incident in air upon the surface of a perfect conductor. Using an electric-field meter, it was determined that the total electric field in the air medium is always zero when measured at a distance of 2 m from the conductor surface. Moreover, no such nulls were observed at distances closer to the conductor. What is the frequency of the incident wave?

8.14 Consider a thin film of soap in air under illumination by yellow light with $\lambda = 0.6 \mu\text{m}$ in vacuum. The film is treated as a planar dielectric slab with $\epsilon_r = 1.72$ and is surrounded on both sides by air. What film thickness would produce strong reflection of the yellow light at normal incidence?

***8.15** A 5 MHz plane wave with electric field amplitude of 10 (V/m) is normally incident in air onto the plane surface of a semi-infinite conducting material with $\epsilon_r = 4$, $\mu_r = 1$,

(b) 确定入射波、反射波和透射波的平均功率密度。

8.3 在 $\epsilon_{r1} = 9$ 的介质中传播的平面波垂直入射到 $\epsilon_{r2} = 4$ 的第二介质上。两种介质均由非磁性、非导电材料制成。入射平面波的磁场由下式给出

$$\mathbf{H}^i = \hat{\mathbf{z}} 2 \cos(2\pi \times 10^9 t - ky) \quad (\text{A/m}).$$

(a) 获得两种介质中电场和磁场的时域表达式。* (b) 确定入射波、反射波和透射波的平均功率密度。

8.4 电场模量为 5 V/m 的 200 MHz 左旋圆极化平面波垂直在空气中入射到 $\epsilon_r = 4$ 的介电介质上，并占据由 $z \geq 0$ 定义的区域。

(a) 假设电场在 $z = 0$ 和 $t = 0$ 处为正最大值，写出入射波的电场相量表达式。(b) 计算反射和透射系数。(c) 写出反射波、透射波和 $z \leq 0$ 区域中总场的电场相量表达式。(d) 确定边界反射并传输到第二介质中的入射平均功率的百分比。

8.5 重复问题 8.4，但用特征为 $\epsilon_r = 2.25$ 、 $\mu_r = 1$ 和 $\sigma = 10^{-4} \text{ S/m}$ 的不良导体替换介电介质。

8.6 电场幅度为 50 V/m 的 50 MHz 平面波垂直在空气中入射到 $\epsilon_r = 36$ 的半无限理想介电介质上。确定：

* (a) Γ ，(b) 入射波和反射波的平均功率密度，以及 (c) 空气介质中从边界到电场强度最接近最小值的距离， $|\mathbf{E}|$ 。

* 8.7 问题 8.6 的空气介质中总电场的最大幅度是多少，它出现在距离边界最近的距离处？

8.8 重复问题 8.6，但将介电介质替换为具有 $\epsilon_r = 1$ 、 $\mu_r = 1$ 和 $\sigma = 2.78 \times 10^{-3} \text{ S/m}$ 的导体。

* 8.9 图 P8.9 所示的三个区域包含理想的电介质。对于介质 1 中的波，法线入射

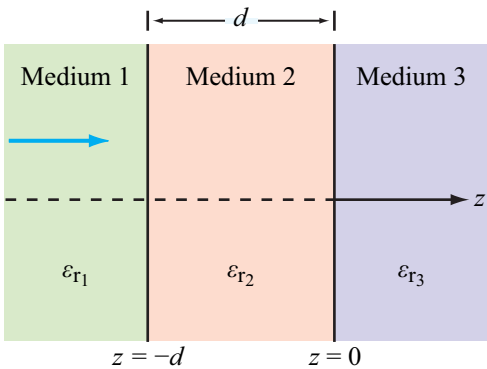


图 P8.9 问题 8.9 至 8.11 的介电层。

$z = -d$ 处的边界， ϵ_{r2} 和 d 的哪种组合不会产生反射？用 ϵ_{r1} 、 ϵ_{r3} 和波的振荡频率 f 表达您的答案。

8.10 对于图 P8.9 所示的配置，使用传输线方程（或史密斯圆图）计算 $\epsilon_{r1} = 1$ 、 $\epsilon_{r2} = 9$ 、 $\epsilon_{r3} = 4$ 、 $d = 1.2 \text{ m}$ 和 $f = 50 \text{ MHz}$ 时 $z = -d$ 处的输入阻抗。还确定结构反射的入射平均功率密度的分数。假设所有介质都是无损且非磁性的。

* 8.11 重复问题 8.10，但互换 ϵ_{r1} 和 ϵ_{r3} 。

8.12 波长 $0.61 \mu\text{m}$ 的橙光 μ 空气中的 m 进入一块玻璃 $\epsilon_r = 1.44$ 。嵌入玻璃中的传感器会呈现什么颜色？颜色的波长范围是紫色 (0.39 至 $0.45 \mu\text{m}$)，蓝色 (0.45 至 $0.49 \mu\text{m}$)，绿色 (0.49 至 $0.58 \mu\text{m}$)，黄色 (0.58 至 $0.60 \mu\text{m}$)，橙色 (0.60 至 $0.62 \mu\text{m}$) 和红色 (0.62 至 $0.78 \mu\text{m}$)。

* 8.13 频率未知的平面波垂直在空气中入射到理想导体的表面。使用电场计，确定在距离导体表面 2 m 处测量时，空气介质中的总电场始终为零。此外，在距离导体更近的地方没有观察到这样的零点。入射波的频率是多少？

8.14 考虑真空中 $\lambda = 0.6 \mu\text{m}$ 黄光照射下空气中的一层肥皂薄膜。该薄膜被视为 $\epsilon_r = 1.72$ 的平面介电板，并且两侧都被空气包围。多大的薄膜厚度会在法向入射时产生强烈的黄光反射？

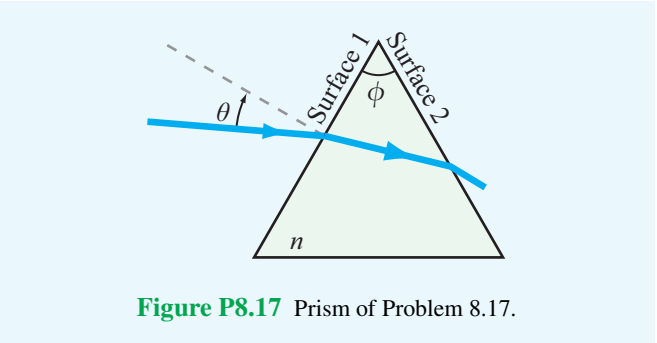
* 8.15 电场幅度为 10 (V/m) 的 5 MHz 平面波垂直在空气中入射到 $\epsilon_r = 4$ 、 $\mu_r = 1$ 的半无限导电材料的平面上，

and $\sigma = 100$ (S/m). Determine the average power dissipated (lost) per unit cross-sectional area in a 2 mm penetration of the conducting medium.

8.16 A 0.5 MHz antenna carried by an airplane flying over the ocean surface generates a wave that approaches the water surface in the form of a normally incident plane wave with an electric-field amplitude of 3,000 (V/m). Seawater is characterized by $\epsilon_r = 72$, $\mu_r = 1$, and $\sigma = 4$ (S/m). The plane is trying to communicate a message to a submarine submerged at a depth d below the water surface. If the submarine's receiver requires a minimum signal amplitude of 0.01 (μ V/m), what is the maximum depth d where successful communication is still possible?

Sections 8-2 and 8-3: Snell's Laws and Fiber Optics

***8.17** A light ray is incident on a prism in air at an angle θ , as shown in Fig. P8.17. The ray is refracted at the first surface and again at the second surface. In terms of the apex angle ϕ of the prism and its index of refraction n , determine the smallest value of θ for which the ray will emerge from the other side. Find this minimum θ for $n = 1.4$ and $\phi = 60^\circ$.



8.18 For some types of glass, the index of refraction varies with wavelength. A prism is made of a material with

$$n = 1.71 - \frac{4}{30} \lambda_0 \quad (\lambda_0 \text{ in } \mu\text{m}),$$

where λ_0 is the wavelength in vacuum, and is used to disperse white light, as shown in Fig. P8.18. The white light is incident at an angle of 50° , the wavelength λ_0 of red light is $0.7 \mu\text{m}$, and that of violet light is $0.4 \mu\text{m}$. Determine the angular dispersion in degrees.

***8.19** The two prisms in Fig. P8.19 are made of glass with $n = 1.5$. What fraction of the power density carried by the ray incident upon the top prism emerges from the bottom prism? Neglect multiple internal reflections.

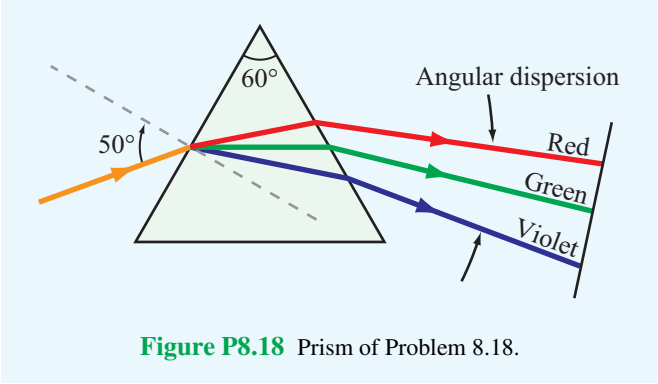


Figure P8.18 Prism of Problem 8.18.

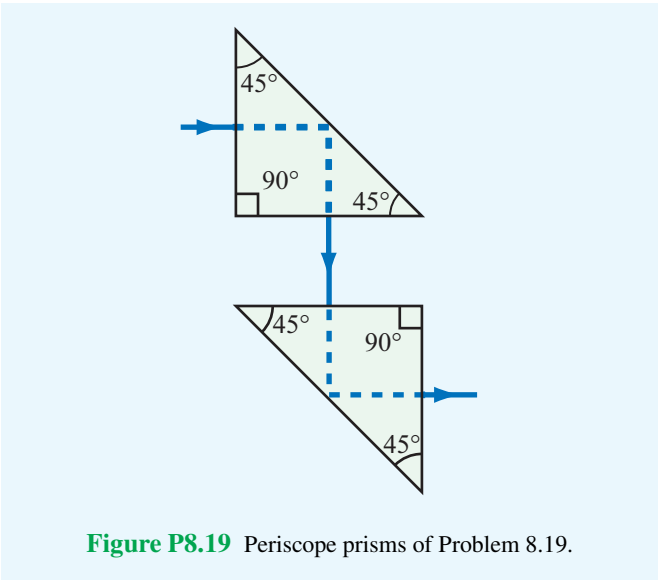


Figure P8.19 Periscope prisms of Problem 8.19.

8.20 A parallel-polarized plane wave is incident from air at an angle $\theta_i = 30^\circ$ onto a pair of dielectric layers, as shown in Fig. P8.20.

- (a) Determine the angles of transmission θ_2 , θ_3 , and θ_4 .
- (b) Determine the lateral distance d .

8.21 A light ray incident at 45° passes through two dielectric materials with the indices of refraction and thicknesses given in Fig. P8.21. If the ray strikes the surface of the first dielectric at a height of 2 cm, at what height will it strike the screen?

***8.22** Figure P8.22 depicts a beaker containing a block of glass on the bottom and water over it. The glass block contains a small air bubble at an unknown depth below the water surface. When viewed from above at an angle of 60° , the air bubble appears at a depth of 6.81 cm. What is the true depth of the air bubble?

和 $\sigma = 100$ (S/m) 。确定导电介质渗透 2 毫米时每单位横截面积消耗 (损失) 的平均功率。

8.16 飞越海洋表面的飞机携带的 0.5 MHz 天线产生的波以垂直入射平面波的形式接近水面，电场幅度为 3,000 (V/m) 。海水的特征为 $\epsilon_r = 72$ 、 $\mu_r = 1$ 和 $\sigma = 4$ (S/m)。这架飞机正试图向潜入水面以下深度 d 的潜艇传达信息。如果潜艇的接收器要求的最小信号幅度为 0.01 (μ V/m)，那么仍然可以成功通信的最大深度 d 是多少？

第 8-2 节和 8-3 节：斯涅尔定律和光纤

***8.17** 光线以 θ 角度入射到空气中的棱镜上，如图P8.17所示。光线在第一表面折射，并在第二表面再次折射。根据棱镜的顶角 ϕ 及其折射率 n ，确定光线从另一侧射出的 θ 的最小值。求 $n = 1.4$ 和 $\phi = 60^\circ$ 的最小值 θ 。

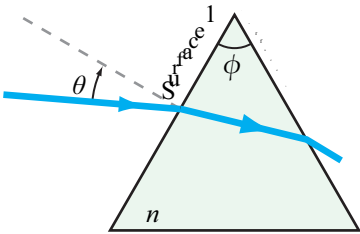


图 P8.17 问题 8.17 的棱镜。

8.18 对于某些类型的玻璃，折射率随波长而变化。棱镜由以下材料制成

$$n = 1.71 - \frac{4}{30} \lambda_0 \quad (\lambda_0 \text{ in } \mu\text{m}),$$

其中 λ_0 为真空中的波长，用于色散白光，如图P8.18所示。白光以 50° 的角度入射，红光的波长 λ_0 为 $0.7 \mu\text{m}$ ，紫光的波长为 $0.4 \mu\text{m}$ 。确定角色散（以度为单位）。

***8.19** 图 P8.19 中的两个棱镜由 $n = 1.5$ 的玻璃制成。入射到顶部棱镜的光线所携带的功率密度从底部棱镜射出的功率密度的几分之几？忽略多重内部反射。

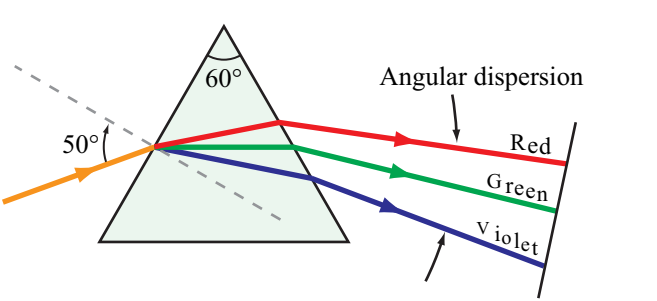


图 P8.18 问题 8.18 的棱镜。

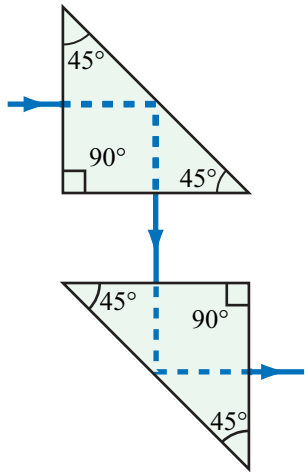


图 P8.19 问题 8.19 的潜望镜棱镜。

8.20 平行偏振平面波从空气以 $\theta_i = 30^\circ$ 角度入射到一对介电层上，如图P8.20所示。

- (a) 确定传输角 θ_2 、 θ_3 和 θ_4 。(b) 确定横向距离 d 。

8.21 以 45° 入射的光线穿过两种介电材料，其折射率和厚度如图 P8.21 所示。如果射线以 2 厘米的高度撞击第一个电介质的表面，那么它会以什么高度撞击屏幕？

***8.22** 图 P8.22 描绘了一个烧杯，底部装有一块玻璃，上面有水。玻璃块在水面以下未知深度处包含一个小气泡。当从上方以 60° 角度观察时，气泡出现的深度为 6.81 厘米。气泡的真实深度是多少？

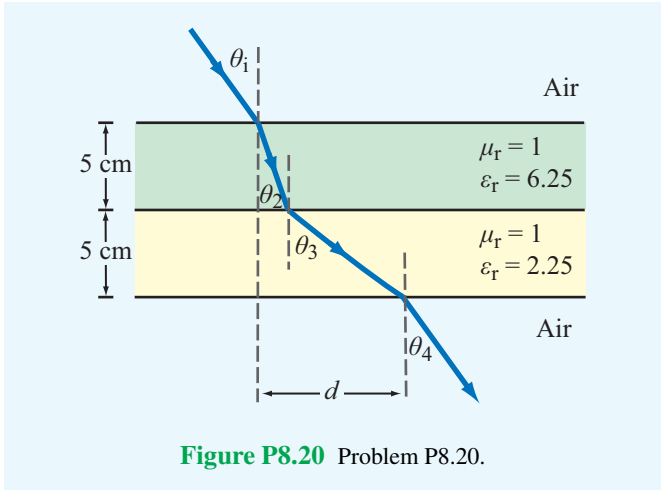


Figure P8.20 Problem P8.20.

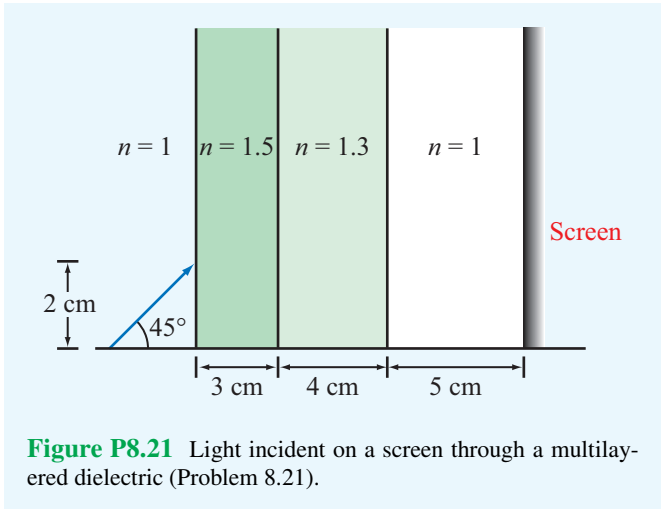


Figure P8.21 Light incident on a screen through a multilayered dielectric (Problem 8.21).

8.23 A glass semicylinder with $n = 1.5$ is positioned such that its flat face is horizontal, as shown in Fig. P8.23, and its horizontal surface supports a drop of oil, as also shown. When light is directed radially toward the oil, total internal reflection occurs if θ exceeds 53° . What is the index of refraction of the oil?

***8.24** A penny lies at the bottom of a water fountain at a depth of 30 cm. Determine the diameter of a piece of paper that when placed to float on the surface of the water directly above the penny, would totally obscure the penny from view. Treat the penny as a point and assume that $n = 1.33$ for water.

8.25 Suppose that the optical fiber of Example 8-5 is submerged in water (with $n = 1.33$) instead of air. Determine θ_a and f_p in that case.

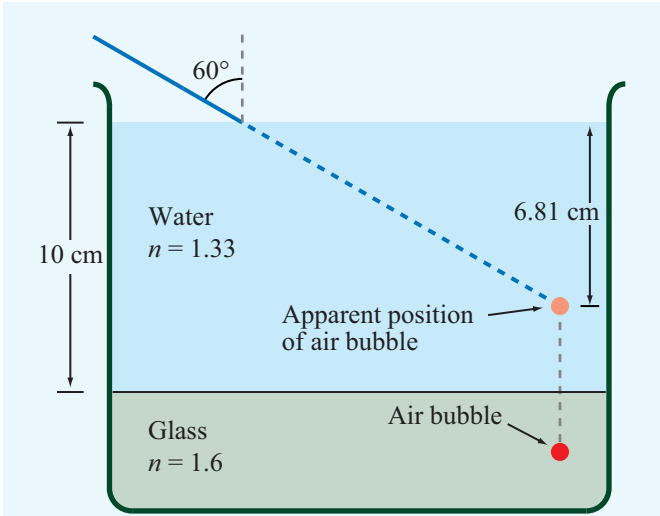


Figure P8.22 Apparent position of the air bubble in Problem 8.22.

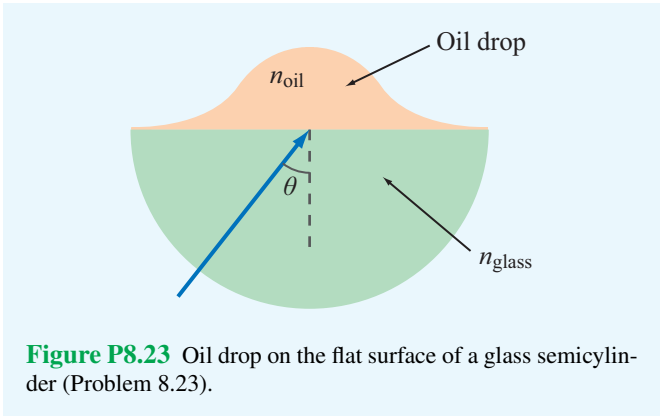


Figure P8.23 Oil drop on the flat surface of a glass semicylinder (Problem 8.23).

***8.26** Equation (8.45) was derived for the case where the light incident upon the sending end of the optical fiber extends over the entire acceptance cone shown in Fig. 8-12(b). Suppose the incident light is constrained to a narrower range extending between normal incidence and θ' , where $\theta' < \theta_a$.

- (a) Obtain an expression for the maximum data rate f_p in terms of θ' .
- (b) Evaluate f_p for the fiber of Example 8-5 when $\theta' = 5^\circ$.

Sections 8-4 and 8-5: Reflection and Transmission at Oblique Incidence

8.27 A 2-km long optical fiber uses a fiber core with $n_f = 1.6$ and a cladding with $n_c = 1.57$. Compute the maximum data

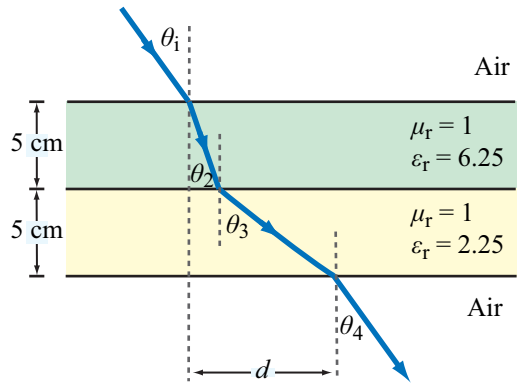


Figure P8.20 Problem P8.20.

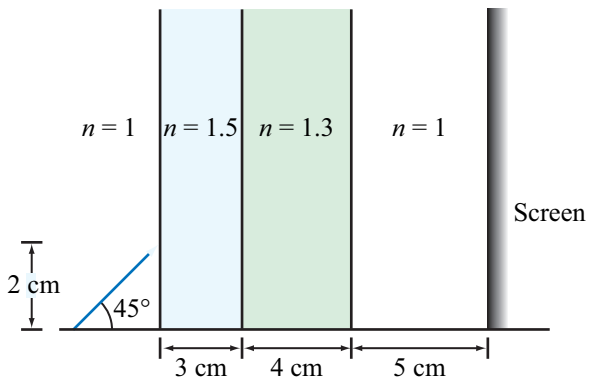


图 P8.21 光通过多层电介质入射到屏幕上 (问题 8.21)。

8.23 将 $n = 1.5$ 的玻璃半圆柱体放置在其平面水平的位置，如图 P8.23 所示，并且其水平表面支撑一滴油，也如图所示。当光径向射向油时，如果 θ 超过 53° ，则会发生全内反射。油的折射率是多少？

*8.24 一枚硬币位于 30 厘米深的喷泉底部。确定一张纸的直径，当这张纸漂浮在硬币正上方的水面上时，会完全遮住硬币。将硬币视为一个点，并假设水的 $n = 1.33$ 。

8.25 假设例 8-5 中的光纤浸入水中 ($n = 1.33$) 而不是空气中。在这种情况下确定 θ_a 和 f_p 。

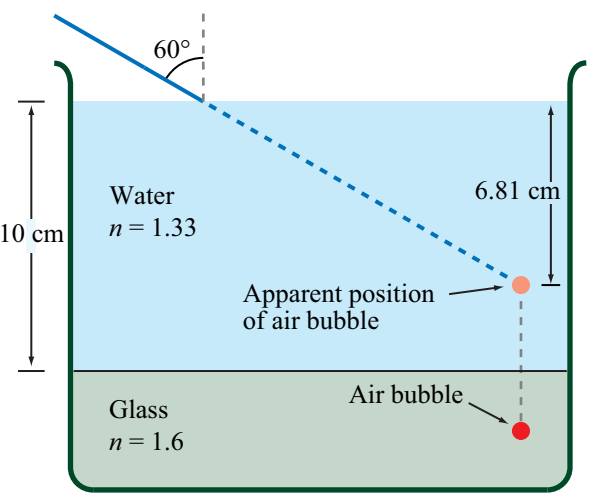


图 P8.22 问题 8.22 中气泡的明显位置。

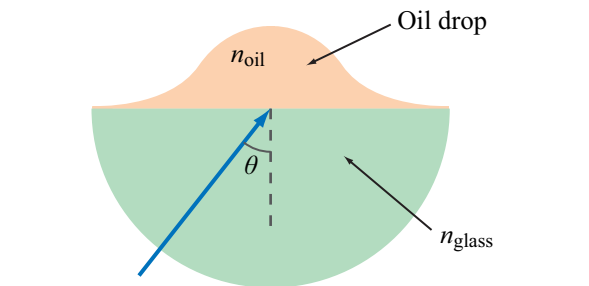


图 P8.23 玻璃半圆柱体平坦表面上的油滴 (问题 8.23)。

*8.26 方程 (8.45) 是针对入射到光纤发送端的光延伸到图 8-12 (b) 所示的整个接收锥体的情况推导的。假设入射光被限制在法向入射和 θ' 之间延伸的较窄范围内，其中 $\theta' < \theta_a$ 。

- (a) 获得以 θ' 表示的最大数据速率 f_p 的表达式。(b) 当 $\theta' = 5^\circ$ 时，计算例 8-5 中光纤的 f_p 。

第 8-4 节和第 8-5 节：斜入射时的反射和透射

8.27 2 公里长的光纤使用 $n_f = 1.6$ 的纤芯和 $n_c = 1.57$ 的包层。计算最大数据

rate and confirm your result using Module 8.2. The operating frequency is 100 THz.

8.28 A 3-km long optical fiber uses a fiber core with $n_f = 1.51$ and a cladding with $n_c = 1.48$. Compute the maximum data rate and confirm your result using Module 8.2. The operating frequency is 300 THz.

8.29 A plane wave in air with

$$\widetilde{\mathbf{E}}_i = \hat{\mathbf{y}}\,20e^{-j(3x+4z)} \quad (\text{V/m})$$

is incident upon the planar surface of a dielectric material with $\epsilon_r = 4$, and it occupies the half-space $z \geq 0$. Determine:

- (a) the polarization of the incident wave,
- ***(b) the angle of incidence,
- (c) the time-domain expressions for the reflected electric and magnetic fields,
- (d) the time-domain expressions for the transmitted electric and magnetic fields, and
- (e) the average power density carried by the wave in the dielectric medium.

8.30 Repeat Problem 8.29 for a wave in air with

$$\widetilde{\mathbf{H}}^i = \hat{\mathbf{y}}\,2 \times 10^{-2}e^{-j(8x+6z)} \quad (\text{A/m})$$

incident upon the planar boundary of a dielectric medium ($z \geq 0$) with $\epsilon_r = 9$.

8.31 A plane wave in air with

$$\widetilde{\mathbf{E}}^i = (\hat{\mathbf{x}}9 - \hat{\mathbf{y}}4 - \hat{\mathbf{z}}6)e^{-j(2x+3z)} \quad (\text{V/m})$$

is incident upon the planar surface of a dielectric material, with $\epsilon_r = 2.25$, and it occupies the half-space $z \geq 0$. Determine:

- ***(a) the incidence angle θ_i ,
- (b) the frequency of the wave,
- (c) the field $\widetilde{\mathbf{E}}^r$ of the reflected wave,
- (d) the field $\widetilde{\mathbf{E}}^t$ of the wave transmitted into the dielectric medium, and
- (e) the average power density carried by the wave into the dielectric medium.

8.32 Natural light is randomly polarized, which means that, on average, half the light energy is polarized along any given direction (in the plane orthogonal to the direction of propagation) and the other half of the energy is polarized along the direction orthogonal to the first polarization direction. Hence, when treating natural light incident upon a planar boundary, we can consider half of its energy to be in the form of parallel-polarized waves and the other half as perpendicularly polarized

waves. Determine the fraction of the incident power reflected by the planar surface of a piece of glass with $n = 1.5$ when illuminated by natural light at 70° .

***8.33** A parallel-polarized plane wave is incident from air onto a dielectric medium with $\epsilon_r = 9$ at the Brewster angle. What is the refraction angle?

8.34 Repeat Problem 8.33 using Module 8.3.

8.35 A perpendicularly polarized wave in air is obliquely incident upon a planar glass–air interface at an incidence angle of 30° . The wave frequency is 600 THz (1 THz = 10^{12} Hz), which corresponds to green light, and the index of refraction of the glass is 1.6. If the electric field amplitude of the incident wave is 50 V/m, determine:

- (a) the reflection and transmission coefficients, and
- (b) the instantaneous expressions for $\mathbf{E}$ and $\mathbf{H}$ in the glass medium.

8.36 Repeat part (a) of Problem 8.35 using Module 8.3.

8.37 Show that the reflection coefficient $\Gamma_\perp$ can be written in the following form:

$$\Gamma_\perp = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)}$$

8.38 Show that for nonmagnetic media, the reflection coefficient $\Gamma_\parallel$ can be written in the following form:

$$\Gamma_\parallel = \frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)}.$$

***8.39** A parallel-polarized beam of light with an electric field amplitude of 10 (V/m) is incident in air on polystyrene with $\mu_r = 1$ and $\epsilon_r = 2.6$. If the incidence angle at the air–polystyrene planar boundary is 50° , determine:

- (a) the reflectivity and transmissivity, and
- (b) the power carried by the incident, reflected, and transmitted beams if the spot on the boundary illuminated by the incident beam is 1 m² in area.

8.40 Repeat part (a) of Problem 8.39 using Module 8.3.

8.41 A 50-MHz right-hand circularly polarized plane wave with an electric field modulus of 30 V/m is normally incident in air upon a dielectric medium with $\epsilon_r = 9$ and occupying the region defined by $z \geq 0$.

- (a) Write an expression for the electric field phasor of the incident wave given that the field is a positive maximum at $z = 0$ and $t = 0$.

使用模块 8.2 评分并确认您的结果。工作频率为100太赫兹。

8.28 3 公里长的光纤使用 $n_f = 1.51$ 的纤芯和 $n_c = 1.48$ 的包层。使用模块 8.2 计算最大数据速率并确认您的结果。工作频率为300太赫兹。

8.29 空气中的平面波

$$\widetilde{\mathbf{E}}_i = \hat{\mathbf{y}}\,20e^{-j(3x+4z)} \quad (\text{V/m})$$

入射到介电材料的平面上， $\epsilon_r = 4$ ，它占据半空间 $z \geq 0$ 。确定：

- (a) 入射波的偏振，
- ***(b) 入射角，
- (c) 反射电场和磁场的时域表达式，(d) 透射电场和磁场的时域表达式，以及 (e) 介电介质中波所携带的平均功率密度。

8.30 对于空气中的波，重复问题 8.29

$$\widetilde{\mathbf{H}}^i = \hat{\mathbf{y}}\,2 \times 10^{-2}e^{-j(8x+6z)} \quad (\text{A/m})$$

入射到 $\epsilon_r = 9$ 的介电介质 ($z \geq 0$) 的平面边界上。

8.31 空气中的平面波

$$\widetilde{\mathbf{E}}^i = (\hat{\mathbf{x}}9 - \hat{\mathbf{y}}4 - \hat{\mathbf{z}}6)e^{-j(2x+3z)} \quad (\text{V/m})$$

入射到介电材料的平面上， $\epsilon_r = 2.25$ ，它占据半空间 $z \geq 0$ 。确定：

- ***(a) 入射角 θ_i ，
- (b) 波的频率，
- (c) 反射波的场 $\widetilde{\mathbf{E}}^r$ ，
- (d) 传输到介电介质中的波的场 $\widetilde{\mathbf{E}}^t$ ，以及 (e) 进入介电介质的波所携带的平均功率密度。

8.32 自然光是随机偏振的，这意味着平均而言，一半的光能沿任何给定方向（在与传播方向正交的平面内）偏振，另一半能量沿与第一个偏振方向正交的方向偏振。因此，当处理入射到平面边界上的自然光时，我们可以认为其能量的一半为平行偏振波，另一半为垂直偏振波。

波浪。确定在 70° 的自然光照射下， $n = 1.5$ 的玻璃平面反射的入射功率的分数。

***8.33** 平行偏振平面波从空气以布儒斯特角入射到 $\epsilon_r = 9$ 的介电介质上。什么是折射角？

8.34 使用模块 8.3 重复问题 8.33。

8.35 空气中的垂直偏振波以 30° 的入射角倾斜入射到平面玻璃-空气界面上。波频率为600 THz（1 THz = 10^{12} Hz），对应于绿光，玻璃的折射率为1.6。如果入射波的电场幅度为50 V/m，则确定：

- (a) 反射系数和透射系数，以及 (b) 玻璃介质中 E 和 H 的瞬时表达式。

8.36 使用模块 8.3 重复问题 8.35 的 (a) 部分。

8.37 证明反射系数 $\Gamma_\perp$ 可以写成以下形式：

$$\Gamma_\perp = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)}$$

8.38 证明对于非磁性介质，反射系数 $\Gamma_\parallel$ 可以写成以下形式：

$$\Gamma_\parallel = \frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)}.$$

***8.39** 电场强度为 10 (V/m) 的平行偏振光束在空气中入射到聚苯乙烯上， $\mu_r = 1$ 和 $\epsilon_r = 2.6$ 。如果空气-聚苯乙烯平面边界处的入射角为 50° ，则确定：

- (a) 反射率和透射率，以及 (b) 如果入射光束照射的边界上的点面积为 1 m²，则入射光束、反射光束和透射光束携带的功率。

8.40 使用模块 8.3 重复问题 8.39 的 (a) 部分。

8.41 电场量为 30 V/m 的 50 MHz 右旋圆极化平面波通常在空气中入射到 $\epsilon_r = 9$ 的介电介质上，并占据由 $z \geq 0$ 定义的区域。

- (a) 假设电场在 $z = 0$ 和 $t = 0$ 处为正最大值，请写出入射波电场相量的表达式。

- (b) Calculate the reflection and transmission coefficients.
- (c) Write expressions for the electric field phasors of the reflected wave, the transmitted wave, and the total field in the region $z \leq 0$.
- (d) Determine the percentages of the incident average power reflected by the boundary and transmitted into the second medium.
- 8.42 Consider a flat 5-mm thick slab of glass with $\epsilon_r = 2.56$.
- *(a) If a beam of green light ($\lambda_0 = 0.52 \mu\text{m}$) is normally incident upon one of the sides of the slab, what percentage of the incident power is reflected back by the glass?
- (b) To eliminate reflections, it is desired to add a thin layer of antireflection coating material on each side of the glass. If you are at liberty to specify the thickness of the antireflection material as well as its relative permittivity, what would these specifications be?

8.43 A perpendicular polarization wave in air is incident upon the surface of a perfect conductor at an incidence angle of 30° . The frequency of the wave is 2 GHz. Use Module 8.1 to determine the location of the nearest electric field maximum from the conductor surface.

8.44 A 2 GHz plane wave in air is incident upon a conducting medium with $\epsilon_r = 4$ and $\sigma = 2 \text{ S/m}$ at an incidence angle of 60° . The wave is parallel-polarized. Use Module 8.4 to compute:

- (a) the reflectivity and transmissivity, and
- (b) the effective refraction angle.

8.45 A 2 GHz plane wave in air is incident upon a conducting medium with $\epsilon_r = 4$ and $\sigma = 2 \text{ S/m}$ at an incidence angle of 60° . The wave is perpendicularly polarized. Use Module 8.4 to compute:

- (a) the reflectivity and transmissivity, and
- (b) the effective refraction angle.

Sections 8-6 to 8-11: Waveguides and Resonators

8.46 Derive Eq. (8.89b).

*8.47 A hollow, rectangular waveguide is to be used to transmit signals at a carrier frequency of 6 GHz. Choose its dimensions so that the cutoff frequency of the dominant TE mode is lower than the carrier by 25% and that of the next mode is at least 25% higher than the carrier.

8.48 A TE wave propagating in a dielectric-filled waveguide of unknown permittivity has dimensions $a = 5 \text{ cm}$ and $b = 3 \text{ cm}$. If the x component of its electric field is given by

$$E_x = -36 \cos(40\pi x) \sin(100\pi y) \\ \cdot \sin(2.4\pi \times 10^{10}t - 52.9\pi z), \quad (\text{V/m})$$

determine:

- (a) the mode number,
- (b) ϵ_r of the material in the guide,
- (c) the cutoff frequency, and
- (d) the expression for H_y .

*8.49 A waveguide filled with a material whose $\epsilon_r = 2.25$ has dimensions $a = 2 \text{ cm}$ and $b = 1.4 \text{ cm}$. If the guide is to transmit 10.5 GHz signals, what possible modes can be used for the transmission?

8.50 For a rectangular waveguide operating in the TE_{10} mode, obtain expressions for the surface charge density $\tilde{\rho}_s$ and surface current density $\tilde{\mathbf{J}}_s$ on each of the four walls of the guide.

*8.51 A waveguide with dimensions $a = 1 \text{ cm}$ and $b = 0.7 \text{ cm}$ is to be used at 20 GHz. Determine the wave impedance for the dominant mode when

- (a) the guide is empty, and
- (b) the guide is filled with polyethylene (whose $\epsilon_r = 2.25$).

8.52 A narrow rectangular pulse superimposed on a carrier with a frequency of 9.5 GHz was used to excite all possible modes in a hollow guide with $a = 3 \text{ cm}$ and $b = 2.0 \text{ cm}$. If the guide is 100 m in length, how long will it take each of the excited modes to arrive at the receiving end?

*8.53 If the zigzag angle θ' is 25° for the TE_{10} mode, what would it be for the TE_{20} mode?

8.54 Measurement of the TE_{101} frequency response of an air-filled cubic cavity revealed that its Q is 4802. If its volume is 64 mm^3 , what material are its sides made of? [*Hint*: See Appendix B.]

8.55 A hollow cavity made of aluminum has dimensions $a = 4 \text{ cm}$ and $d = 3 \text{ cm}$. Calculate Q of the TE_{101} mode for

- *(a) $b = 2 \text{ cm}$, and
- (b) $b = 3 \text{ cm}$.

8.56 A hollow rectangular waveguide with dimensions $a = 2 \text{ cm}$ and $b = 1 \text{ cm}$ is used to propagate signals at 20 GHz. Use Module 8.5 to determine:

- (a) the cutoff frequency of TE_{10} mode, and
- (b) all possible transmission modes.

8.57 A hollow rectangular waveguide with dimensions $a = 4 \text{ cm}$ and $b = 2 \text{ cm}$ is used to propagate signals at 12 GHz. Use Module 8.5 to determine:

- (a) the cutoff frequency of TE_{10} mode, and
- (b) all possible transmission modes.

(b) 计算反射和透射系数。(c) 写出反射波、透射波和 $z \leq 0$ 区域中总场的电场相量表达式。(d) 确定边界反射并传输到第二介质中的入射平均功率的百分比。8.42 考虑一块 5 毫米厚的玻璃板，其 $\epsilon_r =$ 为 2.56。

*(a) 如果一束绿光 ($\lambda_0 = 0.52 \mu\text{m}$) 垂直入射到平板的一侧，则玻璃反射回来的入射功率的百分比是多少？

(b) 为了消除反射，需要在玻璃的每一面添加一层薄薄的减反射涂层材料。如果您可以自由指定抗反射材料的厚度及其相对介电常数，那么这些规格是什么？

8.43 空气中的垂直极化波以 30° 的入射角入射到理想导体的表面。波的频率为 2 GHz。使用模块 8.1 确定距导体表面最近的电场最大值的位置。

8.44 空气中的 2 GHz 平面波以 60° 的入射角入射到 $\epsilon_r = 4$ 和 $\sigma = 2 \text{ S/m}$ 的导电介质上。该波是平行偏振的。使用模块 8.4 计算：

- (a) 反射率和透射率，以及
- (b) 有效折射角。

8.45 空气中的 2 GHz 平面波以 60° 的入射角入射到 $\epsilon_r = 4$ 和 $\sigma = 2 \text{ S/m}$ 的导电介质上。波是垂直偏振的。使用模块 8.4 计算：

- (a) 反射率和透射率，以及 (b) 有效折射角。

第 8-6 节到 8-11 节：波导和谐振器

8.46 推导方程(8.89b)。

*8.47 空心矩形波导用于传输载波频率为 6 GHz 的信号。选择其尺寸，使主 TE 模式的截止频率比载波低 25%，而下一个模式的截止频率至少比载波高 25%。

8.48 在介电常数未知的电介质填充波导中传播的 TE 波的尺寸为 $a = 5 \text{ cm}$ 和 $b = 3 \text{ cm}$ 。如果其电场的 x 分量由下式给出

$$E_x = -36 \cos(40\pi x) \sin(100\pi y) \\ \cdot \sin(2.4\pi \times 10^{10}t - 52.9\pi z), \quad (\text{V/m})$$

决定：

- (a) 模数，(b) 波导中材料的 ϵ_r ,
- (c) 截止频率，以及 (d) H_y 的表达式。

*8.49 填充有 $\epsilon_r = 2.25$ 材料的波导，其尺寸为 $a = 2 \text{ cm}$ 和 $b = 1.4 \text{ cm}$ 。如果要传输10.5GHz信号，可以采用哪些模式来传输？

8.50 对于工作在 TE_{10} 模式的矩形波导，求得波导四个壁上的表面电荷密度 $\tilde{\rho}_s$ 和表面电流密度 $\tilde{\mathbf{J}}_s$ 的表达式。

*8.51 在 20 GHz 下使用尺寸 $a = 1 \text{ cm}$ 和 $b = 0.7 \text{ cm}$ 的波导。确定主模式的波阻抗

- (a) 指南是空的，并且
- (b) 导向器填充有聚乙烯（其 $\epsilon_r = 2.25$ ）。

8.52 使用叠加在频率为 9.5 GHz 的载波上的窄矩形脉冲来激励 $a = 3 \text{ cm}$ 和 $b = 2.0 \text{ cm}$ 的空心波导中的所有可能模式。如果波导长度为 100 m，则每个激发模式需要多长时间才能到达接收端？

*8.53 如果锯齿角 θ' 是 25° 对于 TE_{10} 模式，对于 TE 来说会是什么？₂₀模式？

8.54 对充气立方体空腔的 TE_{101} 频率响应的测量表明其 Q 为 4802。如果其体积为 64 mm^3 ，其侧面由什么材料制成？[提示：请参阅附录 B。]

8.55 铝制空腔的尺寸 $a = 4 \text{ cm}$ ， $d = 3 \text{ cm}$ 。计算 TE_{101} 模式的 Q

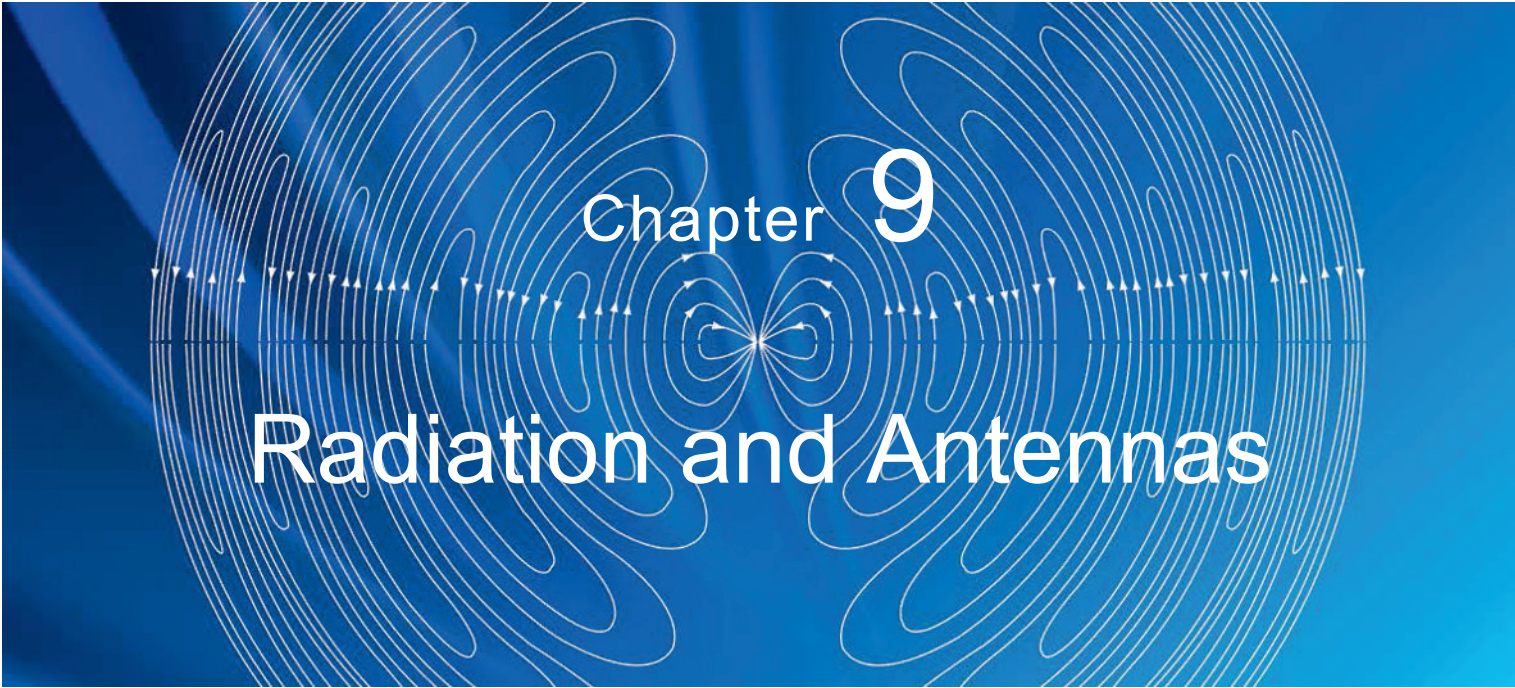
- *(a)b=2厘米，和(
- b)b=3厘米。

8.56 使用尺寸 $a = 2 \text{ cm}$ 和 $b = 1 \text{ cm}$ 的空心矩形波导来传播 20 GHz 的信号。使用模块 8.5 确定：

- (a) TE_{10} 模式的截止频率，以及 (b) 所有可能的传输模式。

8.57 使用尺寸 $a = 4 \text{ cm}$ 和 $b = 2 \text{ cm}$ 的空心矩形波导来传播 12 GHz 信号。使用模块 8.5 确定：

- (a) TE_{10} 模式的截止频率，以及 (b) 所有可能的传输模式。



Chapter Contents

	Overview, 395
9-1	The Hertzian Dipole, 387
9-2	Antenna Radiation Characteristics, 390
9-3	Half-Wave Dipole Antenna, 397
9-4	Dipole of Arbitrary Length, 400
9-5	Effective Area of a Receiving Antenna, 401
9-6	Friis Transmission Formula, 403
TB17	Health Risks of EM Fields, 405
9-7	Radiation by Large-Aperture Antennas, 408
9-8	Rectangular Aperture with Uniform Aperture Distribution, 410
9-9	Antenna Arrays, 412
9-10	<i>N</i> -Element Array with Uniform Phase Distribution, 419
9-11	Electronic Scanning of Arrays, 421
	Chapter 9 Summary, 426
	Problems, 428

Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Calculate the electric and magnetic fields of waves radiated by a dipole antenna.
2. Characterize the radiation of an antenna in terms of its radiation pattern, directivity, beamwidth, and radiation resistance.
3. Apply the Friis transmission formula to a free-space communication system.
4. Calculate the electric and magnetic fields of waves radiated by aperture antennas.
5. Calculate the radiation pattern of multi-element antenna arrays.



章节内容

概述、395 9-1 赫兹偶极子、387 9-2 天线辐射特性、390 9-3 半波偶极子天线、397 9-4 任意长度的偶极子、400 9-5 接收天线的有效面积、401 9-6 弗里斯传输公式、403 TB17 电磁波的健康风险场、405 9-7 大孔径天线辐射、408 9-8 孔径分布均匀的矩形孔径、410 9-9 天线阵列、412 9-10 相位分布均匀的 *N* 元阵列、419 9-11 阵列电子扫描、421 第 9 章总结、426 问题、428

目标

学习本章介绍的材料后，您应该能够：

1. 计算偶极天线辐射的电场和磁场。
2. 通过辐射方向图、方向性、波束宽度和辐射电阻来表征天线的辐射。
3. 将 Friis 传输公式应用于自由空间通信系统。
4. 计算孔径天线辐射波的电场和磁场。
5. 计算多元件天线阵列的辐射方向图。

Overview

An **antenna** is a transducer that converts a guided wave propagating on a transmission line into an electromagnetic wave propagating in an unbounded medium (usually free space), or vice versa. **Figure 9-1** shows how a wave is launched by a hornlike antenna with the horn acting as the transition segment between the waveguide and free space.

Antennas are made in various shapes and sizes (**Fig. 9-2**) and are used in radio and television broadcasting and reception, radio-wave communication systems, cellular telephones, radar systems, anticollision automobile sensors, and many other applications. The radiation and impedance properties of an antenna are governed by its shape, size, and material properties. The dimensions of an antenna are usually measured in units of λ of the wave it is launching or receiving; a 1-m long dipole antenna operating at a wavelength $\lambda = 2$ m exhibits the same properties as a 1-cm long dipole operating at $\lambda = 2$ cm. Hence, in most of our discussions in this chapter, we refer to antenna dimensions in wavelength units.

Reciprocity

The directional function characterizing the *relative* distribution of power radiated by an antenna is known as the **antenna radiation pattern** (or simply the **antenna pattern**). An **isotropic** antenna is a hypothetical antenna that radiates equally in all directions, and it is often used as a reference radiator when describing the radiation properties of real antennas.

► Most antennas are **reciprocal** devices, exhibiting the same radiation pattern for transmission as for reception. ◀

Reciprocity means that, if in the transmission mode a given antenna transmits in direction *A* 100 times the power it transmits in direction *B*, then when used in the reception mode it is 100 times more sensitive to electromagnetic radiation incident from direction *A* than from *B*. All of the antennas shown in **Fig. 9-2** obey the reciprocity law, but not all antennas are reciprocal devices. Reciprocity may not hold for some solid-state antennas composed of nonlinear semiconductors or ferrite materials. Such nonreciprocal antennas are beyond the scope of this chapter; hence, reciprocity is assumed throughout. The reciprocity property is very convenient because it allows us to compute the radiation pattern of an antenna in the transmission mode—even when the antenna is intended to operate as a receiver.

To fully characterize an antenna, one needs to study its radiation properties and impedance. The radiation properties include its directional radiation pattern and the associated polarization state of the radiated wave when the antenna

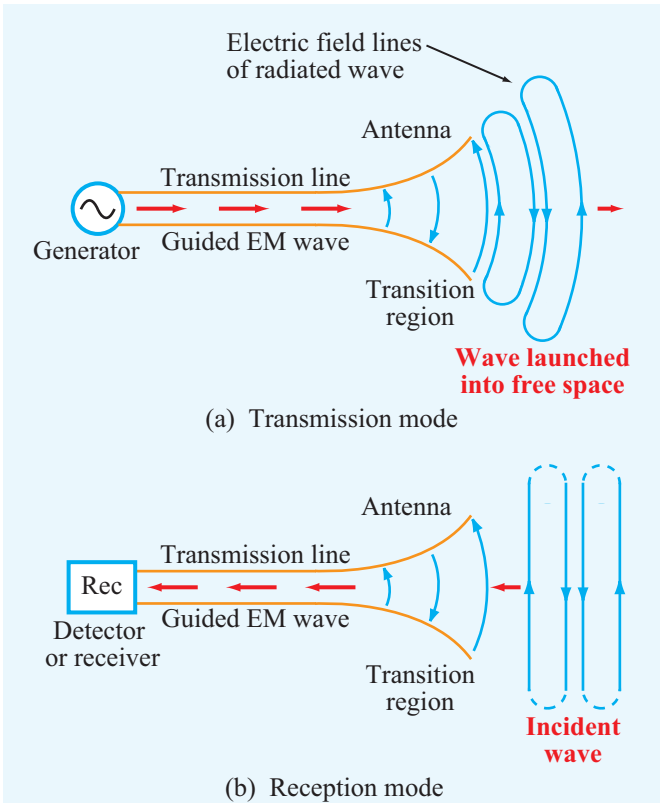


Figure 9-1 Antenna as a transducer between a guided electromagnetic wave and a free-space wave, for both transmission and reception.

is used in the transmission mode, which is also called the **antenna polarization**.

► Being a reciprocal device, an antenna, when operating in the receiving mode, can extract from an incident wave only that component of the wave whose electric field matches the antenna polarization state. ◀

The second aspect, the **antenna impedance**, pertains to the transfer of power from a generator to the antenna when the antenna is used as a transmitter and, conversely, the transfer of power from the antenna to a load when the antenna is used as a receiver, as will be discussed later in Section 9-5. It should be noted that throughout our discussions in this chapter it will be assumed that the antenna is properly matched to the transmission line connected to its terminals, thereby avoiding reflections and their associated problems.

概述

天线是一种换能器，可将传输线上传播的导波转换为在无界介质（通常是自由空间）中传播的电磁波，反之亦然。图 9-1 显示了如何通过喇叭状天线发射波，其中喇叭充当波导和自由空间之间的过渡段。

天线有各种形状和尺寸（图9-2），用于无线电和电视广播和接收、无线电通信系统、蜂窝电话、雷达系统、防碰撞汽车传感器和许多其他应用。天线的辐射和阻抗特性由其形状、尺寸和材料特性决定。天线的尺寸通常以其发射或接收的波的 λ 为单位进行测量；在波长 $\lambda = 2$ m 下工作的 1 米长偶极子天线表现出与在 $\lambda = 2$ cm 下工作的 1 厘米长偶极子天线相同的特性。因此，在本章的大部分讨论中，我们提到的天线尺寸都是以波长为单位的。

互易

表征天线辐射功率相对分布的方向函数称为天线辐射方向图（或简称为天线方向图）。各向同性天线是一种在所有方向上均匀辐射的假想天线，在描述实际天线的辐射特性时通常将其用作参考辐射器。

► 大多数天线都是互易设备，在发射和接收时表现出相同的辐射方向图。 ◀

互易性是指，如果在发射模式下，给定天线在 *A* 方向上发射的功率是在 *B* 方向上发射的功率的 100 倍，那么当用于接收模式时，它对从 *A* 方向入射的电磁辐射的敏感度比从 *B* 方向入射的电磁辐射敏感 100 倍。图 9-2 中所示的所有天线都遵循互易定律，但并非所有天线都是互易器件。对于某些由非线性半导体或铁氧体材料组成的固态天线，互易性可能不成立。这种非互易天线超出了本章的范围；因此，自始至终都假定互易。互易性非常方便，因为它允许我们计算传输模式下天线的辐射方向图——即使天线打算用作接收器。

为了全面表征天线，需要研究其辐射特性和阻抗。辐射特性包括天线的定向辐射图和辐射波的相关极化状态。

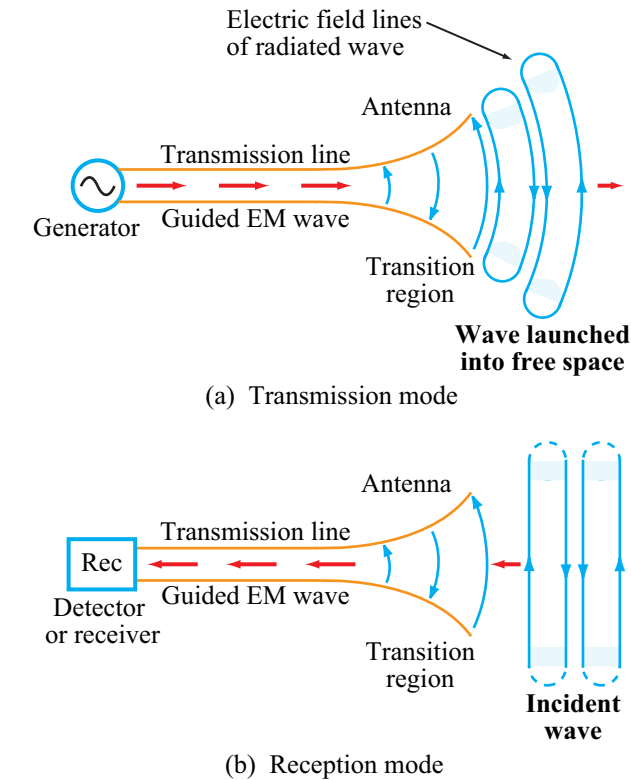


图 9-1 天线作为引导电磁波和自由空间波之间的换能器，用于传输和接收。

用于传输模式，也称为天线极化。

► 作为一种互易装置，天线在接收模式下工作时，只能从入射波中提取电场与天线极化状态匹配的波分量。 ◀

第二个方面，即天线阻抗，涉及当天线用作发射器时从发电机到天线的功率传输，以及相反，当天线用作接收器时从天线到负载的功率传输，稍后将在第 9-5 节中讨论。应该注意的是，在本章的整个讨论中，我们都假设天线与连接到其端子的传输线正确匹配，从而避免反射及其相关问题。